Online Social Networks and Long-Run Career Success*

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Abstract

How does access to online social networks at labor market entry affect young workers' careers? I study this question by exploiting the staggered rollout of Facebook across U.S. colleges in a difference-in-differences design, tracking the career trajectories of nearly 800,000 business and economics students through their public LinkedIn profiles. I show that early access to Facebook leads to more successful careers: exposed students work in higher-paying jobs, are 3.8% more likely to receive early-career promotions, and are 3.4% more likely to reach senior leadership roles in their late 30s. I find that these effects can be attributed to (i) a more successful college-to-work transition, primarily driven by help from college peers in accessing higher-quality early-career jobs, (ii) more stable and productive early-career worker-firm matches, and (iii) intensified early-career sorting into higher-quality and career-supporting firms. The results highlight how access to online social networks during a sensitive period of a worker's career can reduce information frictions, facilitate the college-to-work transition, and generate long-lasting career benefits.

Keywords: online social networks, Facebook, college graduates, labor market entry, career development, leadership, firms, networking

JEL codes: J24, J30, J62, I26, D83, L86

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1 Introduction

How do online social networks affect workers' careers? Platforms such as Facebook and LinkedIn have become an essential part of modern labor markets, with over 3 billion people worldwide using them for work-related purposes (Kemp 2025). Compared to prior technologies, they increase the flow of information and reduce the cost of adding and maintaining social connections (e.g., Bakshy *et al.* 2012; Armona 2025; Armona and Bailey 2025). Given the well-established evidence that social connections are valuable for job search and career advancement (e.g., Simon and Warner 1992; Granovetter 1995; Calvó-Armengol and Jackson 2004; Topa 2011; Schmutte 2015; Pallais and Sands 2016; Rajkumar *et al.* 2022; Cullen and Perez-Truglia 2023), providing workers with access to these platforms could reduce information frictions and improve labor market prospects. This may be especially relevant for younger workers, who have less experience with the job search process and face greater information frictions (e.g., Altonji and Pierret 2001; Pallais 2014). Yet we know surprisingly little about whether online social network access improves young workers' transition into the labor market and how it shapes their careers in the long run.

In this paper, I provide novel evidence on the causal impact of access to online social networking platforms on the initial job placement and career trajectories of young workers. To isolate plausibly exogenous variation, I exploit a natural experiment: the staggered introduction of Facebook across U.S. colleges in the mid-2000s, before it became publicly available in September 2006. As some college cohorts during the rollout phase graduated before their college gained access, the variation allows me to capture the effect of having access to Facebook right before vs. after labor market entry, a sensitive period of a worker's career that can have long-lasting consequences for labor market outcomes (e.g., von Wachter 2020).[1]

I use two main datasets for the analysis. The first comprises the introduction dates of Facebook across the first 842 colleges (Armona 2025), which allows me to identify the academic year in which each U.S. college received access. The second dataset, provided by Revelio Labs, contains detailed CV data from public LinkedIn profiles. The LinkedIn data include two critical pieces of information that administrative data do not combine at this scale. First, the education histories provide the name of the postsecondary institution as well as start and end dates, allowing me to identify which students were directly affected by Facebook's staggered introduction. Second, the employment histories contain information on job titles, job-based earnings, and company names, allowing me to track the career trajectories of nearly 800,000

[1]The labor market entry phase is also the point in time when many universities and career services recommend starting to use online social networking platforms for work-related purposes (e.g., School of Information, San Jose State University 2018).

U.S. business and economics students in great detail.[2] I use the information available on the profiles to measure job mobility and career development (e.g., promotions and the attainment of leadership roles), construct networking indicators, and characterize job and firm choices. This allows me to track how the effects of early access to online social networks propagate through a worker's career from labor market entry up to 19 years after graduation.

To estimate the causal impact of Facebook access at labor market entry on career outcomes, I use a generalized difference-in-differences design. Founded at Harvard in February 2004, Facebook was gradually introduced to other U.S. colleges before opening to the public in September 2006, creating sharp differences in availability across colleges and cohorts. Due to factors such as limited server capacity (Kirkpatrick 2011) and the desire to cultivate a sense of exclusivity (Braghieri *et al.* 2022), the rollout of Facebook was not random but targeted elite and highly selective colleges first. In the main specification, I therefore exclusively rely on the plausibly exogenous variation within each selectivity-by-region cell, narrowing down the comparison to treated and not-yet-treated college cohorts within the same selectivity and region. This strategy accounts for various endogeneity concerns such as spurious correlations with the non-random timing of the Facebook rollout, time-invariant college-specific characteristics (e.g., prestige), and cohort-specific shocks within the same region at colleges of similar selectivity (e.g., fluctuations in labor market entry conditions and returns to education). To address potential biases in staggered difference-in-differences estimation using two-way fixed effects (Goodman-Bacon 2021), I apply the heterogeneity-robust imputation estimator by Borusyak *et al.* (2024).[3]

The main finding of the paper is that access to Facebook at labor market entry generates long-lasting career benefits. Students with early access to Facebook are 3.8% more likely to receive early-career promotions and 3.4% more likely to reach senior leadership positions in their late 30s. They also tend to work in higher-paying jobs, resulting in an increase in cumulative job-based earnings of around 1.6% over the first 19 years after graduation, which is more than half the size of the lifetime earnings loss that college graduates can expect when graduating in a large recession (e.g., von Wachter 2020).

Investigating potential mechanisms behind the long-run career effects, I show that treated students have a more successful college-to-work transition (the first two years after expected labor market entry) at the employment, firm quality, occupation, industry, and job margins. Students are, on average, 1.5% more

[2] I focus on business and economics students because their LinkedIn coverage is by far the highest among all fields and they represent the most popular U.S. undergraduate major.

[3] Appendix F presents the main results using the heterogeneity-robust difference-in-differences estimators of Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020), which differ slightly in their identifying assumptions. The results are very similar.

likely to be employed, 8.9% more likely to begin their careers in high-wage firms, and 7.6% more likely to obtain jobs at firms offering high career support (as indicated by employer ratings on Glassdoor). In addition, treated students are 4.9% (8.6%) more likely to start their careers in high-paying occupations (industries) and 9% more likely to find early-career jobs with high chances of reaching a senior leadership position. They also tend to find their first stable job at a faster rate. The college-to-work transition effects are not explained by graduation timing or postgraduate education choices and are not apparent for lower-quality early-career jobs.

I further show that the improved college-to-work transition primarily reflects more help from college peers in accessing higher-quality early-career jobs. Using a job help proxy based on relational firm mobility patterns, I find that the entire employment effect can be attributed to an increase in early-career job help through the college network, with the largest relative increases in help for jobs at higher-quality firms. I also show that treated students are particularly more likely to choose typical college career paths, suggesting that early access to Facebook ultimately strengthened the core of established college-career pipelines. There is little evidence that targeted employer ads or increased geographic mobility drive a meaningful part of the improvements in the college-to-work transition.

Looking at potential longer-run treatment dynamics, I do not find evidence that the career benefits are attributable to a long-run information advantage in the search for higher-paying jobs. While early Facebook adopters have larger social networks in the long run (Armona and Bailey 2025), the estimated impact on between-firm mobility and promotion patterns is close to zero. This holds even when combining the between-firm mobility outcomes with job help indicators for the college network. In the spirit of traditional job search models (e.g., Burdett 1978), this finding suggests that early access to Facebook did not lead to a permanent increase in the rate of job offer arrivals. This is consistent with students in the control group creating a Facebook account shortly after it became publicly available, reducing differences in the flow of work-related information in the long run.[4]

Instead, I find evidence that part of the career benefits are explained by improved stability and productivity of early-career worker-firm matches, enabling students to climb up the career ladder within their organizations. In the first six years after graduation, treated students are 1% more likely to be in stable worker-firm matches and accumulate 0.35 months more firm tenure on average. The firm tenure effect persists in the longer run. Consistent with more stable and productive early-career worker-firm matches,

[4]Consistent with this finding, Armona and Bailey (2025) show that the positive network size effects decay over time, Facebook was also highly popular among the control group students, and there are no detectable long-run differences in terms of the intensity of Facebook usage between the treatment and control groups.

treated students are 13.0% more likely to receive early-career promotions within their firm, while there is no impact on within-firm demotion patterns.

Part of the documented career benefits can also be explained by intensified sorting into firms that provide better learning opportunities and career support. Consistent with the college-to-work transition effects at the firm quality margin, students with early access to Facebook accumulate 7.1% (3.7%) more work experience at high-wage firms in the first six (twelve) years after graduation. They also accumulate more work experience in firms that offer better career support. Given the recent evidence by Arellano-Bover (2022), Gregory (2023), and Arellano-Bover and Saltiel (2026) that firms differ substantially in the learning opportunities they provide and that heterogeneous human capital accumulation across firms can explain a considerable part of life-cycle differences in career outcomes, the findings suggest that faster on-the-job skill accumulation at higher-quality firms explains part of the career benefits. There is no evidence that a long-run reduction in unemployment risk drives the career benefits or that the career effects differ markedly between men and women.

The findings are robust to a series of sensitivity checks. One key concern with the identification strategy is that the results may be driven by treated students taking away jobs from students in the control group, thereby upward-biasing the treatment effects. To assess the severity of this concern, I predict the control cohorts' exposure to spillovers from the historical career overlap of students across colleges and drop every college with at least one highly exposed control cohort. The results are very similar to the baseline estimates, suggesting that the documented career benefits of early access to Facebook do not simply reflect a reallocation of high-quality jobs but are instead consistent with a reduction in information frictions. Another concern is that the selectivity-by-region cells may not be sufficient to account for Facebook's non-random rollout strategy and may pick up spurious correlations within those cells. To test this, I randomly reshuffle the Facebook introduction dates across colleges within each selectivity-by-region cell. The placebo estimates are tightly clustered around zero, with all statistically significant treatment effects above the 95th percentile of the placebo distribution.

The results of this paper contribute to the large literature on the individual and societal impacts of online social networking platforms. Several studies have documented negative consequences, such as increased mental health problems (Braghieri *et al.* 2022), greater political polarization (Zhuravskaya *et al.* 2020), the spread of misinformation (Allcott and Gentzkow 2017), and the amplification of hate speech (Müller and Schwarz 2021). Other papers highlight potential benefits, such as increased collective action and protest participation (Fergusson and Molina 2019; Enikolopov *et al.* 2020). This paper contributes

by showing that access to online social networks can generate economically meaningful and long-lasting career benefits. There are three more closely related papers in the labor market context. Wheeler *et al.* (2022) show that providing young job seekers with training to use LinkedIn improves their short-run job-finding rate. Armona (2025) exploits the same variation as this paper to show that access to Facebook at labor market entry increases the earnings rank of exposed college cohorts in 2014, providing indirect evidence that the effect stems from stronger social ties among college peers and improved horizontal matching between major and occupation. Armona and Bailey (2025) use the same variation coupled with Facebook data to show that students with early access to Facebook have larger and more diverse networks in the long run, are more likely to sort into areas with higher economic opportunity, and work in higher-paying occupations. I contribute to this literature by showing that access to online social networks at labor market entry can improve workers' initial job placement and shape their long-run career progression. I further improve the understanding of the *dynamics* and *mechanisms* behind the labor market effects of early access to Facebook documented so far.

The findings of this paper also speak to the literature on the role of college networks in labor market outcomes. Social connections among college peers have been shown to affect early-career job placement (Marmaros and Sacerdote 2002; Zhu 2025; Gompers *et al.* 2026), channel students into top jobs (Kramarz and Thesmar 2013; Zimmerman 2019; Michelman *et al.* 2022; Fischer *et al.* 2025; Hampole *et al.* 2026), mitigate the negative consequences of job loss due to layoffs (Ost *et al.* 2026), and affect managerial decision-making (Lerner and Malmendier 2013; Shue 2013). This paper highlights that access to online social networks for college students increases the *value* of the college network as an information system for accessing early-career jobs and *shifts* their early-career job search toward it.

Finally, this paper contributes to the literature on the role of labor market entry in long-run career success. Kahn (2010), Oreopoulos *et al.* (2012), Altonji *et al.* (2016), Schwandt and von Wachter (2019), and von Wachter (2020) show that graduating in a recession hinders a successful transition to the labor market, reducing the likelihood of working in high-quality jobs and harming workers' careers in the long run. Arellano-Bover (2024) shows that starting a career at a large firm benefits workers' long-run labor market success. This paper shows that online social network access at labor market entry can facilitate the transition into the labor market and improve initial job placement, with lasting effects on career progression.

The rest of the paper is organized as follows. Section 2 provides a short history of Facebook. Section 3 describes the data. Section 4 discusses the empirical strategy. Section 5 presents the results. Section 6 concludes.

2 Background

The early expansion of Facebook. Founded in February 2004 at Harvard, Facebook started as an exclusive social networking platform for college students, requiring a verified *.edu* email address to create an account and connect with other students from the same university. Over the following years, Facebook was gradually rolled out to other U.S. colleges and to universities abroad until it became publicly available in September 2006. Appendix Figure A1 shows the rollout of Facebook across all selective U.S. colleges offering four-year business and economics degrees by academic year and college selectivity. The rollout was not random but initially focused on highly selective colleges, driven by limited server capacity (Kirkpatrick 2011), a deliberate strategy to launch in college networks where students already knew each other well offline (Aral 2021), and the desire to cultivate a sense of exclusivity (Braghieri *et al.* 2022).

During the expansion period, Facebook was highly popular among college students, with around 85% of undergraduates creating an account (Arrington 2005) and nearly three-quarters of users logging in daily (Hass 2006). Compared to prior technologies, such as email and instant messaging, Facebook increased the flow of information and reduced the cost of adding and maintaining social connections (e.g., Ellison *et al.* 2007; Bakshy *et al.* 2012). Students used it primarily to keep in touch with friends and learn more about classmates (Lampe *et al.* 2006, 2008), with each additional year of early access increasing the number of Facebook connections within the college network by 13.6% (Armona 2025).[5] Since alumni could sign up with their *.edu* email address, these networks also extended to former students. Early access to Facebook thus primarily enabled students to improve their connectedness within the college network before entering the labor market.

Facebook in a work-related context. While Facebook never fully specialized as a work-related online social networking platform, its features and immense popularity supported work-related activities, such as job search, professional networking, and recruiting, from the very beginning. As early as April 2004, Facebook allowed a small set of employers to advertise jobs and products to specific colleges (Grynbaum 2004), and in April 2006, it enabled students to list their current workplace and introduced workplace networks (Facebook 2006), which around 20–25% of early-adopter U.S. students used to report their employment history (Armona and Bailey 2025). Features addressing job seekers and employers more directly only followed in 2007: *Facebook Marketplace*, allowing users to post job ads within their networks (Facebook 2008), and *Facebook Pages*, enabling employers to create public company profiles (Facebook

[5] Braghieri *et al.* (2022) show that Facebook's introduction also came at a cost by worsening the mental health of some college students.

2007). Over time, Facebook became the most important online social networking platform for job-seeking activities: in 2011, 48% of U.S. job seekers used Facebook for job search and 16% obtained a job referral from a Facebook friend (Jobvite 2011).[6] On the firm side, among U.S. employers using or planning to use online social networking platforms for recruiting, 36% used Facebook in 2008 (Jobvite 2009), a number that increased to 68% in 2021 (Jobvite 2021).

Facebook's rollout and timeline have two important implications for students with early access. First, when they entered the labor market, Facebook's work-related features were still in their infancy, with most employers not yet actively using the platform for recruiting or screening. For their early-career job search, they therefore primarily had access to a larger, more closely connected college network and the supply-side-focused work-related platform features, but were not necessarily exposed to vastly different demand-side information channels.[7] Second, as Facebook became publicly available in September 2006, students from the control group also received access to the technology shortly after their labor market entry. From then on, students with early access differed from the control group mainly in their head start: the larger online network they had accumulated before labor market entry (Armona and Bailey 2025), their longer experience with the platform when the later work-related features were introduced, and the jobs they had obtained at labor market entry. I examine the relevance of these channels and dynamics in Section 5.2.

3 Data

To assess the career effects of access to Facebook at labor market entry, this paper primarily combines four data sources: LinkedIn profile data provided by the company Revelio Labs, the introduction dates of Facebook across the first 842 colleges, the number of awards and demographic composition of college graduation classes from the National Center for Education Statistics, and institutional college characteristics including Barron's selectivity index. This section provides an overview of the data sources, key variables, and descriptive statistics.

3.1 LinkedIn Data

Information on the career trajectories of U.S. students and firm characteristics comes from a snapshot of public LinkedIn data from November 2024. With more than 1 billion members and 67 million companies,

[6] The second-largest platform during this period was LinkedIn, with substantially smaller usage numbers (26% and 9%) due to its lower adoption rate.

[7] The only exception at this stage was targeted employer ads. Section 5.2.1 discusses their relevance in greater detail.

LinkedIn is currently the world's largest professional networking site (LinkedIn 2024). In addition to its extensive global coverage, LinkedIn is particularly popular among U.S. college graduates (Auxier and Anderson 2021). Users create profiles that resemble CVs and contain detailed information about employment history and educational background, with the primary goal of searching for jobs, building professional networks, and increasing their visibility to potential employers (Evsyukova *et al.* 2025). Firms also create profiles that include information about firm characteristics to advertise open positions.

The LinkedIn data comes from the company Revelio Labs. Revelio Labs is a workforce intelligence company that provides an extensive human resource database from various online sources. In the case of LinkedIn, Revelio Labs scrapes information from public user and company profiles and converts it into a structured database using proprietary algorithms.

The employment history section of public LinkedIn profiles contains information on job titles, start and end dates of employment, employer names, and job locations. The educational background section contains information on the postsecondary institution, degree, field of study, and start and end dates. The key advantage of using public LinkedIn profile data is that most of these variables are typically unavailable in public or administrative datasets at this level of detail and scale. The start and end dates of each stage in the CV allow me to construct a retrospective yearly career panel for each student up to 19 years after graduation.[8] At the end of the career panel, students are typically in their early 40s, when career trajectories have largely stabilized (Chetty *et al.* 2020). To ensure adequate coverage of the underlying student population, I focus on business and economics graduates, who have the highest LinkedIn coverage among all fields and represent the most popular U.S. undergraduate major. Below, I describe the key variables derived from the public LinkedIn data.

Job-based earnings. A frequently used measure of workers' career trajectories is earnings (e.g., Bronson and Thoursie 2019; von Wachter 2020). While LinkedIn profiles do not report individual-level earnings, I can use Revelio Labs' job-based earnings imputation to track whether students work in typically higher- or lower-paying jobs over their careers. To generate job-based earnings, Revelio Labs employs a machine learning model that predicts the annual salary of each job stage using information on job title, company name, job location, tenure, and calendar year. The model is trained on publicly available online job postings and visa application data. Dorn *et al.* (2025) shows that the imputed earnings align closely with official earnings data from the American Community Survey and serve as a reliable proxy for job-based earnings. To make the job-based earnings measure comparable across years and cohorts, I

[8]To construct the yearly career panel, I take a snapshot on June 30 of each year, similar to Armona and Bailey (2025).

convert it into real 2024 dollars using the U.S. Consumer Price Index (CPI).

Job-based promotions. To assess career-ladder steps more formally, I use the job-based earnings measure to construct a proxy for promotions. Following Bronson and Thoursie (2019), I define promotions as large relative real wage growth jumps of at least 10 log points. The measure compares the real annual wage growth of a worker to the predicted real annual wage growth the worker would have obtained had they remained in their previous position.[9] Since I do not observe within-job earnings differentials, this measure of promotions essentially captures job-based promotions identified through changes in job titles, firms, or geographic location. Appendix Table A1 and Figure A2 investigate whether the job-based earnings measure by Revelio Labs, combined with the "relative wage growth jump" methodology in Bronson and Thoursie (2019), is capable of detecting promotion patterns in the LinkedIn data. Appendix Table A1 exploits the availability of job titles on LinkedIn profiles to assess the most frequent job title changes coinciding with a promotion. The most common job title transitions closely reflect standard moves up the career ladder. For example, the most common promotion event in the early-career stage reflects a change in job title from "Staff Accountant" to "Senior Accountant," in the mid-career stage from "Financial Analyst" to "Senior Financial Analyst," and in the later-career stage from "Project Manager" to "Senior Project Manager." Appendix Figure A2 further supports the suitability of the promotion measure by showing that the distribution of the imputed real annual wage growth (in log points) closely mirrors the individual-level wage growth distribution in Bronson and Thoursie (2019). The majority of the real annual wage growth distribution is concentrated around zero. Consistent with their evidence, the distribution has a pronounced right tail, indicative of promotion events, with one-tenth of the observations having imputed real annual wage growth above 10%.

Leadership positions. One key advantage of the employment history on LinkedIn profiles is the availability of job titles, which I can use to identify leadership positions as an additional measure of workers' career development that does not rely on the job-based earnings imputation.[10] To do so, I use the keyword-based approach from Hampole *et al.* (2026), which maps job titles into the following managerial career pipeline: individual contributor, manager, director, vice president, senior vice president, and c-suite. Following Hampole *et al.* (2026), I define senior leadership positions as director-level to c-suite-level positions. To validate the mapping, Appendix Table A2 presents basic descriptive statistics for each category in the managerial career pipeline, while Appendix Table A3 lists the most commonly mapped

[9] Since the job-based earnings measure predicts the same earnings trajectory for *all* workers in a given job, this definition coincides with their methodology, which uses coworkers' earnings to predict counterfactual wage growth.

[10] Leadership positions are considered major career steps for workers, coming with a significant increase in responsibility, earnings, and prestige (e.g., Gibbons and Waldman 1999).

job titles. Average job-based annual earnings increase monotonically along the career pipeline, from an average of $84,503 for individual contributor roles to $169,990 for director-level jobs and up to $249,912 for C-suite positions. Consistent with the literature (Lean In and McKinsey & Company 2020; Hampole *et al.* 2026), the share of female U.S. business graduates gradually decreases with each step on the career ladder. In addition, Appendix Table A3 demonstrates that the most commonly mapped job titles closely align with each concept in the managerial pipeline.[11]

Firm choice. Job stages on LinkedIn profiles also contain information on the company name, with Revelio Labs linking each company to the corresponding LinkedIn company profile and Glassdoor amenity ratings. This unique feature of the data allows me to investigate the role of firm choice across various dimensions, which has been shown to be an important driver of workers' career trajectories (e.g., Oreopoulos *et al.* 2012; von Wachter 2020; Arellano-Bover 2024). To analyze firm choice, I focus on several measures of firm quality. Starting with firm pay as the first quality dimension, I combine information on company names with the job-based earnings imputation to classify high-wage and non-high-wage firms following the approach in Oreopoulos *et al.* (2012), Haltiwanger *et al.* (2018), and von Wachter (2020). To do so, I use information from *all* LinkedIn profiles and companies that employ college students in my estimation sample and compute the average median log earnings at the firm level, net of year fixed effects. I then split the employment-weighted firm-earnings distribution into quartiles and define high-wage firms as all firms in the upper quartile of the distribution and non-high-wage firms as all firms below this threshold.[12]

As workers also highly value non-pecuniary aspects of work (e.g., Mas and Pallais 2017; Sorkin 2018; Sockin 2022; Maestas *et al.* 2023; Caldwell *et al.* 2026), I extend the firm quality measures to non-wage attributes using the information on Glassdoor employer ratings.[13] To do so, I use the same employment-weighted quartile split procedure after calculating the average employer rating, net of year fixed effects, to create firm quality proxies for the following dimensions: overall rating, compensation and benefits, career opportunities, culture and values, senior leadership, and work-life balance. Appendix Table A4 shows the coverage, distribution, and relation to the firm pay quality proxy in more detail. I merge the firm amenity characteristics to slightly more than two-thirds of the observed firm choices in the career panel. The upper quartile threshold for the different dimensions is around 3.5 to 4 on the Glassdoor

[11] For example, the most frequent terms for the C-suite category are "President" and "CEO," the most common terms for director-level positions are "Partner" and "Controller," and the most common terms for individual contributor positions are "Account Executive" and "Senior Accountant."

[12] Appendix Table A4 shows the coverage and distribution of the firm pay variable in more detail. Using alternative thresholds produces similar results.

[13] For a detailed discussion on the coverage and validation of the Glassdoor amenity ratings in the U.S., see Sockin (2022).

rating scale from 1 to 5.[14] Looking at the correlation between the different firm quality margins provides several important insights. First, the firm pay and amenity variables are weakly positively correlated (correlations of 0.04–0.1), in line with recent evidence by Sockin (2022) and Arellano-Bover and Saltiel (2026). Second, there is a strong positive correlation between the different Glassdoor amenity rating dimensions (0.63–0.87), indicating that firms tend to offer amenities in a bundle rather than focusing on a single dimension.

Occupational and industry choice. While LinkedIn profiles do not contain occupational or industry codes, Revelio Labs infers them using proprietary algorithms. For 2018 SOC occupation codes, they draw primarily on job titles, reported skills, and job descriptions. For NAICS industry codes, they use the employers' primary industry classification in external corporate datasets, supplemented by firm descriptions. This allows me to analyze occupational and industry choice as further important determinants of career trajectories (e.g., Keane and Wolpin 1997; Krueger and Summers 1988). Similar to the firm quality proxies, I use job-based earnings imputation to construct measures of occupational and industry quality. To do so, I compute the average median log earnings for each 5-digit SOC occupation and each 3-digit NAICS industry in the Revelio Labs data between 2000 and 2024 and perform employment-weighted quartile splits of both earnings distributions. I then classify the upper quartile of each distribution as high-wage occupations, mainly reflecting white-collar professional and managerial jobs (e.g., top executives, financial managers, engineers, and lawyers), and high-wage industries (e.g., professional and technical services, securities and investments, and insurance), respectively.

Steep career ladder early-career jobs. The detailed employment history on LinkedIn profiles also enables me to assess whether students sort into early-career jobs with steep or flat career prospects. To do this, I follow the idea in Almar *et al.* (2025) and predict the long-run wage growth and likelihood of attaining a senior leadership position for each early-career job. To construct the measures, I aggregate early-career jobs at the occupation-by-firm level and perform a leave-one-cohort-out prediction of both measures 15 years later.[15] I then split both occupation-by-firm-level career prospect distributions at the upper quartile to define early-career jobs with steep career prospects.

Job help in the college network. LinkedIn profiles do not contain explicit information on whether students shared work-related information or helped one another to access jobs. Nevertheless, the panel structure and the availability of firm names allow me to construct a proxy for the use of social ties in the job

[14] Also here, using alternative thresholds for the classification of high-quality firms in the various amenity dimensions produces similar results.

[15] For occupation-by-firm cells with fewer than 30 observations, I collapse the career cells at the 5-digit SOC level to avoid small-sample bias. I define an early-career job as any job in the first two years after expected graduation.

search process based on relational firm mobility patterns, following Gee *et al.* (2017). Suppose I observe two college peers in the data. If one of them starts working at a firm where the other has already worked for at least one year, this is taken as evidence that the first has received help from the college peer to access the new job at the connected firm. This measure has been validated by Gee *et al.* (2017) and is considered the state-of-the-art approach to measuring relational job mobility (Rajkumar *et al.* 2022).

Degree, field of study, and postsecondary institution. The educational background section on LinkedIn profiles contains information on degree, field of study, and educational institution, enabling me to merge the Facebook introduction dates with each student's career outcomes. To convert the raw text strings into standardized educational background characteristics, I follow the keyword approach in Dorn *et al.* (2025). For educational degrees, I map the raw degree input into six education levels using a keyword approach based on several online sources on typical degree abbreviations (YourDictionary 2022; StudentNews Group 2024; Best Universities 2024). The six levels are as follows: high school, associate, bachelor, master, professional degree, and doctorate. To classify field of study, I use the Classification of Instructional Programs (CIP) taxonomy provided by National Center for Education Statistics (2024a). As field-of-study entries on LinkedIn profiles are closely related to the CIP classification, I can map most of them to the corresponding CIP codes using regular expressions. For the most common non-mapped entries, I apply manual mapping. To standardize information about educational institutions, I use the clustering algorithm provided by Revelio Labs, which also accounts for common abbreviations. To further improve the clustering algorithm's accuracy, I refine the mapping by manually removing the most frequent false-positive cases from the estimation sample.

3.2 Further Data Sources

Facebook introduction dates. The second primary data source comes from Armona (2025) and contains information on Facebook's introduction dates for the first 842 postsecondary institutions. While the exact introduction date of Facebook is only known for all selective U.S. colleges up to August 2005, I can use the remaining selective U.S. colleges as a control group. They did not have access to Facebook before the academic year 2005–2006. Using this information, I can determine for all selective U.S. colleges the academic year in which they received access to Facebook and merge the treatment information with the LinkedIn profiles.

IPEDS data. The third data source comes from the Integrated Postsecondary Education Data System (IPEDS) and contains detailed information on the demographic composition and degree counts of each

graduation class at a U.S. postsecondary institution (National Center for Education Statistics 2024b). I use this information to investigate the representativeness of the LinkedIn sample and to merge the number of business and economics bachelor's degrees awarded to each graduation cohort.

Mobility Report Cards data. To merge institutional characteristics of all four-year colleges to the LinkedIn profiles, I utilize the Mobility Report Cards (MRC) data from the Equality of Opportunity Project (Chetty *et al.* 2020). The MRC dataset provides information on the location of postsecondary institutions and their selectivity, measured by Barron's selectivity index (Barron's Educational Series 2008).

3.3 Data Merge, Sample Selection, and Representativeness of LinkedIn Profiles

To construct the final estimation sample, I start by linking the Facebook introduction dates, IPEDS data, and MRC data using the unitid identifier. Since I focus on undergraduate business and economics students at selective U.S. colleges, I limit the sample to selective four-year colleges that offered an undergraduate business or economics degree between 2000 and 2005, resulting in 1,242 potential colleges.[16] In the next step, I merge the college-level information into the LinkedIn profiles using the information on college names. Appendix Table A5 documents the sampling procedure in detail. I can match 1,234 colleges (99.4%) to at least one LinkedIn profile reporting a bachelor's degree in business and economics, resulting in 951,663 profiles. To ensure that the estimation sample does not contain any incomplete or stale profiles, I drop 20,559 observations reporting an invalid undergraduate graduation date (2.2%).[17] I also drop 115,653 observations (12.4%) that do not report any employment for more than half of the career panel.[18] Following Overgoor *et al.* (2020), I further restrict the sample to colleges with average LinkedIn coverage rates between 0.25 and 1.25 to maintain representative coverage across college cohorts.[19] This refinement leaves a final estimation sample with 1,165 colleges and 797,344 profiles. The sample selection steps ensure that only sufficiently complete LinkedIn profiles and sufficiently representative college cohorts enter the final estimation sample. Appendix Table A5 shows that this refinement has little overall impact on the sample's composition in terms of age, gender, or college selectivity.

To assess the representativeness and coverage of the final estimation sample, I compare basic descriptive

[16] I define business and economics majors following the methodology used by U.S. News & World Report. Business majors are defined as any major with the two-digit CIP code 52, excluding 52.06. Economics majors are classified as any major with the CIP codes 45.06, 01.103, 03.0204, 51.2007, 52.06, 30.3901, and 30.4001 (Brooks 2025).

[17] I define a valid undergraduate graduation date as any end date of the education spell that is not missing, strictly after the indicated start date, and implies a study duration of under 10 years.

[18] The employment filter ensures that LinkedIn profiles that did not indicate an early-career employment history and did not update their LinkedIn profiles in recent years are not included in the estimation sample.

[19] Alternative thresholds do not affect the results.

statistics with information from the IPEDS data. Appendix Figure A3 indicates that the final estimation sample covers 47.2% of all U.S. business and economics students during that time period. Consistent with other studies that use a LinkedIn sample of U.S. college graduates (e.g., Dorn *et al.* 2025), females are slightly underrepresented, with 41.8% of the LinkedIn sample compared to 47.3% in IPEDS. The shares of students coming from colleges located in different U.S. regions and from colleges in different selectivity tiers are closely aligned across both data sets. Very selective colleges are slightly overrepresented in the LinkedIn data (51.6%) compared to IPEDS (48.0%).[20]

3.4 Descriptive Statistics

Table 1 presents descriptive statistics on the career trajectories of U.S. business and economics students, measured at the person-by-year level. In the first six years after expected graduation, 77.8% of students report being employed, with average job-based real annual earnings of around $76,000 (2024 USD). Consistent with the well-documented evidence on life cycle employment and earnings dynamics (e.g., Rubinstein and Weiss 2006), the employment share of U.S. business and economics students quickly rises and stabilizes at around 94% with the average real annual earnings reaching $122,000, 13–19 years after expected graduation. Taking a closer look at job mobility patterns, Table 1 shows that the annual promotion rate in the sample is around 8%, with roughly two-thirds of the promotions occurring between firms and one-third within firms. Consistent with previously documented evidence on mobility patterns within- and across-firms (e.g., Belzil and Bognanno 2008; Frederiksen *et al.* 2016), promotions occur more frequently than lateral job moves (3.6% to 4.2%) and demotions (4.5% to 5.9%), but are not the only type of job transition in the data. Investigating firm choice, Table 1 indicates that the share of students reporting to work at high-wage firms starts at 25.7% in the first six years after graduation and rises to 34.6% in the later career stage. Looking at top jobs, the share of U.S. business and economics students holding a senior leadership position starts at 10% in the first six years of their careers and increases up to 26.9% 13–19 years after expected graduation.[21]

How important are college peers for job finding and career development? Following the literature (e.g., Gompers *et al.* 2026), I define college peers as students with overlapping college experience: between three

[20]While the main analysis utilizes the LinkedIn sample composition for estimation, Appendix G shows that the results are very similar when reweighting the sample in terms of demographic and college characteristics using the official statistics from the IPEDS data.

[21]This pattern closely aligns with the findings in Hampole *et al.* (2026) on the dynamics of holding senior leadership positions for MBA graduates, with the levels shifted downwards as the sample here comprises undergraduate students.

classes below and three classes above the respective cohort.[22] The relational firm mobility job help proxy by Gee *et al.* (2017) shows that in the first 12 years, roughly 14% of U.S. business and economics students received job-related information from their college peers to obtain their current job. While the share declines to 11.6% in the later career stage, the evidence suggests that more than 12.5% of the reported jobs in the career panel were obtained through college peers. This underscores their relevance in the job search process. Interacting the job help proxy with firm quality and career development outcomes suggests that college peers are also an important source for career advancement. Between 4.3% and 5.4% of students used their college network to obtain a job at a high-wage firm, 0.7% to 1.3% to obtain a promotion at a connected firm, and around 0.2% to transition into a senior leadership position. The relevance of college peers for career advancement declines slowly over time.

Table 1: Career Outcomes Over Time

	1–6 Years	7–12 Years	13–19 Years
Panel A: Career Development			
Employment	0.778	0.939	0.943
Job-Based Real Annual Earnings (2024, USD)	76,015	106,687	121,547
Job-Based Promotion (≥10 log pts.)	0.082	0.089	0.074
Between-Firm Promotions (≥10 log pts.)	0.058	0.057	0.044
Within-Firm Promotions (≥10 log pts.)	0.024	0.032	0.030
Lateral Moves	0.036	0.042	0.037
Job-Based Demotion (≤ −10 log pts.)	0.045	0.059	0.059
High-Wage Firm	0.257	0.327	0.346
Senior Leadership Positions	0.099	0.197	0.269
Panel B: College Network Outcomes			
Job via College Network	0.137	0.144	0.116
High-Wage Firm Job via College Network	0.053	0.054	0.043
Promotion via College Network (≥10 log pts.)	0.013	0.011	0.007
Senior Leadership Transition via College Network	0.002	0.002	0.002
Observations	4,784,064	4,784,064	5,581,408

Notes: Descriptive statistics of career outcomes over time reported at the person*year level. Each value represents the mean outcome pooled over one of the following time intervals after expected graduation: 1–6 Years, 7–12 Years, 13–19 Years. Sample: all business and economics undergraduates expected to graduate from 2000 to 2005 at selective 4-year colleges with valid LinkedIn profile and reasonable college-level coverage (1,165 colleges). Job-based real annual earnings are set to zero in years without employment. Job mobility outcomes start at year 2 after expected graduation. *Data source:* Revelio Labs.

[22]The college peer networks are constructed from the LinkedIn profiles of business and economics students from the expected graduation cohorts 1994 to 2010, so that the full window of three classes below and above is observed for every cohort in the estimation sample.

4 Empirical Strategy

4.1 Identification

To isolate the causal effect of access to Facebook at labor market entry on career outcomes, I exploit the sharp and staggered rollout of Facebook across U.S. colleges in a generalized difference-in-differences design. The variation in treatment timing allows me to compare the career outcomes of business and economics students from the same college across treated and untreated cohorts. I account for common cohort differences and spurious correlations with the Facebook rollout using not-yet-treated cohorts from colleges in the same region and selectivity tier.

The baseline model in a two-way fixed effects (TWFE) setting can be specified as follows:

$$Y_{fs,c,t} = \alpha_1 Treat_{fs,c} + \delta_{fs,\text{tier,region}} + \phi_c + \epsilon_{fs,c,t}\,, \tag{1}$$

where $Y_{fs,c,t}$ represents various career outcome measures for freshman cohort fs at college c, t years after expected graduation. $Treat_{fs,c}$ is a binary variable indicating whether freshman cohort fs from college c had access to Facebook before the *expected* graduation date, avoiding a potentially endogenous relation between treatment and the actual graduation date.[23] Since Facebook became public in 2006, the treatment indicator essentially captures the intent-to-treat (ITT) effect of having access to Facebook before vs. after labor market entry. ϕ_c denotes college fixed effects, accounting for time-invariant differences across colleges, such as prestige. $\delta_{fs,\text{tier,region}}$ represents cohort-by-college-tier-by-region fixed effects, narrowing down the comparison in the difference-in-differences setting to college cohorts within the same region and selectivity cell.[24] As the rollout of Facebook was not random, the key idea behind $\delta_{fs,\text{tier,region}}$ is to absorb all systematic differences across college cohorts that might be spuriously correlated with the treatment timing and exclusively rely on the plausibly exogenous variation within these cells. Examples of potential spurious correlations absorbed by $\delta_{fs,\text{tier,region}}$ are cohort-specific shocks in career outcomes within the same region at colleges of similar selectivity (e.g., fluctuations in labor market entry conditions and returns to education) or differential cohort-specific trends across those cells.

Consistent with the interpretation that the remaining variation is plausibly exogenous, Appendix

[23] I follow Armona and Bailey (2025) and define access to Facebook before the *expected* graduation date as having access to Facebook at the respective U.S. college before June 30 of the expected graduation year (fourth year of the undergraduate).

[24] Since all but one elite college received early access to Facebook, I group elite and highly competitive institutions (Barron's tiers 1 and 2) into a single category for the fixed-effects estimation. To still allow for differential trends across cohorts within the broader group of highly selective colleges, I include linear cohort trends for Ivy Plus colleges, other elite colleges, and all highly competitive-by-region interactions. Appendix G shows that the results are robust to dropping Ivy Plus and other elite colleges.

Table A6 shows that the correlation between the Facebook rollout and various potential confounders, such as college characteristics, student and major composition, as well as local labor market entry conditions, vanishes after incorporating the fixed-effects structure.[25]

A growing body of literature emphasizes that in staggered adoption designs, the conventional two-way fixed effects (TWFE) model requires not only the parallel-trends assumption but also treatment-effect homogeneity to produce consistent estimates (Goodman-Bacon 2021; Sun and Abraham 2021; Roth *et al.* 2023). Specifically, Goodman-Bacon (2021) shows that the estimate of the TWFE model is a weighted average of all possible 2×2 difference-in-differences comparisons, including comparisons of already-treated units to newly-treated units. If the treatment effect varies over time or by adoption cohort, this biases the treatment effect obtained with TWFE. Given that Facebook initially targeted more selective universities, it is possible that earlier adoption cohorts experienced different treatment effects than later cohorts, calling into question the strong assumption of treatment effect homogeneity.

To ensure that the results are not driven by this bias, I estimate the treatment effect in the spirit of equation (1) using the imputation heterogeneity-robust estimator proposed by Borusyak *et al.* (2024). The idea behind the imputation approach is to first regress the outcome $Y_{fs,c,t}$ on the group ϕ_c and period $\delta_{fs,\text{tier},\text{region}}$ fixed effects using the not-yet-treated observations only. The estimated fixed effects are then used to impute the untreated potential outcome of each treated observation, $\hat{Y}_{fs,c,t}(0)$, and to estimate the aggregate treatment effect based on the difference between the observed and the imputed outcome of the treated observations. This procedure effectively avoids using already-treated units as controls for newly-treated ones and is robust to heterogeneous treatment effects. To account for potential correlations in the error term across cohorts within the same college, I cluster standard errors at the college level (Athey and Imbens 2022). For computational efficiency, I estimate the regressions at the college-by-cohort-by-year level (as noted in equation (1)), weighting each cell by the corresponding number of LinkedIn profiles.

4.2 Threats to Identification

The core identifying assumption is that, in the absence of treatment, the career outcomes of students in treated colleges would have evolved in parallel to those in not-yet-treated colleges of similar selectivity located in the same U.S. region *and* did evolve in parallel before the treatment. To assess the plausibility of the parallel trends assumption for each career outcome, Appendix B shows estimates of an event-study

[25]The main results in Figure 1 are very similar when including these characteristics as controls in the regression (not shown here).

version of equation (1) (Figures B1 to B7), where the event refers to the first freshman cohort expected to be introduced to Facebook at the respective college. Almost all pre-treatment coefficients of the more than 50 event studies are individually and jointly insignificant at the 5% level, supporting the parallel trends assumption in this setting. There are also no signs of significant spillover (anticipation) effects to the last cohort expected to graduate before Facebook's introduction at the respective college. I discuss the event-study estimates in greater detail in Section 5.

Another key concern is that access to Facebook may affect the probability of having a LinkedIn account or change the composition of the sample. In this case, estimated treatment effects might simply reflect sample selection bias. Reassuringly, Appendix Table A7 shows that Facebook access at labor market entry does not affect the likelihood of having a LinkedIn account. Facebook access also does not alter the sample composition with respect to various background and pre-treatment characteristics such as gender, age, being a first-time student, or listing prior work experience on the profile.

Lastly, early access to Facebook might affect the completeness of LinkedIn profiles, even if the propensity to have a LinkedIn account and sample composition remain unaffected. Since the sample restrictions exclude incomplete profiles from the baseline sample, this could challenge a causal interpretation of the results. To assess the severity of this concern, Appendix Table A8 estimates the treatment effect on each incomplete-profile restriction, starting with all LinkedIn profiles at the colleges in the estimation sample. Reassuringly, early Facebook access affects neither the LinkedIn coverage before any sample restriction nor the share of profiles that pass the complete employment history restriction or all incomplete-profile restrictions jointly, indicating that treated and untreated students have LinkedIn profiles of similar completeness.[26] I present further evidence supporting the identifying assumptions in Section 5.3.

5 Results

5.1 Career Development

Baseline estimates. Does access to Facebook at labor market entry affect the career development of U.S. students? Figure 1 investigates this question by tracking their career trajectories up to 19 years after graduation, with the corresponding point estimates and control group means reported in Appendix Table A9. Panel A shows the results using the job-based earnings measure. In the first six years after

[26] The only sample restriction that can be predicted by the treatment is the likelihood of having a valid graduation date. However, the estimated effect size is rather small (0.3 percentage points), and Appendix Figure A4 shows that the main results are robust to the broader LinkedIn sample without this or any incomplete-profile restrictions.

expected graduation, students with early access to Facebook earn on average $1,500 more than the control group, with the earnings effect increasing to $1,867 in the years 7–12.[27] Comparing the treatment effect to the average job-based annual earnings of the control group indicates a relative effect size of 2.0% ($75,464) and 1.8% ($105,642), respectively, suggesting a moderate but economically meaningful shift in average annual earnings. While the earnings effect is fading over time (see years 13–19), the results for job-based earnings provide clear evidence that the initial career trajectories of students with early access to Facebook have been positively affected. The positive earnings effects in the first six years stem from both an increase in labor supply and higher pay, while the effect in years 7–12 exclusively comes from students working in higher-paying jobs (see Appendix Figure A5).[28]

Figure 1, Panel B, uses the relative wage growth jump methodology by Bronson and Thoursie (2019) to analyze how early access to Facebook affects promotion patterns. Consistent with the positive earnings effect, the figure shows that students with early access to Facebook are 0.3 percentage points more likely to receive promotions in the first six years after graduation. Comparing the effect size to the control group's average annual promotion rate (8.0%) indicates an economically meaningful relative effect size of 3.8%.[29] While the positive effect on the annual job-based promotion rate fades out rather quickly (see years 7–12 and 13–19), Appendix Figure A7 shows that the positive early-career impact translates into a persistent *cumulative* edge in the number of promotions up to 12 years after graduation.[30]

The positive earnings and promotion effects raise the question of whether access to Facebook enabled some students to reach top jobs in the economy. Panel C of Figure 1 investigates this by examining the senior leadership measure initially proposed in Hampole *et al.* (2026). Consistent with the intuition that reaching a senior leadership position takes time, there is no impact in the first years after expected graduation. However, Facebook access at labor market entry increases the likelihood of attaining a senior leadership position in the longer run. 13–19 years after graduation, students with early access to Facebook are 0.9 percentage points more likely to reach senior leadership positions, a 3.4% increase relative to the control group mean. This suggests that while the average earnings and promotion effects are fading out over time, early access to Facebook enabled some students to reach top jobs in the economy through a

[27]The estimated earnings effect 7–12 years after expected graduation is consistent with Armona (2025), who finds that each year of Facebook access during college increases average earnings in 2014 by $972 ($1,280 in 2024 dollars).

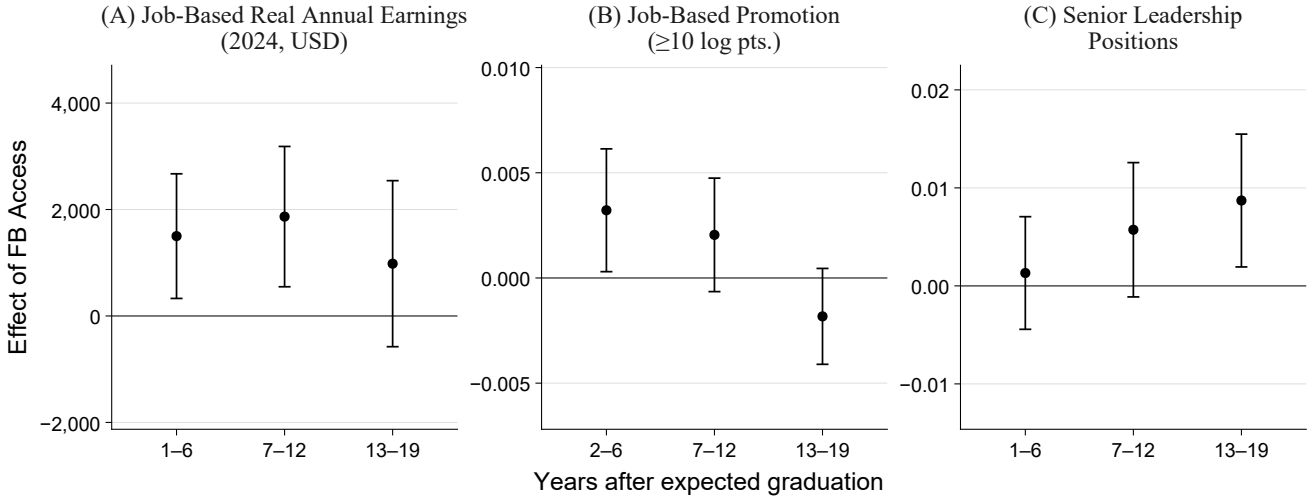
[28]The labor supply and earnings dynamics are very similar to the documented career patterns in the literature on graduating in a recession for college graduates (e.g., Oreopoulos *et al.* 2012; Altonji *et al.* 2016; von Wachter 2020).

[29]Appendix Figure A6 shows that the results are robust to alternative specifications of the promotion threshold.

[30]There is no impact on lateral job mobility and demotions (see Appendix Figure A8), suggesting that the positive earnings effects are not explained by treated students being less likely to receive a demotion or by frequent small positive wage-growth jumps due to increased lateral job mobility.

persistent effect on their career progression.[31] Consistent with this interpretation, there is some evidence that early access to Facebook has a positive long-run impact on the likelihood of attaining a high-paying job and of having a steep promotion profile (see Appendix Table A10). Taken together, the results in Figure 1 provide evidence that access to Facebook at labor market entry generated long-lasting career benefits and facilitated economic upward mobility to top jobs for the exposed students.

Figure 1: Facebook Access and Career Development



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. Job-based real annual earnings: job-based annual earnings (2024 USD), set to zero in years without employment. Appendix Table A9 reports the point estimates, the average outcomes for the not-yet-treated college cohorts, and the number of observations. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Event-study graphs and gender heterogeneity. To test for parallel trends and to study treatment effect dynamics in terms of the time exposed to Facebook during college, Appendix Figure B1 depicts event-study estimates for the outcomes in Figure 1, in which the event refers to the first freshman cohort expected to be exposed to Facebook's introduction. All but one of the event-study graphs show that the pre-treatment coefficients are close to zero and jointly insignificant at the 10% level.[32] The findings are generally consistent with the estimation strategy successfully isolating the plausibly exogenous variation of the Facebook rollout and with the parallel trends assumption. There is also no evidence of a significant dip or positive jump in the career outcomes of the last cohort expected to graduate before Facebook's introduction, indicating that there are no spillover (anticipation) effects in this setting. Looking at the

[31] Appendix Figure A9 repeats the estimation using management occupations (SOC 11), top executives (SOC 11-1), and board members as alternative definitions of top jobs. While the concepts differ slightly in the kinds of jobs they capture, the results for management and top executive occupations are very similar. There is no impact on the likelihood of becoming a board member.

[32] The only exception is Appendix Figure B1, Panel E (job-based promotion 7–12 years after expected graduation), where the pre-treatment coefficients are jointly significant, but the treatment effect in Figure 1 is not.

post-treatment coefficients indicates that the treatment effects are stronger for cohorts with at least seven months of Facebook exposure during college, supporting the idea that greater exposure during the final year of college and at the start of the job search period facilitates students' ability to capture the technology's labor market benefits.[33] Appendix Figure C1 shows that the career benefits documented in Figure 1 are shared equally between men and women, suggesting that early access to Facebook did not meaningfully affect gender gaps in career outcomes.

Cumulative earnings. What does having early access to Facebook imply for the cumulative earnings of U.S. students? Appendix Table A11 investigates this question using the cumulative job-based earnings in the first 19 years after graduation. Assuming a 5% discount rate, the estimated increase in cumulative job-based earnings from early access to Facebook is around $19,035, equivalent to a 1.6% increase relative to the control group. This gain is 56% as large as the lifetime earnings loss that college graduates can expect when graduating in a large recession (e.g., von Wachter 2020), suggesting that early Facebook access had an economically meaningful impact on the cumulative earnings of the exposed U.S. business and economics students.

5.2 Mechanisms

Having shown that access to Facebook at labor market entry has a long-run positive impact on students' career trajectories, I next examine potential mechanisms. To do so, I split the treatment dynamics into two stages: the college-to-work transition and the period after it.

5.2.1 The College-to-Work Transition

During the college-to-work transition, students may rely on improved information flows from their larger and more connected online social networks or the early work-related platform features to support the early-career job search process.[34] As an extensive literature emphasizes that a successful college-to-work transition and better initial job placement are key to long-run career development (e.g., Kahn 2010; Oreopoulos *et al.* 2012; Altonji *et al.* 2016; Schwandt and von Wachter 2019; von Wachter 2020; Arellano-Bover 2024), this mechanism could explain part of the documented career benefits.

[33] This finding is consistent with Armona (2025), who documents that longer exposure during college produces larger earnings effects. Nevertheless, the estimated post-treatment effects should be interpreted with caution, as the composition of colleges and the number of cohorts already exposed to the technology differ between the two post-treatment groups.

[34] As discussed in Section 2, the majority of employers were not yet actively using Facebook during the labor market entry phase of the first exposed college cohorts, so that most demand-side information channels (e.g., employer screening, recruiters) were not at play. The main exception, targeted employer ads, is discussed below.

Early employment and quality of firms and jobs. Table 2 investigates this channel by examining early-career employment and job-placement outcomes in the first two years after expected labor market entry.[35] Column 1 shows that early Facebook access increases the probability of finding a job during the expected labor market entry phase by 1 percentage point (1.5%), providing first evidence of an improvement in the college-to-work transition.

To assess whether the *job quality* margin was also affected, Columns 2–7 focus on several indicators of high-quality entry-level jobs. Starting with firm quality, Columns 2 and 3 show that students with early access to Facebook are 1.8 and 1.2 percentage points more likely to start their careers at high-wage firms and at firms highly rated for career support, respectively. Compared to the control group, this implies a relative effect size of 8.9% and 7.6%, respectively, suggesting an economically meaningful shift toward high-quality firms. Consistent with the career benefits documented in Figure 1, students with early access to Facebook are also more likely to sort into firms that have a high rating in terms of compensation and benefits (see Appendix Table A12), while there is no effect on the other Glassdoor dimensions (overall rating, culture and values, senior leadership, and work-life balance).[36]

Table 2: Facebook Access and the College-to-Work Transition

		Indicators of High-Quality Early-Career Job Placement						
	Employment	High-Wage Firm	High-Career-Support Firm	High-Wage Occupation	High-Wage Industry	Steep Wage Growth	Sr. Leadership Track	Time to First Stable Job
FB Access	0.010*	0.018***	0.012**	0.012*	0.021***	0.002	0.018***	-0.054*
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)	(0.004)	(0.023)
Dep. Var. Mean	0.654	0.203	0.158	0.246	0.245	0.159	0.201	1.47
Observations	13,926	13,926	13,910	13,926	13,919	13,926	13,926	6,963
College FE	X	X	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-wage firm: an indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Person-years at employers without a Glassdoor rating are excluded, and person-years without employment are coded as zero. High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. High-wage industry: indicator for working in an industry in the upper employment-weighted quartile of the average median log earnings distributions at the 3-digit NAICS level. Steep wage growth job: an indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after, above the 75th percentile. Time to first stable job: years from expected graduation to the first active employment spell after expected graduation lasting at least 4 months (Kramarz and Skans 2014). Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

[35] Varying the definition of the labor market entry phase does not affect the results.

[36] Appendix Table A13 shows that the results are not driven by firms with missing information on amenities.

Columns 4 and 5 show that treated students are also 1.2 and 2.1 percentage points (4.9% and 8.6%) more likely to start working in high-paying occupations and industries, suggesting that early access to Facebook also improved early-career outcomes at the occupational and industry margins.[37] To also consider the long-run perspective of early-career jobs on career development, Columns 6 and 7 investigate sorting into early-career jobs associated with *steep career prospects* in terms of wage growth and the likelihood of reaching a senior leadership position. While Column 6 indicates that early access to Facebook did not increase the likelihood of obtaining an early-career job with a steep wage-growth profile, Column 7 shows that students are 1.8 percentage points (9%) more likely to access entry-level jobs with a high chance of reaching a senior leadership position later in their careers. I do not find any evidence that the college-to-work transition effect is apparent for *lower-quality* early-career jobs (see Appendix Table A14). Column 8 further shows that treated students find their first stable job after expected graduation 0.054 years (two-thirds of a month, 3.7%) faster than the control group. Although the effect is small in economic magnitude, the finding is consistent with early access to Facebook leading to an information advantage in the search for early-career jobs. Taken together, the findings suggest that early access to Facebook primarily enabled students to quickly find *high-quality* early-career jobs after labor market entry, improving their initial job placement and shifting their career prospects in the short as well as long run.

Consistent with the event-study estimates of career development outcomes presented in Section 5.1, Appendix Figure B2 shows that the improvement in the initial job placement outcomes is particularly pronounced among treated students with at least seven months of expected Facebook access, while the employment effect is similar for students with more and less exposure. Also, all the pre-treatment coefficients are jointly insignificant at the 5% level, supporting the parallel trends assumption. Appendix Figure C2 indicates that the college-to-work transition benefits are shared equally between men and women.

Graduation timing and postgraduate education choice. Appendix Table A15 investigates to what extent the college-to-work transition effects are driven by graduation timing and postgraduate education choices. Looking at the reported undergraduate study duration and likelihood of studying in the first two years after expected graduation, the effects are close to zero and insignificant. This suggests that students without early access to Facebook did not strategically delay their labor market entry by staying in college. I also find no impact on the likelihood of obtaining a postgraduate degree or on the type of postgraduate degree, suggesting that the documented treatment effects are not mediated through postgraduate education

[37] Appendix Figure A10, Panel A, shows that the occupation margin is primarily driven by a shift toward business and financial occupations. Panel B shows that the industry margin is primarily driven by a shift toward the financial activities sector.

decisions.

Early-career job help in the college network. While the previous results indicate that Facebook access at labor market entry facilitated the college-to-work transition, it remains unclear where this effect comes from. As described in Section 2, treated students could increase their connectedness within the college network, maintain those connections after exiting college at a lower cost, and use the early-stage platform features of Facebook to quickly exchange information. The improved transition may therefore stem from students sharing work-related information and helping one another in the early-career job search process. Consistent with this hypothesis, early survey evidence by Ellison *et al.* (2007) suggests that U.S. students expected Facebook to be useful for obtaining job referrals and learning more about jobs from their social network. To examine this channel, Table 3 decomposes the college-to-work transition effect in Table 2 into early-career jobs at firms *with* and *without* a college peer already working there, using the relational firm mobility job-help proxy by Gee *et al.* (2017).

Panel A, Column 1, investigates whether early access to Facebook increased the likelihood that students find an early-career job at a firm connected to their college network. The treatment effect is positive and statistically significant at 1.1 percentage points (10.4%). At the same time, Panel B, Column 1, indicates that there is no treatment effect on the likelihood that students find an early-career job without such a connection. This suggests that the entire early-career job finding effect in Table 2, Column 1, is concentrated at firms connected to the college network, consistent with improved access to work-related information and job help from college peers. In line with this, Column 8 shows that early access to Facebook shifts the likelihood of finding the first stable job through the college network, as opposed to without it, by 1.7 percentage points.

To investigate whether students also used the college network more intensively to find high-quality entry-level jobs, Columns 2–7 combine the job help proxy with the high-quality early-career job indicators, so that each outcome indicates a job of the respective type at a firm where a college peer already worked. Columns 2 and 3, Panel A, show that treated students are 0.7 percentage points (17.5%) more likely to obtain a job at a connected high-wage firm and 0.7 percentage points (14.6%) more likely to obtain a job at a connected highly career-supporting firm. Comparing the coefficients to the estimated effect sizes in Columns 2 and 3 of Panel B (1.1 and 0.5 percentage points) suggests that 40 to 60 percent of the positive impact of early access to Facebook on the likelihood of obtaining a job at a high-quality firm comes from jobs at firms connected to the college network. While the college network seems to be slightly less relevant for the improvements at the occupational and industry margins (Columns 4 and

5), Column 7 shows that the college network is also relevant for accessing jobs associated with a high chance of reaching senior leadership positions (0.8 vs. 1.0 percentage points) at connected firms. Notably, while Column 6 indicates that there is no aggregate treatment effect on the likelihood of attaining a steep wage growth job, treated students are more likely to obtain these jobs through their college network (0.7 percentage points) than without (−0.4 percentage points). This provides further evidence that early access to Facebook ultimately *shifted* students' early-career job search toward their college network and strengthened established college-firm hiring pipelines, a pattern that is also evident at the lower-quality early-career job margin (see Appendix Table A16) and when the college-to-work transition effect is split by established hiring pipelines instead of the job help proxy (see Appendix Table A17).

Table 3: Facebook Access and Early-Career Job Help in the College Network

	Any Job	High-Wage Firm	High-Career-Support Firm	High-Wage Occupation	High-Wage Industry	Steep Wage Growth	Sr. Leadership Track	First Stable Job
Panel A: Early-Career Job via College Network								
FB Access	0.011***	0.007***	0.007***	0.004	0.006**	0.007***	0.008***	0.017***
	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.004)
Dep. Var. Mean	0.106	0.040	0.048	0.041	0.055	0.027	0.046	0.330
Observations	13,926	13,926	13,910	13,926	13,919	13,926	13,926	6,963
Panel B: Early-Career Job without College Network								
FB Access	-0.002	0.011**	0.005	0.009	0.015***	-0.004	0.010*	-0.017***
	(0.005)	(0.004)	(0.004)	(0.005)	(0.004)	(0.005)	(0.004)	(0.004)
Dep. Var. Mean	0.548	0.163	0.110	0.205	0.190	0.132	0.155	0.670
Observations	13,926	13,926	13,910	13,926	13,919	13,926	13,926	6,963
College FE	X	X	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Panel A: early-career job at a firm with a college peer already working there. Panel B: early-career job at a firm without a college peer already working there. In Columns 1–7, the two panels partition the corresponding outcome in Table 2, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. Column 1 (Any Job) is the partition of the Employment outcome shown in Column 1 of that table. High-wage firm: indicator for working at a firm in the upper employment-weighted quartile of the average median log earnings distributions at the firm level. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Person-years at employers without a Glassdoor rating are excluded, and person-years without employment are coded as zero. High-wage occupation: indicator for working in an occupation in the upper employment-weighted quartile of the average median log earnings distributions at the 5-digit SOC level. High-wage industry: indicator for working in an industry in the upper employment-weighted quartile of the average median log earnings distributions at the 3-digit NAICS level. Steep wage growth job: indicator for working in an early-career job with a leave-one-cohort-out 15-year wage growth prediction above the 75th percentile. Sr. leadership track job: indicator for working in an early-career job with a leave-one-cohort-out prediction of reaching a senior leadership position 15 years after above the 75th percentile. Column 8 (First Stable Job): indicator for the first active employment spell after expected graduation lasting at least 4 months being at a firm with (Panel A) or without (Panel B) a college peer already working there. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Appendix Figures B3 and B4 show the corresponding event-study estimates, supporting the parallel trends assumption and indicating that this pattern is particularly pronounced among students with at least seven months of expected access to Facebook during college. Appendix Figure C3 also shows

that the effect is evident for both men and women. Notably, splitting the treatment effect on job help within the college network by the gender of college peers indicates that women, in particular, tended to receive more opposite-sex job help, especially for jobs at high-wage firms. The pattern is consistent with the evidence that early access to Facebook reduced gender homophily (Armona and Bailey 2025) and facilitated women's access to otherwise male-dominated networks.

Which type of college peer is responsible for the improved college-to-work transition of treated students? Appendix Figure A11 investigates this question, decomposing the early-career job help effect into below-cohort, same-cohort, and above-cohort college peer ties. The results indicate that the effects are primarily driven by above-cohort college peer ties, consistent with the idea that they can provide the most useful work-related information without competing for the same jobs.[38]

Peer similarity in early-career choices. While the previous results indicate that early access to Facebook increased job help within the college network, it is ex ante unclear whether this enabled treated students to explore more uncommon career tracks or intensified their entry into typical college career paths. Appendix Table A18, Panels A and B, investigate this question by decomposing the employment effect in Table 2 into jobs in the top five occupations (industries) held by prior college graduates or at firms with an established college-firm hiring pipeline, versus all other jobs. The results indicate that treated students are exclusively more likely to make typical college career choices at the occupation, industry, and firm margins. This suggests that, although treated students had more information on the set of career options, early access to Facebook ultimately strengthened the core of established college-career pipelines. Consistent with this observation, Panel C indicates that treated students also make more similar early-career choices than their same-cohort peers at the occupation and industry margins.

Employer ads. Another potential reason behind the improved college-to-work transition is that students with early access to Facebook may have engaged more actively with employer ads, which were introduced as early as April 2004 (Grynbaum 2004). If targeted ads increased awareness of jobs and companies among treated students, this could also have facilitated their search for early-career jobs. However, there are two reasons to believe this channel is unlikely to explain a meaningful part of the college-to-work transition effects in Table 2. First, targeted ads were only temporarily available to a very small set of firms during the college-to-work transition phase of the early Facebook adopters.[39]

[38]The positive below-cohort effect likely reflects peers who held a job at the firm during college (e.g., internships). The small positive same-cohort effect is consistent with less job help due to competition for the same entry-level jobs.

[39]To be precise, targeted ads were available to a small set of firms during April 2004 (Grynbaum 2004). The importance and adoption of targeted ads changed with the official launch of *Facebook Ads* in November 2007 (Facebook 2007), shortly after the college-to-work transition phase of the early Facebook adopters, which is here defined as the period until June 2007.

Second, survey evidence from the late 2000s indicates that only 12% of U.S. students actively engaged with employer ads during their first job search (National Association of Colleges and Employers 2009), suggesting that student engagement with this feature was low.

Geographic mobility. Lastly, Armona and Bailey (2025) show that students with early access to Facebook were more likely to move away from their hometown and to live in denser areas with higher economic opportunity. As more urban areas provide access to higher-paying jobs and higher-quality firms (e.g., Glaeser and Maré 2001; Dauth *et al.* 2022; Card *et al.* 2025), increased sorting into stronger labor markets might also explain part of the positive college-to-work transition effect. Appendix Table A19 investigates the relevance of this channel using the job location information reported on LinkedIn profiles.[40] Consistent with Armona and Bailey (2025), I find that treated students are particularly more likely to find jobs in the three largest metropolitan areas outside the MSA of their college (New York, Los Angeles, and Chicago). However, accounting for geographic job sorting only explains 2% to 12% of the documented college-to-work transition effects on the high-quality firm, occupation, industry, and job margins, suggesting that most of the improvement in initial job placement occurred within local labor markets.

5.2.2 After the College-to-Work Transition

The evidence so far indicates that part of the documented career benefits stem from a more successful college-to-work transition in which the college network played a central role. Another part may reflect longer-run treatment dynamics that unfold after labor market entry.

Long-run job search, firm mobility, and job help in the college network. All U.S. business and economics students received access to Facebook in the long run. Part of the documented career benefits for the early adopters could nevertheless be explained by a long-run increase in access to information about job opportunities, facilitating their search for higher-paying jobs.[41] There are three reasons why this might be the case. First, treated students may continue to receive more information about job opportunities as they had more time to build their social network and were less likely to lose their college ties when transitioning into the labor market. Relatedly, Armona and Bailey (2025) show that students with early access to Facebook have larger online social networks and are more closely connected to their college peers in the long run. As workers rely heavily on their social networks for information about job opportunities

[40] I use geographic information at the metropolitan statistical area (MSA) level of the job, which is the most fine-grained harmonized location measure available in the Revelio Labs data.

[41] In traditional job search models (e.g., Burdett 1978), this channel would reflect a permanent (or slowly decaying) increase in the job offer arrival rate, allowing workers to draw from better job offers in the wage distribution.

(e.g., Caldwell and Harmon 2019) and larger networks predict a greater arrival rate of job offers from other firms, this channel could explain part of the long-run career benefits. Second, students with early access to Facebook may be using the later-introduced work-related platform features, such as Facebook Marketplace, Pages, or Ads, more intensively to acquire job-related information. Third, due to a potentially higher visibility on the platform, students with early access to Facebook might receive more job-related information from employers and recruiters, who were increasingly using the platform.

Panel A in Figure 2 investigates the joint significance of these channels by examining the likelihood of receiving a between-firm promotion (Appendix Table A20 reports the point estimates). If early access to Facebook increases access to information about job opportunities in the long run, job search models would predict an increase in the annual rate of between-firm promotions. However, Panel A shows that the estimated treatment effect on between-firm promotions is close to zero and insignificant across the entire career panel. This suggests that a long-run increase in the job offer arrival rate is not driving the documented career benefits. Treatment effects are also close to zero for any between-firm switches and for between-firm switches with no immediate promotion (see Appendix Figure A12). Panels B and C test more directly the hypothesis that early access to Facebook leads to a long-run improvement in access to job opportunities through the more closely connected college network by interacting the between-firm promotion measure with the job help proxy for college peers. Similar to the findings for between-firm promotions, the results are close to zero and insignificant.[42]

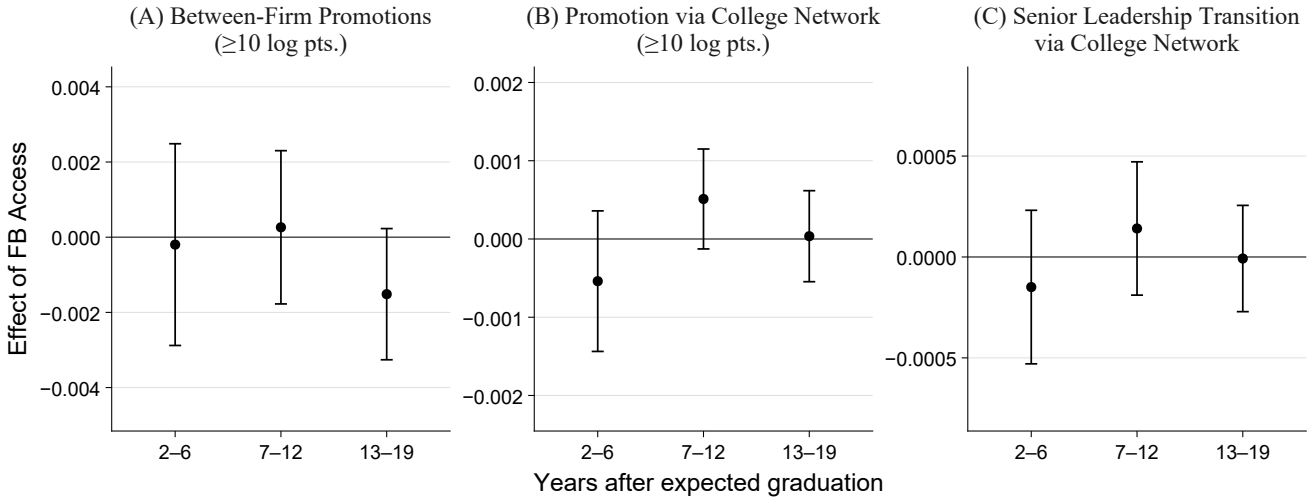
While the results in Figure 2 may appear surprising at first, they are in line with evidence by Armona and Bailey (2025) that the social network size advantage of the early adopters of Facebook decreased over time, that Facebook was also quickly adopted by the control group after it became publicly available, and that there was no long-run difference in the overall platform activity between treatment and control group. Overall, the results are consistent with differences in the job offer arrival rate decaying quickly once the control group received access to Facebook.

Appendix Figure B5 shows the corresponding event-study estimates, indicating that the parallel trends assumption is generally supported. The estimated treatment effects are largely similar for students with more and less exposure to Facebook during the last year of college, apart from somewhat larger (but not always significant) estimates for students with at least seven months of access 7–12 years after expected graduation. Looking at gender differences in Appendix Figure C4 indicates that this effect is particularly driven by men, suggesting that they might have accrued the benefits of an information advantage in the

[42] Appendix Figure A12 also shows that the results for moves to connected firms with no promotion are close to zero.

search for higher-paying jobs slightly longer than women.

Figure 2: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via college network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior leadership transition via college network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Appendix Table A20 reports the point estimates, the average outcomes for the not-yet-treated college cohorts, and the number of observations. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Match quality, skill accumulation, and long-run firm sorting. While the previous results show little to no impact on between-firm promotions, another potential reason behind the documented career benefits could be that treated students use their initial information advantage during the college-to-work transition to sort into higher-quality worker-firm matches. This would enable them to accelerate their human capital accumulation and move up the career ladder within their organizations. A higher-quality worker-firm match in this context could reflect two things. First, students with early access to Facebook might have an easier time finding *stable* and *productive* worker-firm matches. Second, they might increasingly *sort* into firms that provide better learning opportunities and career support, even if the stability and productivity of the match remains unchanged.

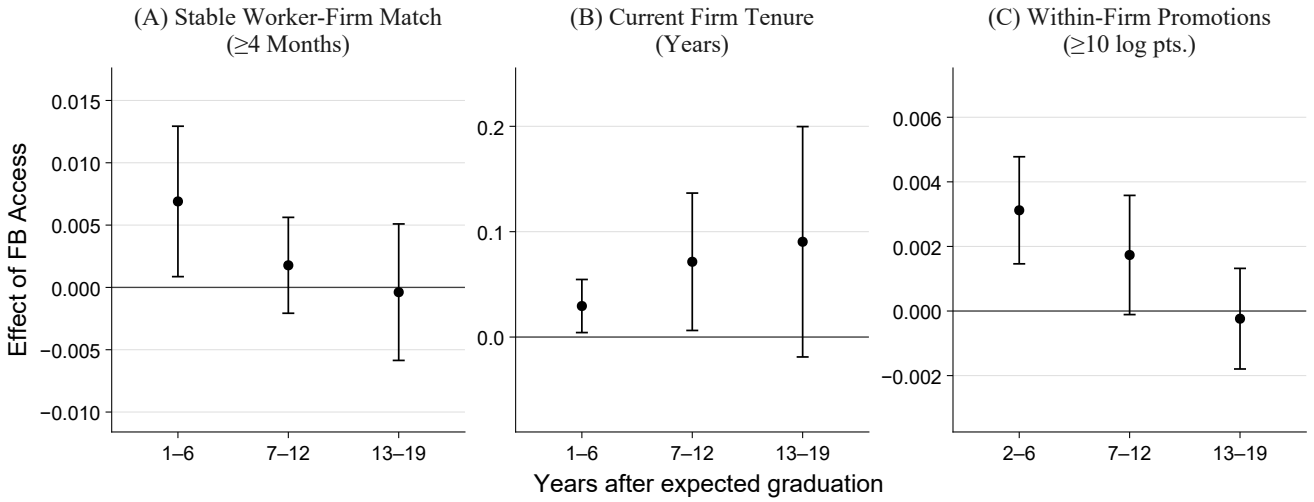
Starting with the stable and productive worker-firm match channel, Figure 3, Panel A, analyzes the fraction of students reporting to work at their current employer for at least 4 months as a measure of early-career worker-firm match stability (Appendix Table A21 reports the point estimates and control group means).[43] The figure shows that in the first six years after expected graduation, students with early access to Facebook are 0.7 percentage points (1%) more likely to report working at their employer for at least 4 months, while there is no impact on the likelihood of working in an unstable worker-firm match (see Appendix Figure A13). While the estimates for years 7–12 and 13–19 show that the positive effect

[43]The results for alternative early-career worker-firm match stability thresholds are very similar (not shown here).

is fading over time, this could be driven by the well-documented fact that firm tenure increases with experience (e.g., Topel and Ward 1992).

To investigate whether worker-firm match stability also improved in the longer run, Panel B therefore shows the results for current firm tenure. Consistent with the evidence on match stability, the estimate for years 1–6 indicates a positive effect on firm tenure in the first six years after expected graduation. Compared to the average firm tenure of the control group (1.64 years or 19.68 months), students with early access to Facebook stay on average 0.029 (0.35) more years (months) with their current employer. The estimate for years 7–12 shows that the firm tenure effect persists in the longer run and increases over time, reaching 0.07 (0.85) years (months) 7–12 years after graduation. While the firm tenure effect becomes insignificant in the long run, the findings on match stability and firm tenure provide evidence that treated students were more likely to find stable early-career worker-firm matches that persisted over time.

Figure 3: Facebook Access and Worker-Firm Match Stability and Productivity



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 4 months. Current firm tenure: years at current employer. Appendix Table A21 reports the point estimates, the average outcomes for the not-yet-treated college cohorts, and the number of observations. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

As a more stable worker-firm match does not automatically imply that treated students benefited in terms of career development, Panel C of Figure 3 examines whether early access to Facebook also improved the productivity of the worker-firm match, using the probability of within-firm promotions as the outcome. The figure shows that in the first six years after graduation, students with early access to Facebook are 0.3 percentage points (13%) more likely to obtain an early-career within-firm promotion, with the effect fading out in the long run. There is no positive impact on within-firm demotions and only a slight positive impact on within-firm lateral moves (see Appendix Figure A13). Taken together, the results suggest that early

access to Facebook enabled students to secure stable, productive worker-firm matches, allowing them to accumulate firm-specific human capital and advance within their organizations.

Appendix Figure B6 shows the corresponding event-study estimates, documenting jointly insignificant pre-treatment coefficients at the 5% level and supporting the parallel trends assumption. The post-treatment coefficients indicate that the worker-firm match stability results are similar for students with more and less exposure to Facebook during the final year of college, while the within-firm promotion patterns are stronger for students with at least seven months of Facebook access. There are no statistically significant differences across gender (see Appendix Figure C5).

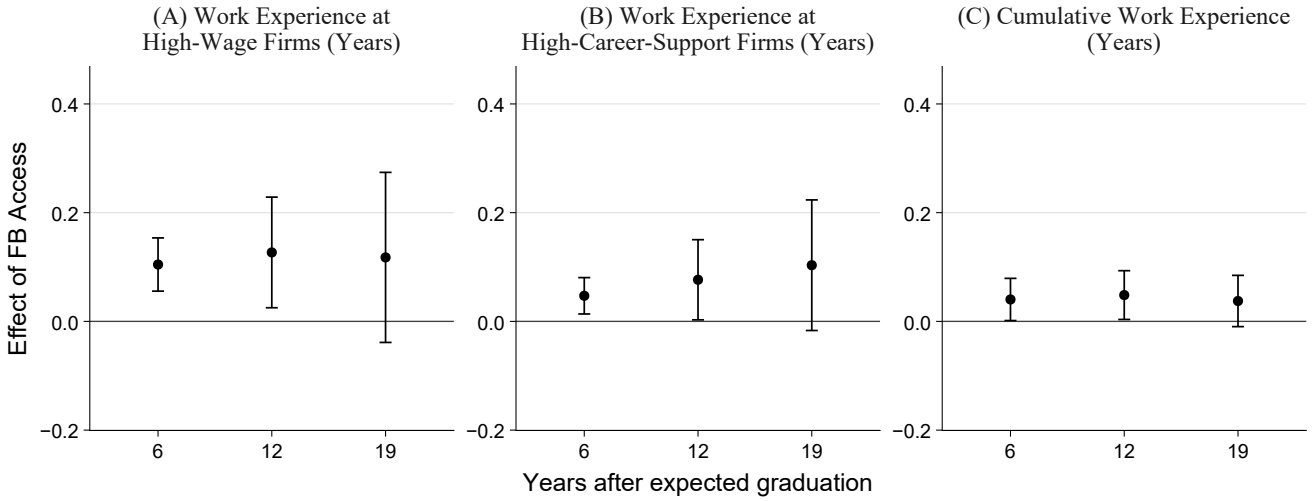
While the previous results provide evidence of improved stability and productivity of worker-firm matches, another reason for the positive career benefits could be intensified sorting into firms that offer better learning opportunities and career support. Theoretical work by Rosen (1972) and Gibbons and Waldman (2006), as well as more recent empirical evidence by Arellano-Bover (2022), Gregory (2023), and Arellano-Bover and Saltiel (2026) emphasizes that firms differ substantially in the learning opportunities they provide, rationalizing a sizable fraction of life-cycle differences in career outcomes. The college-to-work transition results already indicate that early access to Facebook enabled students to start their careers in higher-quality firms and jobs. Longer-run sorting into training-intensive and career-supporting firms could therefore be a complementary mechanism that explains the documented career effects. Figure 4, Panel A, uses the accumulated work experience at high-wage firms as a first proxy for sorting into firms that provide better learning opportunities.[44] Consistent with the college-to-work transition results, the figure shows that in the first six years after graduation, treated students accumulated on average 0.105 years (1.26 months) more work experience at high-wage firms, a 7.1% increase compared to the control group (1.47 years, see Appendix Table A22). The effect remains of similar size in the longer run and stays significant up to 12 years after graduation.

To capture the effect of sorting into firms that provide better learning opportunities and career support more directly, Panel B analyzes accumulated work experience in high-career-support firms, using Glassdoor's career opportunity rating dimension. Consistent with the work experience effects for high-wage firms, the figure shows that in the first six years after graduation, treated students also accumulate, on average, 0.047 years more work experience in high-career-support firms (5.5% relative to the baseline). In contrast to the constant effect for high-wage firms, the results at 12 and 19 years indicate that the effect on

[44]While the findings of the literature on heterogeneous skill accumulation across firms suggest that it is difficult to predict learning opportunities based on observable firm characteristics, Arellano-Bover and Saltiel (2026) provides some evidence that there is a weak positive correlation between firm pay and learning opportunities.

accumulated work experience in career-supporting firms increases over time. The results for both types of firms are thus consistent with the idea that an acceleration of on-the-job skill accumulation due to intensified early-career sorting into firms that provide better learning opportunities and career support explains part of the documented career benefits.

Figure 4: Facebook Access, Firm Sorting, and Human Capital Accumulation



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. Work experience at high-career-support firms: cumulative years employed at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Cumulative work experience: total years of employment. Appendix Table A22 reports the point estimates, the average outcomes for the not-yet-treated college cohorts, and the number of observations. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

The event-study estimates show that the accumulated work experience effect in higher-quality firms is particularly pronounced for students with longer exposure to Facebook during the last year of college, and generally support the parallel trends assumption in this setting (individual and joint insignificance at the 10% level, see Appendix Figure B7). The effects do not differ markedly between men and women (see Appendix Figure C6).

General labor market experience and long-run unemployment risk. Another potential reason behind the documented career benefits could be that students with early access to Facebook accumulated more general labor market experience and faced a lower risk of unemployment in the long run. Figure 4, Panel C, investigates this channel using cumulative work experience. Consistent with the documented positive early-career employment effects, students with early access to Facebook accumulate 0.040 years (0.48 months) more work experience in the first six years after expected graduation. The effect remains positive and significant 12 years after graduation (0.048 years) before slowly fading out in the long run. However, the relative effect sizes are small (0.9% and 0.5%, respectively, see Appendix Table A22),

suggesting that general labor market experience contributes little to the documented career effects.[45] Consistent with these results, Appendix Figure A5, Panel A, shows that the positive employment effects documented in Table 2 quickly fade out over time, suggesting that a long-run reduction in unemployment risk does not explain the documented career effects.

5.3 Robustness

This section presents a battery of robustness checks addressing concerns related to remaining spurious correlations with the Facebook rollout, spillover effects, multiple hypothesis testing, the choice of heterogeneity-robust estimator, the sample composition on LinkedIn, and empirical design choices for elite colleges.

Randomization inference. A key concern with the main empirical strategy is that even within narrow selectivity-by-region comparisons, there might be remaining unobservable spurious correlations with the timing of the Facebook rollout, biasing the main estimates. For example, Facebook may have (deliberately or not) expanded early to colleges *within* the same selectivity and U.S. region where career outcomes of U.S. business and economics cohorts were already on a differential trend. To formally test this, I implement a randomization inference test in which I randomly reshuffle the introduction year of Facebook across colleges within each selectivity-by-region cell. The test preserves the total number of treated colleges per cell and the overall distribution of treatment timing. Appendix D presents the results. The placebo estimates are densely distributed around zero, and virtually all statistically significant effects from the main specification fall above the 95th percentile of the placebo distribution. The remaining variation in the Facebook rollout is therefore unlikely to reflect spurious correlations.

Spillover analysis. A further concern is that the results are driven by spillovers. In particular, treated students might displace students in the control group from jobs, biasing the treatment effect upward and violating the stable-unit treatment value assumption (SUTVA). To assess the severity of this concern, I predict each potential control cohort's exposure to spillovers based on the historical career overlap of U.S. business and economics students across colleges prior to the Facebook rollout. I then drop all colleges that have at least one potential control cohort with the highest exposure score from the estimation. Reassuringly, Appendix E shows that the main patterns are very similar to the baseline estimates. Taken together with the previous findings, the results suggest that the treatment effects do not simply reflect a reallocation of jobs

[45] This is especially true in light of the well-documented evidence of decreasing returns to general labor market experience over a worker's career (e.g., Mincer 1974; Rubinstein and Weiss 2006).

within the economy. They are rather consistent with the idea that access to Facebook reduces information frictions about job opportunities.

Multiple hypothesis testing. Given the large number of main results, some statistically significant findings could simply arise by chance. Appendix Table A23 reports adjusted p-values controlling for the false discovery rate following Benjamini and Hochberg (1995). The main findings remain statistically significant after the correction (the majority at the 5% level and the rest at the 10% level), suggesting that the key results are unlikely to reflect false positives.

Further robustness. Appendix F examines whether the main effects are sensitive to the choice of heterogeneity-robust estimator, presenting results using the estimators of Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfœuille (2020) as two alternatives. The main patterns are very similar, indicating that the results are not driven by the choice of estimator. Appendix G reweights the analysis sample by gender and cohort size using IPEDS enrollment data to ensure representativeness in baseline demographics and college characteristics. The results are nearly identical to the baseline estimates, indicating that the main findings are not driven by LinkedIn's sample composition. Appendix G also excludes Ivy Plus and other elite colleges, which are among the earliest adopters and can only be modeled with linear cohort trends. The estimates remain similar, suggesting that the main findings are not driven by empirical design choices for elite colleges and that the career benefits also accrue to non-elite colleges.

6 Conclusion

More than 3 billion people worldwide report using online social networks such as Facebook or LinkedIn for work-related purposes (Kemp 2025), yet evidence on the labor market impacts of these platforms remains scarce. This paper investigates how providing young workers with access to Facebook during a sensitive period in their careers – the labor market entry phase – affects initial job placement and long-run career progression. To do so, I exploit the staggered introduction of Facebook across U.S. colleges as a natural experiment and trace the careers of exposed students through their public LinkedIn profiles. I document that access to Facebook at labor market entry generates long-lasting career benefits. U.S. business and economics students with early access to Facebook are 3.8% more likely to receive early-career promotions and 3.4% more likely to reach senior leadership positions in their late 30s. They also tend to work in higher-paying jobs, resulting in an economically meaningful increase in cumulative job-based earnings of around 1.6% over the first 19 years after graduation. I further document that the career benefits do not

differ markedly between men and women.

A considerable part of the long-run career benefits can be explained by a more successful college-to-work transition, in which the college network played a central role in providing access to higher-quality early-career jobs and shifting students toward typical college career paths. Looking at longer-run treatment dynamics, I further show that part of the career benefits can be explained by improved stability and productivity of early-career worker-firm matches, enabling students to climb the career ladder within their organizations, and by intensified early-career sorting into higher-quality and career-supporting firms. I do not find evidence that the career benefits are attributable to a long-run information advantage in the search for higher-paying jobs or a long-run reduction in unemployment risk. Overall, the findings of this paper highlight that access to online social networks during a sensitive period of a worker's career can reduce information frictions, facilitate the transition into the labor market, and generate long-lasting career benefits.

What can we learn from the findings about the role of online social networking platforms in job search and career development today? The use of online social networking platforms, such as Facebook, has increased considerably in recent decades, spreading from a small subset of college students to over 70% of U.S. adults by 2025 (Pew Research Center 2025). With a higher job-offer arrival rate across the entire population, traditional job search models (e.g., Burdett 1978) would predict that the career benefits of using online social networking platforms diminish as individuals increasingly hear about and compete for the same jobs.[46] Nevertheless, three pieces of evidence suggest that some of these benefits persist today: (i) the finding of this paper that the career benefits do not simply reflect a reallocation of high-quality jobs but are consistent with reduced information frictions and improved worker-firm match quality, (ii) the well-documented evidence of persistent information frictions in the labor market (e.g., Conlon *et al.* 2018; Heller and Kessler 2024), with job seekers continuing to be inaccurately informed about potential job opportunities (e.g., Belot *et al.* 2019; Jäger *et al.* 2024; Caldwell *et al.* 2026), and (iii) empirical evidence by Wheeler *et al.* (2022) on positive short-run labor market effects of LinkedIn training for young job seekers in more recent years. Therefore, career services and unemployment agencies should actively encourage the use of online social networking platforms as a low-cost tool for job search and career development that can mitigate search frictions and improve prospects for workers in sensitive periods of their careers.

While the results in this paper improve our understanding of the relevance of online social networks for job finding and career development of young workers, several important questions for future research

[46]This is especially true for a single online social networking platform, as more social networking platforms have nowadays gained popularity and provide work-related information (e.g., Facebook, LinkedIn, X).

remain. In particular, examining (i) how the career effects differ for less-educated workers and workers at older ages, (ii) the general equilibrium effects, and (iii) the relative importance of the different channels (e.g., referrals, recruiters, job ads, company profiles) and platforms (e.g., Facebook, LinkedIn, X) is key to fully understanding the labor market impacts of online social networking platforms today.

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Appendix A: Additional Tables and Figures

Table A1: Most Common Job Title Transitions for Promotion Events

Years Since Graduation	Top 10 Job Title Transitions (Promotion Events)
2-6 Years	Staff Accountant -> Senior Accountant, Financial Analyst -> Senior Financial Analyst, Analyst -> Associate, Associate -> Vice President, Consultant -> Senior Consultant, Accountant -> Senior Accountant, Account Executive -> Senior Account Executive, Account Executive -> Account Executive, Associate -> Associate, Associate -> Senior Associate
7-12 Years	Financial Analyst -> Senior Financial Analyst, Associate -> Vice President, Manager -> Senior Manager, Attorney -> Attorney, Staff Accountant -> Senior Accountant, Associate -> Partner, Associate -> Associate, Assistant Vice President -> Vice President, Project Manager -> Senior Project Manager, Senior Accountant -> Accounting Manager
13-19 Years	Project Manager -> Senior Project Manager, Vice President -> Senior Vice President, Associate -> Partner, Controller -> Chief Financial Officer, Assistant Controller -> Controller, Manager -> Senior Manager, Vice President -> President, Vice President -> Director, Attorney -> Partner, General Manager -> General Manager

Notes: Most common job title transitions for identified promotion events by years since graduation. Promotions are identified by large relative real annual wage growth jumps of at least 10 log points following Bronson and Thoursie (2019). *Data source:* Revelio Labs.

Table A2: Managerial Pipeline Validation – Earnings

Variable	Individual Contributor	Manager	Director	Vice president	Senior vice president	C-Suite
Job-based annual earnings	84,503	102,305	169,990	190,472	209,206	249,912
Share female	0.453	0.416	0.411	0.306	0.260	0.248
Observations	6,197,965	3,634,123	1,452,676	667,344	406,809	390,372

Notes: Descriptive statistics at the person–year level by managerial pipeline position, based on person–years with employment. Sample: all U.S. business & economics undergraduates expected to graduate from 2000 to 2005 at selective 4-year colleges with a valid LinkedIn profile and reasonable college-level coverage (1,165 colleges). *Data source:* Revelio Labs

Table A3: Most Common Job Titles – Leadership Pipeline

Category	Top 10 job titles
Individual Contributor	Account Executive, Senior Accountant, Associate, Accountant, Financial Advisor, Attorney, Financial Analyst, Sales Representative, Staff Accountant, Senior Financial Analyst
Manager	Project Manager, Manager, Account Manager, Operations Manager, Sales Manager, Store Manager, Branch Manager, Accounting Manager, Senior Manager, Marketing Manager
Director	Partner, Controller, Director, Director of Operations, Assistant Vice President, Marketing Director, Director of Finance, Director of Marketing, Director of Sales, Director Of Operations
Vice President	Vice President, Principal, Regional Vice President, Vice President of Sales, Vice President of Operations, Senior Director, VP, Vice President Operations, Vice President of Finance, VP of Operations
Senior Vice President	General Manager, Managing Director, Executive Director, Managing Partner, Senior Vice President, Executive Vice President, Assistant General Manager, Senior Managing Director, Managing Principal, GM
C-Level	President, CEO, Chief Financial Officer, Chief Executive Officer, Chief Operating Officer, CFO, COO, President & CEO, President/CEO, General Counsel

Notes: Most common job titles by leadership pipeline category. *Data source:* Revelio Labs.

Table A4: Coverage, Distribution, and Correlation of Permanent Firm Characteristics

	Coverage (%)		Distribution (emp.-wtd.)			Correlation matrix						
	LinkedIn	Est. sample	p_{25}	Median	p_{75}	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Firm pay	100.0	99.8	10.70	11.00	11.26	1.00	0.04	0.05	0.05	0.10	0.07	0.10
(2) Overall rating	43.8	69.6	3.34	3.65	3.94	0.04	1.00	0.77	0.84	0.73	0.84	0.69
(3) Career opportunities	43.2	69.0	3.01	3.35	3.63	0.05	0.77	1.00	0.77	0.65	0.79	0.73
(4) Culture & values	43.1	68.9	3.13	3.51	3.87	0.05	0.84	0.77	1.00	0.76	0.87	0.68
(5) Work-life balance	43.2	69.0	3.14	3.50	3.85	0.10	0.73	0.65	0.76	1.00	0.75	0.63
(6) Senior leadership	43.2	68.9	2.87	3.17	3.48	0.07	0.84	0.79	0.87	0.75	1.00	0.69
(7) Comp. & benefits	43.2	69.0	3.02	3.41	3.73	0.10	0.69	0.73	0.68	0.63	0.69	1.00

Notes: Descriptive statistics on the coverage, distribution, and correlation of all permanent firm characteristics. LinkedIn: coverage of the respective firm characteristic among all companies on LinkedIn that employ college students in the main estimation sample weighted by the average LinkedIn employee size. Est. sample: coverage of the respective firm characteristic among all person-year observations indicating an employer in the main estimation sample. Distribution columns: p_{25}, median, and p_{75} of each permanent firm characteristic among all companies on LinkedIn that employ college students in the main estimation sample weighted by the average LinkedIn employee size. Firm pay: average median log annual earnings, net of year fixed effects. Glassdoor amenity ratings are measured on a scale from 1–5: overall rating, career opportunities, culture & values, work-life balance, senior leadership, comp. & benefits. All Glassdoor amenity ratings are averaged across years, net of year fixed effects. Correlation matrix: Pearson correlations at the firm level between the different firm quality measures. *Data source:* Revelio Labs and Glassdoor.

Table A5: Sample Selection

Filtering step	Observations	Number of Colleges	Cumulative	Age	Female	At Least Very Selective College
All LinkedIn profiles, U.S. business and economics undergraduates	951,663	1,234	1.00	18.42	0.436	0.503
Valid undergraduate graduation date	931,104	1,234	0.978	18.41	0.435	0.504
Complete employment history	815,451	1,233	0.857	18.41	0.426	0.510
Reasonable college-level coverage	797,344	1,165	0.838	18.40	0.425	0.516

Notes: Each row reports the number of LinkedIn profiles remaining after the corresponding filtering step is applied. The final estimation sample covers all U.S. business & economics undergraduates expected to graduate from 2000 to 2005 at selective 4-year colleges with a valid LinkedIn profile and reasonable college-level coverage (1,165 colleges). Female: average predicted probability that the profile belongs to a woman (provided by Revelio Labs). *Data source:* Revelio Labs

Table A6: Treatment Timing and Potential Confounders

	Raw	Baseline
College characteristics		
Bachelor's degrees awarded (log)	0.046***	0.032
	(0.004)	(0.047)
Region (ref.: West)		
Midwest	0.019	—
	(0.012)	
Northeast	0.067***	—
	(0.014)	
South	0.004	—
	(0.011)	
Selectivity (ref.: Less competitive)		
Competitive	0.027**	—
	(0.010)	
Very competitive	0.046***	—
	(0.012)	
Highly competitive	0.045**	—
	(0.014)	
Elite	0.124***	—
	(0.016)	
Student and major composition		
Share female (in %)	-0.0008	-0.001
	(0.0004)	(0.001)
Share White (in %)	-0.002***	0.0006
	(0.0002)	(0.0010)
Share computer science (in %)	-0.018***	-0.0010
	(0.002)	(0.002)
Local labor market entry conditions		
College-state unemployment rate	-0.009	-0.005
	(0.006)	(0.009)
Joint F-test	50.72 [<0.001]	0.465 [0.803]
R^2	0.049	0.843
Observations	6,963	6,963
College FE		X
Cohort*Tier*Region FE		X

Notes: *p<0.05; **p<0.01; ***p<0.001. OLS regressions of the treatment indicator on potential confounders, estimated at the college*cohort level. Raw: no fixed effects. Baseline: fixed-effects structure of equation (1). The region and selectivity indicators are absorbed by the fixed effects. Bachelor's degrees awarded (log): log number of bachelor's degrees awarded by the college in the expected graduation year. Share female, share White, and share computer science (in %): shares of these bachelor's degrees awarded to women, to White students, and in computer and information sciences (CIP 11). College-state unemployment rate: unemployment rate in the college's state in the expected graduation year. Joint F-test: test of joint significance of the five time-varying characteristics reported in both columns, with p-values in brackets. Regressions are weighted by the number of LinkedIn profiles per college*cohort cell. Standard errors are clustered at the college level. *Data source:* Revelio Labs, IPEDS, BLS, and Barron's.

Table A7: Facebook Access, LinkedIn Sign-Up, and Balancing

	Coverage			Age at Entry	Female	First-Time Student	Prior Work Experience
	Overall	Male	Female				
FB Access	0.010	0.009	0.011	0.008	−0.004	0.0001	−0.005
	(0.011)	(0.014)	(0.012)	(0.024)	(0.006)	(0.004)	(0.005)
Dep. Var. Mean	0.457	0.509	0.402	18.42	0.420	0.886	0.236
Observations	6,963	6,779	6,890	6,963	6,963	6,963	6,963
College FE	X	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year or college*cohort*year*gender level. Coverage: fraction of students in the LinkedIn estimation sample compared to the official graduation counts in IPEDS. Coverage regressions are weighted using IPEDS graduation counts. Age at entry: imputed age at college entry. First-time student: indicator for students without a tertiary degree (e.g., an associate degree) before the bachelor's degree. Prior work experience: indicator for a first job on the profile that started before January 1 of the junior year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs and IPEDS.

Table A8: Facebook Access and LinkedIn Profile Completeness

	Coverage	Share of Profiles Meeting the Sample Restriction		
	(All Profiles)	Valid Graduation Date	Complete Employment History	All Restrictions (Final Sample)
FB Access	0.011	-0.003*	-0.002	-0.002
	(0.012)	(0.001)	(0.004)	(0.004)
Dep. Var. Mean	0.535	0.979	0.870	0.854
Observations	6,963	6,963	6,963	6,963
College FE	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort level. Each cell reports a coefficient from a separate DiD regression. Sample: all LinkedIn profiles of business and economics students expected to graduate from 2000 to 2005 at the 1,165 colleges of the estimation sample, before any sample restriction. Coverage: number of LinkedIn profiles relative to the official graduation counts in IPEDS. Valid graduation date: share of profiles reporting a valid undergraduate graduation date. Complete employment history: share of profiles reporting employment in at least 10 of the 19 years after expected graduation, i.e., for more than half of the career panel (see Appendix Table A5). All restrictions (final sample): share of profiles meeting all restrictions and entering the estimation sample. Column 1 is weighted using IPEDS graduation counts. Columns 2 to 4 are weighted by the number of LinkedIn profiles in the college cohort before any sample restriction. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs and IPEDS.

Table A9: Facebook Access and Career Development

	1–6 Years	7–12 Years	13–19 Years
Panel A: Job-Based Real Annual Earnings (2024, USD)			
FB Access	1,500*	1,867**	982
	(597)	(672)	(796)
Dep. Var. Mean	75,464	105,642	120,922
Observations	41,778	41,778	48,741
Panel B: Job-Based Promotion (≥10 log pts.)			
FB Access	0.003*	0.002	-0.002
	(0.001)	(0.001)	(0.001)
Dep. Var. Mean	0.080	0.086	0.075
Observations	34,815	41,778	48,741
Panel C: Senior Leadership Positions			
FB Access	0.001	0.006	0.009*
	(0.003)	(0.003)	(0.003)
Dep. Var. Mean	0.100	0.196	0.267
Observations	41,778	41,778	48,741
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. Panel A: job-based annual earnings (2024 USD), set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A10: Facebook Access, High-Paying Jobs, and Steep Promotion Profiles

	High-Paying Job		Steep Promotion Profile	
	Top 25%	Top 10%	≥2 Promotions	≥3 Promotions
FB Access	0.008*	0.0002	0.008	0.009**
	(0.004)	(0.002)	(0.005)	(0.003)
Dep. Var. Mean	0.244	0.096	0.355	0.137
Observations	48,741	48,741	48,741	48,741
College FE	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the years 13–19 after expected graduation. High-paying job: indicator for job-based annual earnings in the top 25% or top 10% of the distribution among all graduates in the sample of the same expected graduation cohort in the same year after expected graduation (thresholds between $149,000 and $182,000 for the top 25% and between $226,000 and $259,000 for the top 10%, in 2024 USD). Steep promotion profile: indicator for at least two or at least three job-based promotions since the second year after expected graduation, which roughly corresponds to the top 37% and the top 15% of all graduates 13–19 years after expected graduation (between 31% and 41%, and between 11% and 18%, depending on the cohort). Job-based annual earnings are set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A11: Facebook Access and Cumulative Earnings

	$\delta = 0\%$	$\delta = 2.5\%$	$\delta = 5\%$
FB Access	27,074*	22,533*	19,035**
	(11,365)	(8,990)	(7,308)
Dep. Var. Mean	1,933,096	1,520,993	1,223,438
Observations	6,963	6,963	6,963
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort level. The present discounted value of earnings is the discounted sum of average annual earnings over the 19-year observation window: $\text{PDV}(\delta) = \sum_{t=1}^{19} \bar{w}_{j,c,t}/(1+\delta)^{t-1}$, where $\bar{w}_{j,c,t}$ is the average job-based annual earnings of college j, cohort c, at year t after graduation, and δ is the discount rate. Job-based annual earnings are set to zero in years without employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A12: Facebook Access and the College-to-Work Transition – Other High-Amenity Firm Dimensions

	Overall	Comp. & Benef.	Culture	Leadership	Work-Life
Panel A: Employed at High-Amenity Firm					
FB Access	0.0007	0.011**	0.004	0.005	0.004
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)
Dep. Var. Mean	0.100	0.139	0.113	0.123	0.096
Observations	13,912	13,910	13,910	13,910	13,910
Panel B: Early-Career High-Amenity Firm Job via College Network					
FB Access	0.001	0.004	0.002	0.003*	0.0003
	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)
Dep. Var. Mean	0.017	0.027	0.020	0.029	0.014
Observations	13,912	13,910	13,910	13,910	13,910
College FE	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. High-amenity firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor rating distribution for the given dimension (among firms with at least two ratings). Person-years at employers without a Glassdoor rating are excluded, and person-years without employment are coded as zero. The career opportunities dimension is shown separately in Tables 2, 3, and A22. Overall: overall rating on Glassdoor. Comp. & Benef.: compensation and benefits. Culture: company culture. Leadership: senior leadership. Work-Life: work-life balance. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A13: Facebook Access and Missing Glassdoor Rating – Balancing Check

	Overall	Comp. & Benef.	Career Opps.	Culture	Leadership	Work-Life
FB Access	0.002	0.002	0.002	0.003	0.003	0.002
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Dep. Var. Mean	0.208	0.212	0.212	0.213	0.212	0.212
Observations	13,926	13,926	13,926	13,926	13,926	13,926
College FE	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Missing amenity: indicator for working at an employer with no Glassdoor rating in the respective dimension. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A14: Facebook Access and the College-to-Work Transition – Other Job Types

	Non-High-Wage Firm	Non-High-Career-Support Firm	Non-High-Wage Occupation	Non-Steep Wage Growth	Non-Sr. Leadership Track
FB Access	-0.008	-0.002	-0.002	0.008	-0.007
	(0.005)	(0.006)	(0.006)	(0.005)	(0.005)
Dep. Var. Mean	0.450	0.405	0.406	0.484	0.442
Observations	13,926	13,910	13,926	13,926	13,926
College FE	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Non-High-Wage Firm: indicator for working at a firm not in the upper employment-weighted quartile of the average median log earnings distribution at the firm level. Non-High-Career-Support Firm: indicator for working at a firm not in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Person-years at employers without a Glassdoor rating are excluded, and person-years without employment are coded as zero. Non-High-Wage Occupation: indicator for working in an occupation not in the upper employment-weighted quartile of the average median log earnings distribution at the 5-digit SOC level. Non-Steep Wage Growth: indicator for an early-career job with a leave-one-cohort-out 15-year wage growth prediction at or below the 75th percentile. Non-Sr. Leadership Track: indicator for an early-career job with a leave-one-cohort-out prediction of reaching senior leadership 15 years after at or below the 75th percentile. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A15: Facebook Access and Education Outcomes

			Postgraduate Degree			
	Study Duration	In College	Any	Business & Economics	MBA	PhD
FB Access	0.003	-0.006	0.0008	0.002	-0.0008	-0.002
	(0.013)	(0.003)	(0.006)	(0.006)	(0.005)	(0.002)
Dep. Var. Mean	3.83	0.111	0.328	0.261	0.147	0.014
Observations	6,963	13,926	6,963	6,956	6,963	6,963
College FE	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Study Duration: total years from enrollment to completion of the bachelor's degree. In College: indicator whether students are enrolled in a degree program on June 30 of the first two years after expected graduation. Postgraduate Degree: reporting a master's, PhD, or professional degree as the highest degree on the LinkedIn profile. PhD: reporting a doctorate as the highest degree on the LinkedIn profile. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A16: Facebook Access and Early-Career Job Help in the College Network – Other Job Types

	Non-High-Wage Firm	Non-High-Career-Support Firm	Non-High-Wage Occupation	Non-Steep Wage Growth	Non-Sr. Leadership Track
Panel A: Early-Career Job via College Network					
FB Access	0.004	0.007**	0.008***	0.005	0.003
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Dep. Var. Mean	0.066	0.076	0.064	0.080	0.061
Observations	13,926	13,910	13,926	13,926	13,926
Panel B: Early-Career Job without College Network					
FB Access	-0.012*	-0.009	-0.010	0.004	-0.010
	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)
Dep. Var. Mean	0.383	0.329	0.341	0.404	0.381
Observations	13,926	13,910	13,926	13,926	13,926
College FE	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Complement outcomes to Table 3. Panel A: lower-quality early-career job at a firm with a college peer already working there. Panel B: the same job at a firm without a college peer already working there. The two panels partition the corresponding outcome in Table A14, so the Panel A and Panel B coefficients sum, up to rounding, to the coefficient reported there. College peers: students up to 3 classes above and below the expected graduation cohort. Job via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). Non-High-Career-Support Firm: indicator for working at a firm not in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Person-years at employers without a Glassdoor rating are excluded, and person-years without employment are coded as zero. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A17: Facebook Access, Prior Hires from College, and the College-to-Work Transition

	Employment	High-Wage Firm	High-Career-Support Firm	High-Wage Occupation	High-Wage Industry	Steep Wage Growth	Sr. Leadership Track	First Stable Job
Panel A: Firm with Prior Hire from College								
FB Access	0.013**	0.011***	0.010**	0.007**	0.009**	0.007***	0.011***	0.014**
	(0.004)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)	(0.005)
Dep. Var. Mean	0.174	0.065	0.077	0.068	0.085	0.043	0.069	0.239
Observations	13,926	13,926	13,910	13,926	13,919	13,926	13,926	6,963
Panel B: Firm without Prior Hire from College								
FB Access	-0.003	0.007	0.002	0.005	0.012**	-0.005	0.006	-0.014**
	(0.005)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)	(0.005)
Dep. Var. Mean	0.480	0.138	0.081	0.179	0.160	0.116	0.132	0.761
Observations	13,926	13,926	13,910	13,926	13,919	13,926	13,926	6,963
College FE	X	X	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Prior hire from college: at least one graduate of the same college from the three expected graduation cohorts two to four years earlier worked at the firm in the first or second year after expected graduation. Jobs without a firm identifier are included in Panel B. In Columns 1–7, the two panels partition the corresponding outcome in Table 2, so the coefficients sum, up to rounding, to the coefficient reported there. Column 8 (First Stable Job): indicator for the first active employment spell after expected graduation lasting at least 4 months being at a firm with (Panel A) or without (Panel B) a prior hire from college, estimated at the college*cohort level. The two shares sum to one. See Table 2 for the definitions of the job characteristics. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A18: Facebook Access, Career Similarities, and the College-to-Work Transition

	Occupation	Industry	Firm
Panel A: Typical College Career Choice			
FB Access	0.016***	0.012*	0.013**
	(0.004)	(0.005)	(0.004)
Dep. Var. Mean	0.176	0.235	0.174
Observations	13,926	13,926	13,926
Panel B: Atypical College Career Choice			
FB Access	-0.007	-0.002	-0.003
	(0.006)	(0.006)	(0.005)
Dep. Var. Mean	0.478	0.419	0.480
Observations	13,926	13,926	13,926
Panel C: Same Career Choice as Same College-Cohort Peers			
FB Access	0.003**	0.004*	0.0004
	(0.0009)	(0.002)	(0.0005)
Dep. Var. Mean	0.030	0.065	0.005
Observations	13,807	13,746	13,808
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Prior graduates from college: graduates of the same college from the three expected graduation cohorts two to four years earlier, in the first or second year after expected graduation. Panel A: job in one of the five occupations (Column 1) or industries (Column 2) held most often by prior graduates from college, following Altonji *et al.* (2016), or at a firm with a prior hire from college (Column 3, see Table A17). Panel B: all other jobs. Panel C: probability that two randomly drawn employed graduates of the same college and expected graduation cohort work in the same 5-digit SOC occupation (Column 1), 3-digit NAICS industry (Column 2), or firm (Column 3), computed for college cohorts with at least two such graduates using the Simpson index. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A19: Facebook Access, the College-to-Work Transition, and the Role of Job Location

	Employment by Job Location			Accounting for Geographic Job Sorting				
	Outside College's MSA	Outside College's MSA: Large Metro Area	Outside College's MSA: Three Largest MSAs	High-Wage Firm	High-Career-Support Firm	High-Wage Occupation	High-Wage Industry	Sr. Leadership Track
FB Access	0.002	0.009	0.007*	0.016***	0.012**	0.011*	0.020***	0.017***
	(0.005)	(0.006)	(0.003)	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)
Share Explained by Geographic Job Sorting				12%	2%	5%	6%	4%
Dep. Var. Mean	0.419	0.282	0.061	0.203	0.158	0.246	0.245	0.201
Observations	13,926	13,926	13,926	13,926	13,910	13,926	13,919	13,926
College FE	X	X	X	X	X	X	X	X
Cohort*Tier*Region FE	X	X	X	X	X	X	X	X
Job Location FE				X	X	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, pooled over the first two years after expected graduation. Job location is defined at the level of metropolitan statistical areas (MSAs, 2009 delineation published by the U.S. Census Bureau), at the state level outside MSAs, and at the country level abroad. Columns 1–3: indicators for working outside the college's MSA (or non-metropolitan part of the college's state), outside it in a large metropolitan area with at least one million residents in 2010 (Ingram and Franco 2014), and outside it in one of the three largest MSAs by 2010 population (New York, Los Angeles, and Chicago). Non-employed students are coded as zero. Columns 4–8 account for geographic job sorting by residualizing the high-quality job indicators of Table 2 that showed significant treatment effects at the 5% level (see the definitions there): I estimate job-location fixed effects among employed students in the not-yet-treated college cohorts (conditional on the baseline fixed effects and trends), subtract each employed student's job-location fixed effect net of its mean in that group, and run the regression on the residualized indicators. Share explained by geographic job sorting: one minus the ratio of the residualized estimate to the corresponding estimate in Table 2. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs, U.S. Census Bureau.

Table A20: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network

	2–6 Years	7–12 Years	13–19 Years
Panel A: Between-Firm Promotions (≥10 log pts.)			
FB Access	-0.0002	0.0003	-0.002
	(0.001)	(0.001)	(0.0009)
Dep. Var. Mean	0.057	0.055	0.045
Observations	34,815	41,775	48,741
Panel B: Promotion via College Network (≥10 log pts.)			
FB Access	-0.0005	0.0005	0.00004
	(0.0005)	(0.0003)	(0.0003)
Dep. Var. Mean	0.012	0.010	0.007
Observations	34,815	41,778	48,741
Panel C: Senior Leadership Transition via College Network			
FB Access	-0.0001	0.0001	-0.000008
	(0.0002)	(0.0002)	(0.0001)
Dep. Var. Mean	0.002	0.002	0.002
Observations	34,815	41,778	48,741
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. College peers: students up to 3 classes above and below the expected graduation cohort. Promotion via College Network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017), coupled with an immediate promotion. Senior Leadership Transition via College Network: indicator for a transition from a non-senior leadership to a senior leadership role at a firm where a college peer had already worked for at least one year. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A21: Facebook Access and Worker-Firm Match Stability and Productivity

	1–6 Years	7–12 Years	13–19 Years
Panel A: Stable Worker-Firm Match (≥4 Months)			
FB Access	0.007*	0.002	-0.0004
	(0.003)	(0.002)	(0.003)
Dep. Var. Mean	0.717	0.890	0.901
Observations	41,778	41,778	48,741
Panel B: Current Firm Tenure (Years)			
FB Access	0.029*	0.071*	0.090
	(0.013)	(0.033)	(0.056)
Dep. Var. Mean	1.64	4.03	6.06
Observations	41,778	41,778	48,741
Panel C: Within-Firm Promotions (≥10 log pts.)			
FB Access	0.003***	0.002	-0.0002
	(0.0008)	(0.0009)	(0.0008)
Dep. Var. Mean	0.023	0.031	0.030
Observations	34,815	41,775	48,741
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. Stable worker-firm match: indicator equal to one if the individual is employed at the same firm for at least 4 months. Current firm tenure: years at current employer. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A22: Facebook Access, Firm Sorting, and Human Capital Accumulation

	6Y	12Y	19Y
Panel A: Work Experience at High-Wage Firms (Years)			
FB Access	0.105***	0.127*	0.118
	(0.025)	(0.052)	(0.080)
Dep. Var. Mean	1.47	3.43	5.84
Observations	6,963	6,963	6,963
Panel B: Work Experience at High-Career-Support Firms (Years)			
FB Access	0.047**	0.077*	0.103
	(0.017)	(0.038)	(0.061)
Dep. Var. Mean	0.848	1.81	2.99
Observations	6,955	6,948	6,954
Panel C: Cumulative Work Experience (Years)			
FB Access	0.040*	0.048*	0.038
	(0.020)	(0.023)	(0.024)
Dep. Var. Mean	4.51	10.15	16.81
Observations	6,963	6,963	6,963
College FE	X	X	X
Cohort*Tier*Region FE	X	X	X

Notes: *p<0.05; **p<0.01; ***p<0.001. Imputation DiD using Borusyak *et al.* (2024), estimated at the college*cohort*year level. Each cell reports a coefficient from a separate DiD regression, estimated at 6, 12, and 19 years after expected graduation. Work experience at high-wage firms: cumulative years employed at a firm in the upper quartile of the average median log earnings distribution. High-career-support firm: indicator for working at a firm in the upper employment-weighted quartile of the Glassdoor career opportunities rating distribution (among firms with at least two ratings). Work experience at high-career-support firms: cumulative years employed at a high-career-support firm. Cumulative work experience: total years of employment. Dep. Var. Mean: average outcome for the not-yet-treated college cohorts. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Table A23: Multiple Hypothesis Testing Corrections

	Coefficient	Unadj. p-value	BH (1995) p-value
Panel A: Career Development (Figure 1)			
Job-Based Real Annual Earnings [1–6Y]	1,500*	0.012	0.036
Job-Based Real Annual Earnings [7–12Y]	1,867**	0.006	0.019
Job-Based Real Annual Earnings [13–19Y]	982	0.217	0.283
Job-Based Promotion (≥10 log pts.) [2–6Y]	0.003*	0.031	0.066
Job-Based Promotion (≥10 log pts.) [7–12Y]	0.002	0.137	0.191

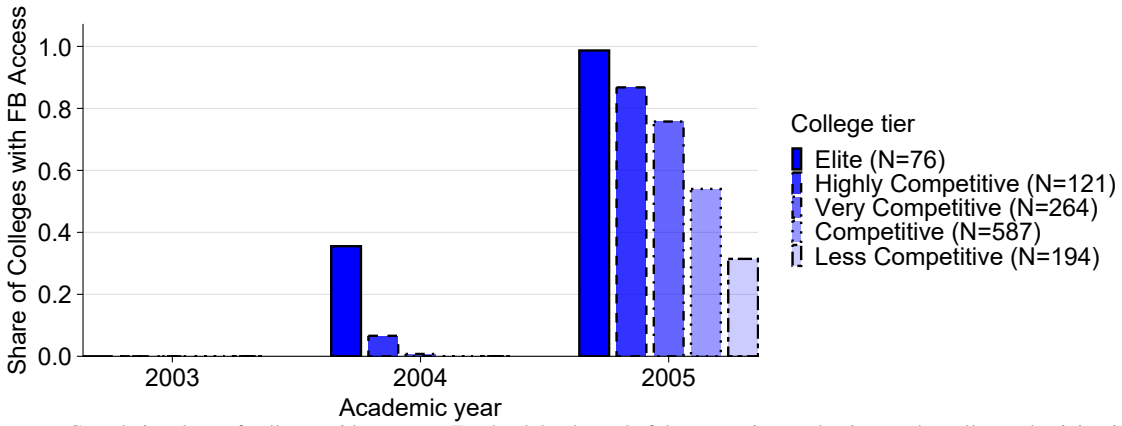
	Coefficient	Unadj. p-value	BH (1995) p-value
Job-Based Promotion (≥10 log pts.) [13–19Y]	-0.002	0.116	0.170
Senior Leadership Positions [1–6Y]	0.001	0.654	0.740
Senior Leadership Positions [7–12Y]	0.006	0.101	0.160
Senior Leadership Positions [13–19Y]	0.009*	0.012	0.036
Panel B: College-to-Work Transition (Table 2)			
Employment [1–2Y]	0.010*	0.023	0.053
High-Wage Firm [1–2Y]	0.018***	0.00003	0.0003
High-Career-Support Firm [1–2Y]	0.012**	0.001	0.006
High-Wage Occupation [1–2Y]	0.012*	0.022	0.053
High-Wage Industry [1–2Y]	0.021***	0.00003	0.0003
Steep Wage Growth [1–2Y]	0.002	0.574	0.662
Sr. Leadership Track [1–2Y]	0.018***	0.00002	0.0003
Time to First Stable Job (Years)	-0.054*	0.020	0.051
Panel C: Early-Career Job Help in the College Network (Table 3)			
Job Help × Any Job [1–2Y]	0.011***	0.00005	0.0003
Job Help × High-Wage Firm [1–2Y]	0.007***	0.00002	0.0003
Job Help × High-Career-Support Firm [1–2Y]	0.007***	0.0002	0.001
Job Help × High-Wage Occupation [1–2Y]	0.004	0.069	0.122
Job Help × High-Wage Industry [1–2Y]	0.006**	0.002	0.007
Job Help × Steep Wage Growth [1–2Y]	0.007***	0.000007	0.0002
Job Help × Sr. Leadership [1–2Y]	0.008***	0.000003	0.0002
Job Help × First Stable Job	0.017***	0.00004	0.0003
No Job Help × Any Job [1–2Y]	-0.002	0.752	0.835
No Job Help × High-Wage Firm [1–2Y]	0.011**	0.003	0.011
No Job Help × High-Career-Support Firm [1–2Y]	0.005	0.166	0.222
No Job Help × High-Wage Occupation [1–2Y]	0.009	0.096	0.156
No Job Help × High-Wage Industry [1–2Y]	0.015***	0.0008	0.004
No Job Help × Steep Wage Growth [1–2Y]	-0.004	0.377	0.462
No Job Help × Sr. Leadership [1–2Y]	0.010*	0.012	0.036
No Job Help × First Stable Job	-0.017***	0.00004	0.0003

	Coefficient	Unadj. p-value	BH (1995) p-value
Panel D: Firm Mobility and Long-Run Job Help in the College Network (Figure 2)			
Between-Firm Promotion (≥10 log pts.) [2–6Y]	-0.0002	0.886	0.920
Between-Firm Promotion (≥10 log pts.) [7–12Y]	0.0003	0.800	0.857
Between-Firm Promotion (≥10 log pts.) [13–19Y]	-0.002	0.089	0.152
Job Help × Promotions [2–6Y]	-0.0005	0.240	0.306
Job Help × Promotions [7–12Y]	0.0005	0.117	0.170
Job Help × Promotions [13–19Y]	0.00004	0.904	0.920
Job Help × Sr. Leadership [2–6Y]	-0.0001	0.442	0.520
Job Help × Sr. Leadership [7–12Y]	0.0001	0.402	0.483
Job Help × Sr. Leadership [13–19Y]	-0.000008	0.954	0.954
Panel E: Worker-Firm Match Stability and Productivity (Figure 3)			
Stable Worker-Firm Match (≥4 Months) [1–6Y]	0.007*	0.025	0.056
Stable Worker-Firm Match (≥4 Months) [7–12Y]	0.002	0.366	0.458
Stable Worker-Firm Match (≥4 Months) [13–19Y]	-0.0004	0.891	0.920
Firm Tenure [1–6Y]	0.029*	0.022	0.053
Firm Tenure [7–12Y]	0.071*	0.032	0.066
Firm Tenure [13–19Y]	0.090	0.105	0.161
Within-Firm Promotion (≥10 log pts.) [2–6Y]	0.003***	0.0002	0.001
Within-Firm Promotion (≥10 log pts.) [7–12Y]	0.002	0.065	0.119
Within-Firm Promotion (≥10 log pts.) [13–19Y]	-0.0002	0.765	0.835
Panel F: Firm Sorting and Human Capital Accumulation (Figure 4)			
Exp. at High-Wage Firm [6Y]	0.105***	0.00003	0.0003
Exp. at High-Wage Firm [12Y]	0.127*	0.015	0.040
Exp. at High-Wage Firm [19Y]	0.118	0.140	0.191
Exp. at High-Career-Support Firm [6Y]	0.047**	0.006	0.019
Exp. at High-Career-Support Firm [12Y]	0.077*	0.042	0.078
Exp. at High-Career-Support Firm [19Y]	0.103	0.092	0.153
Work Experience [6Y]	0.040*	0.042	0.078
Work Experience [12Y]	0.048*	0.034	0.069

Work Experience [19Y]	0.038	0.119	0.170

Notes: Unadjusted and adjusted p-values for each coefficient in the main results section. Adjusted p-values are computed using the Benjamini and Hochberg (1995) adjustment procedure. *Data source:* Revelio Labs.

Figure A1: Facebook Rollout across Selective U.S. Colleges with a Business and Economics Undergraduate Degree by College Selectivity Tier



Notes: Cumulative share of colleges with access to Facebook by the end of the respective academic year, by college selectivity tier. College selectivity is measured using Barron's selectivity index (Barron's Educational Series 2008). N: number of colleges per selectivity tier. *Data source:* Facebook introduction dates from Armona (2025).

Figure A2: Imputed Real Annual Wage Growth Distribution

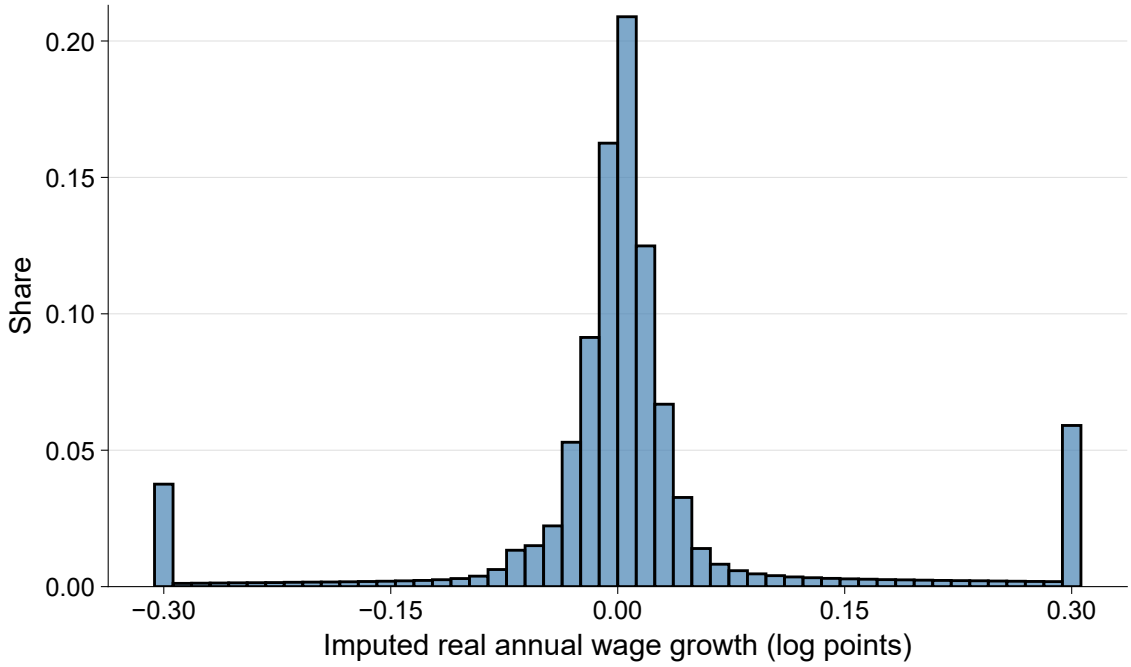
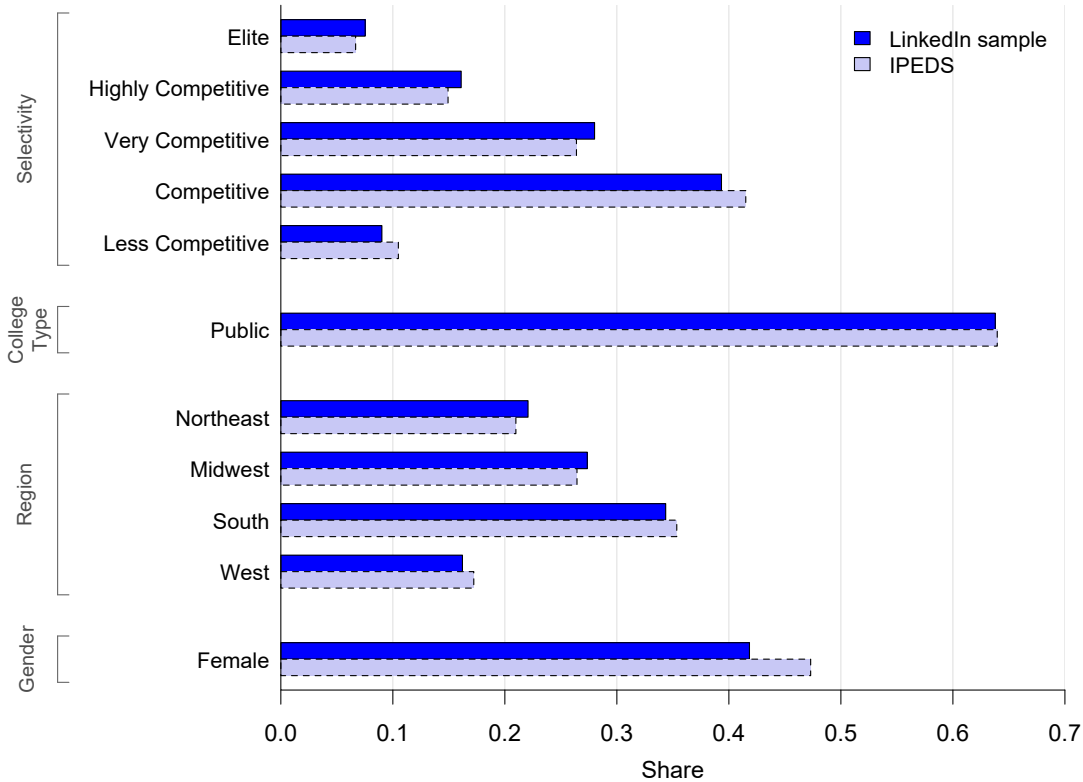
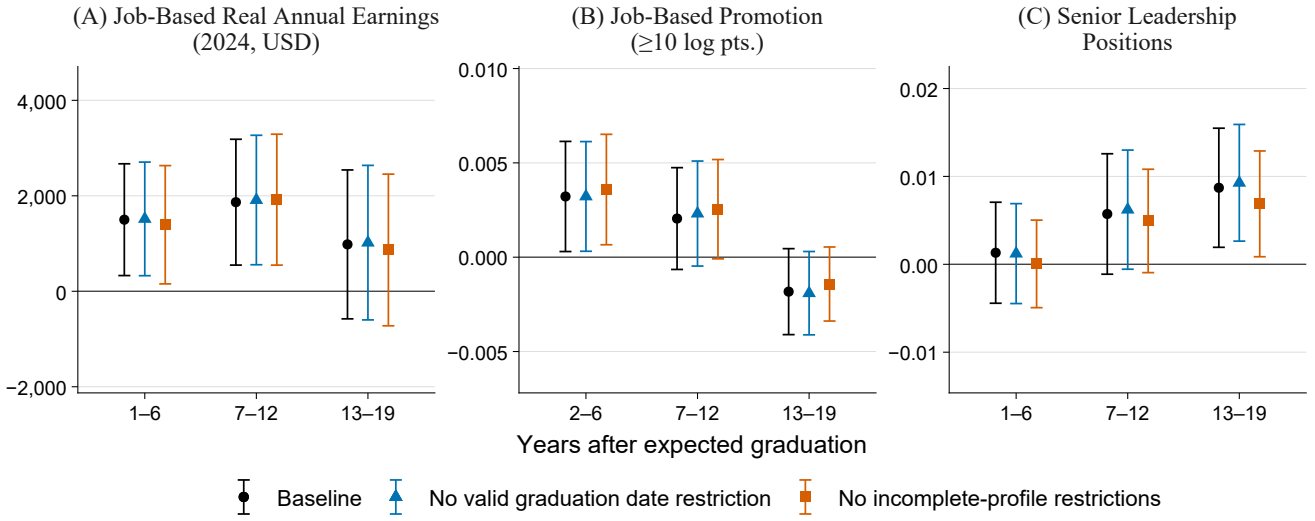


Figure A3: Sample Comparison – Revelio Labs vs. IPEDS



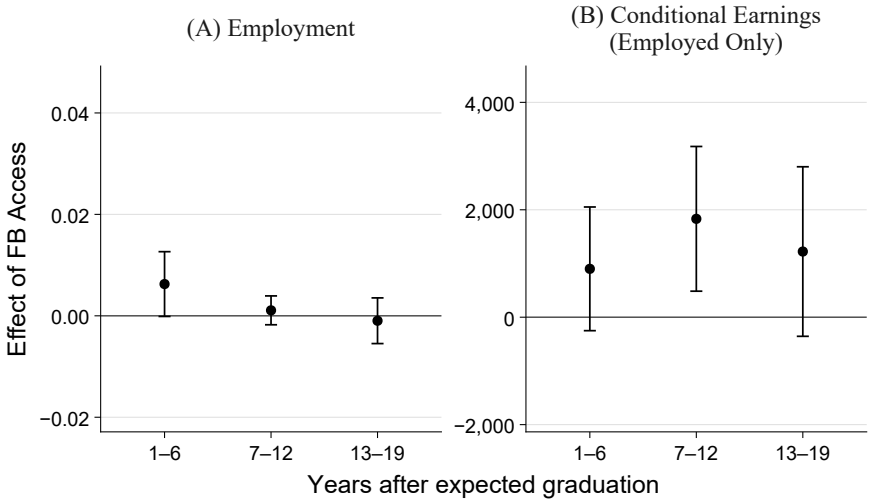
Notes: Descriptive statistics for various samples. LinkedIn sample (actual graduation): all business & economics graduates from the 1,165 colleges in the graduation cohorts from 2000 to 2005 with a valid LinkedIn profile and reasonable college-level coverage (N = 811,178). LinkedIn sample (expected graduation): all business & economics students from the 1,165 colleges in the expected graduation cohorts from 2000 to 2005 with a valid LinkedIn profile and reasonable college-level coverage (N = 797,344). IPEDS: all business & economics graduates from the 1,165 colleges in the graduation cohorts from 2000 to 2005 in the IPEDS data (N = 1,688,064). *Data source:* Revelio Labs and IPEDS.

Figure A4: Facebook Access and Career Development – Robustness to the Incomplete-Profile Restrictions



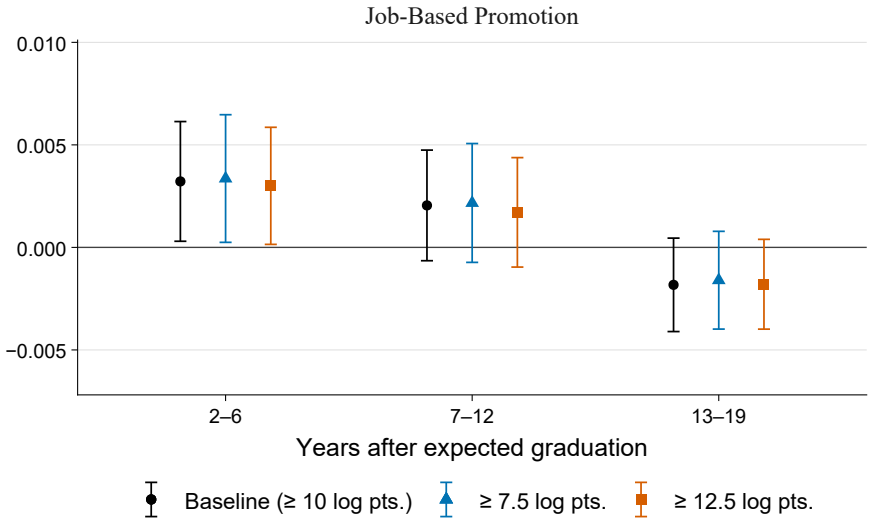
Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 1, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. Baseline: estimation sample of Figure 1 (797,344 LinkedIn profiles). No valid graduation date restriction: sample without the restriction to profiles with a valid graduation date (810,872 profiles). No incomplete-profile restrictions: sample without the restrictions to profiles with a valid graduation date and a complete employment history (929,430 profiles). All samples cover the 1,165 colleges of the estimation sample. See the notes of Figure 1 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A5: Facebook Access, Employment, and Conditional Earnings



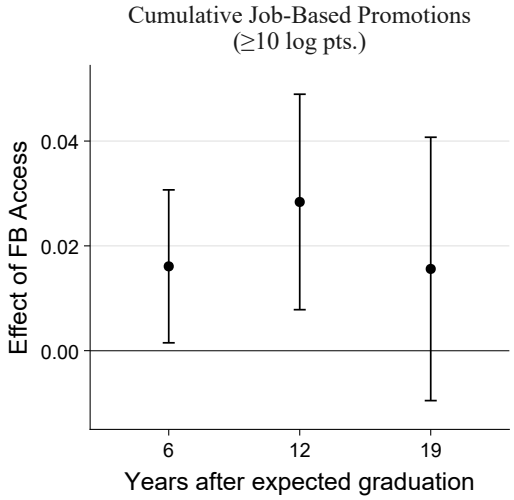
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Employment: share of graduates with a recorded employment spell. Conditional earnings: average job-based annual earnings (2024 USD) conditional on employment, weighted by the number of employed graduates. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A6: Facebook Access and Alternative Promotion Thresholds



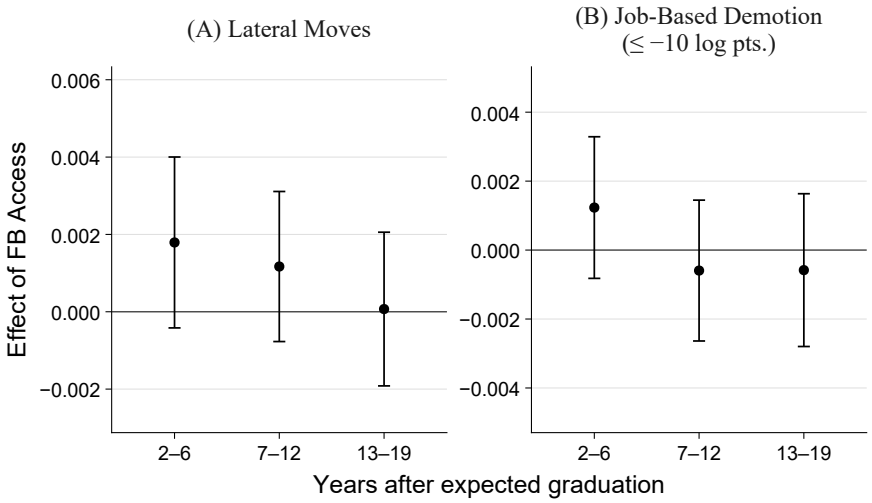
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Baseline: job-based promotion measure of Figure 1, which requires a wage growth jump of at least 10 log points. The two other series use a threshold of 7.5 and of 12.5 log points. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A7: Facebook Access and Cumulative Promotions



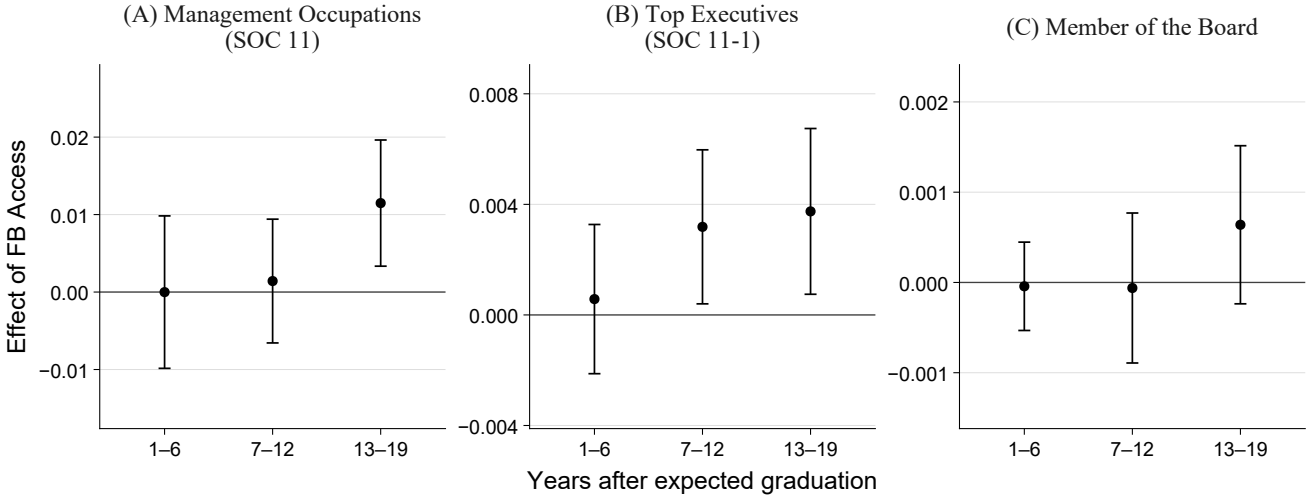
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort level. Outcome: cumulative number of job-based promotions (at least 10 log points, as in Figure 1) from the second year up to 6, 12, and 19 years after expected graduation, measured as the sum of the annual promotion rates of the college cohort. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A8: Facebook Access, Lateral Moves, and Demotion



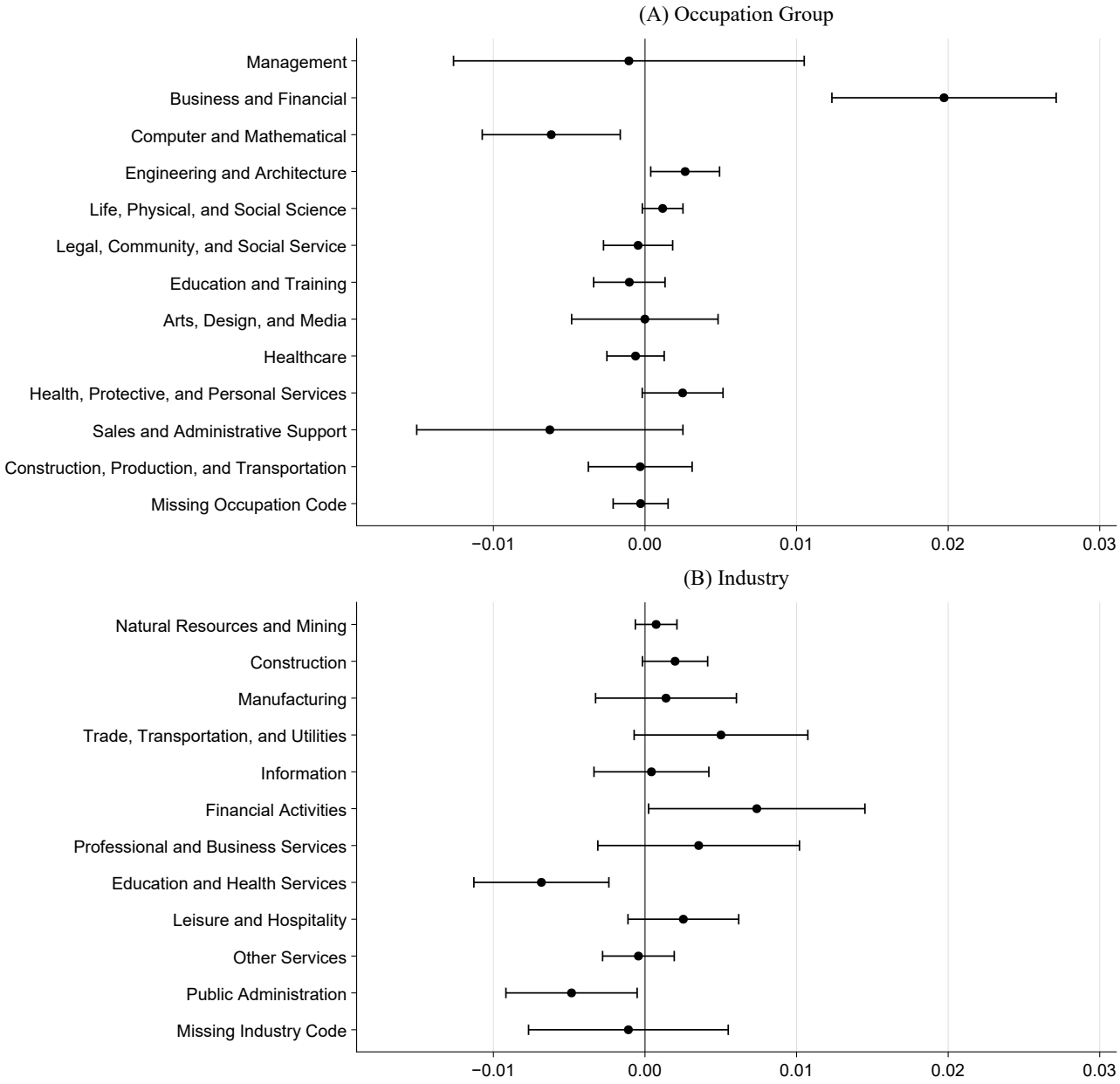
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Lateral moves: share of graduates transitioning to a job with relative real annual wage growth between −10 and +10 log points. Job-based demotion: share of graduates transitioning to a job with relative real annual wage growth below −10 log points. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A9: Facebook Access and Alternative Top Jobs



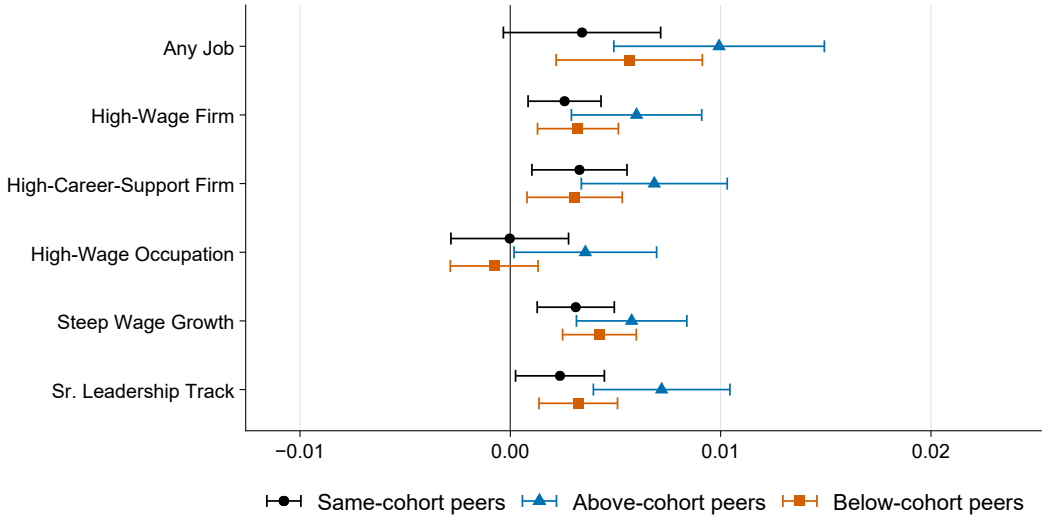
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. Management occupations (top executives): indicator for working in an occupation of SOC major group 11 (minor group 11-1). Member of the board: indicator for working as a board member or member of the board of directors using a keyword search on job titles. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A10: Facebook Access and Occupation and Industry Choice at Labor Market Entry



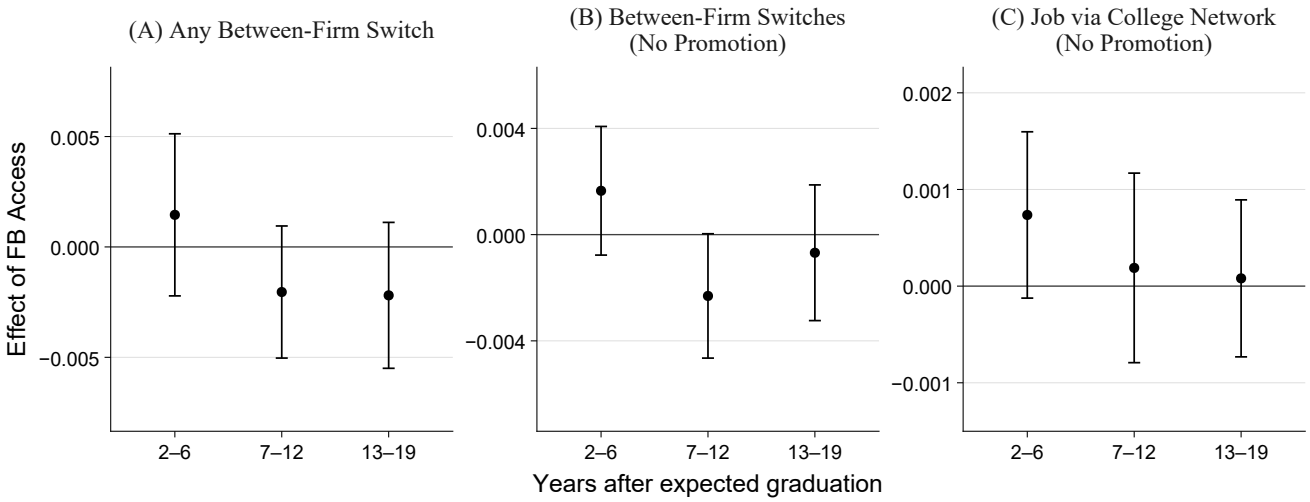
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level, pooled over the first two years after expected graduation. Outcome: indicator for working in an occupation (Panel A) or industry (Panel B) of the respective group, set to zero in years without employment. Within each panel, the groups partition the Employment outcome of Table 2, so that the coefficients sum to its Column 1 coefficient. Groups are ordered by their SOC and NAICS codes. Panel A: occupation categories of Deming and Noray (2020), based on 2-digit SOC codes (Legal, Community, and Social Service: SOC 21 and 23; Health, Protective, and Personal Services: SOC 31–39; Sales and Administrative Support: SOC 41 and 43; Construction, Production, and Transportation: SOC 45–55). Panel B: NAICS supersectors of the U.S. Bureau of Labor Statistics. Missing occupation (industry) code: employed in a job without an occupation (industry) code. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A11: Facebook Access and Early-Career Jobs via College Network by Peer Type



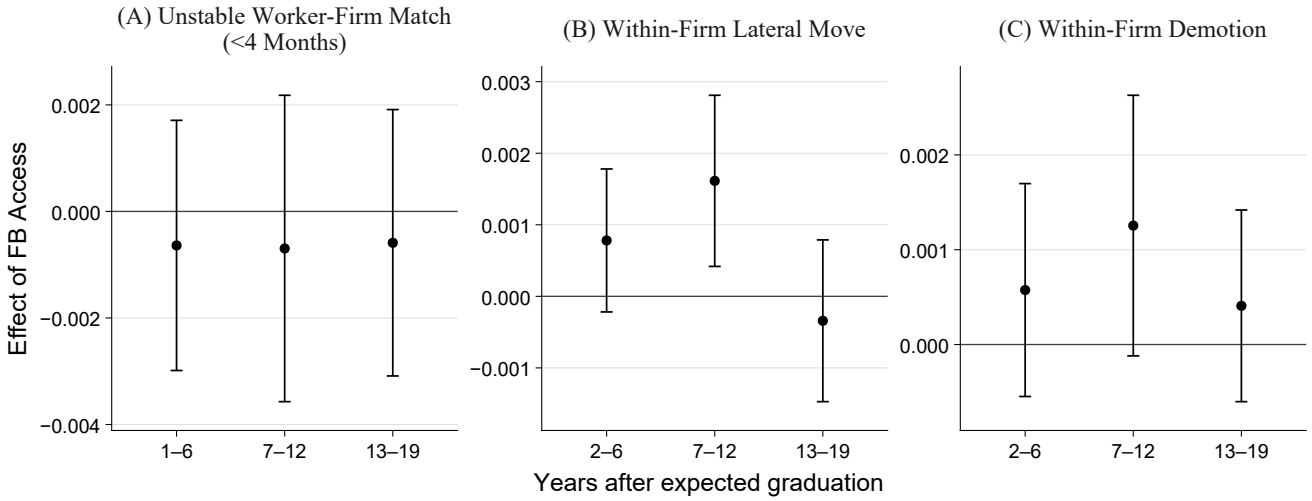
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the first two years after expected graduation. Same-cohort peers: students in the expected graduation cohort. Above-cohort peers: students 1 to 3 classes above the expected graduation cohort. Below-cohort peers: students 1 to 3 classes below the expected graduation cohort. Any job: indicator for joining a firm where a college peer of the respective type had already worked for at least one year, following Gee *et al.* (2017). See the notes of Table 3 for the definitions of the job characteristics. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A12: Facebook Access and Between-Firm Job Changes



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Any between-firm switch: share of graduates who change employers in the given period. Between-firm switches (no promotion): share of graduates who change employers without a relative real annual wage growth jump of at least 10 log points. Job via college network (no promotion): share of graduates who change to a firm where a college peer had already worked, without a relative real annual wage growth jump of at least 10 log points. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure A13: Facebook Access, Job Stability, and Within-Firm Mobility



Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. The within-firm outcomes start in year 2, as within-firm transitions are only defined for ongoing employment spells. Unstable worker-firm match: share of graduates with a current employer tenure of less than 4 months. Within-firm lateral move: share of graduates who transition to a different role at the same employer with a relative real annual wage growth jump between −10 and +10 log points. Within-firm demotion: share of graduates who transition to a role at the same employer with a relative real annual wage growth jump below −10 log points. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Appendix B: Event-Study Specification

To assess the plausibility of the parallel trends assumption and examine potential treatment dynamics in terms of the time exposed to Facebook during college, I estimate an event-study version of equation (1), where the expected graduation year of the first freshman cohort exposed to the introduction of Facebook at a given college is referred to as the *event*. The event-study model in a two-way fixed effects (TWFE) setting can be specified as follows:

$$Y_{fs,c,t} = \sum_{k=-4}^{-2} \mathbf{1}\{t_{fs,c} - t_c = k\}\,\beta_k + \mathbf{1}\{t_{fs,c} \geq t_c,\, M_{fs,c} \leq 6\}\,\beta^{\leq 6} + \mathbf{1}\{t_{fs,c} \geq t_c,\, M_{fs,c} \geq 7\}\,\beta^{\geq 7} + \delta_{fs,\text{tier,region}} + \phi_c + \epsilon_{fs,c,t}, \tag{2}$$

where t_c corresponds to the *event* in college c, $t_{fs,c}$ to the expected graduation year of freshman cohort fs in college c, and $M_{fs,c}$ to the number of months freshman cohort fs in college c had access to Facebook before 30 June of its expected graduation year. The remaining components of equation (2) are the same as in the baseline model. The pre-treatment coefficients in the event-study specification allow me to investigate nonlinear pre-trends in the career outcome variables between treatment and control cohorts. To ensure that they are not biased by the strong assumption of treatment effect homogeneity, I follow the recommended approach by Borusyak *et al.* (2024) and estimate them in a dynamic TWFE specification in the spirit of equation (2) on the not-yet-treated observations *only*. To allow for separate identification of the dynamic treatment effect coefficients β_k and the cohort fixed effects in a setup that uses only not-yet-treated cohorts as comparison groups, I follow the recommendation by Schmidheiny and Siegloch (2023) and bin the lower endpoint of the event-time window ($k \leq -5$ with $k = -4$). Event time $k = -1$ is omitted as the reference category.

Crucially, the post-treatment coefficients in equation (2) are not obtained from the same dynamic TWFE specification. They are imputation estimates as described in Section 4, which draw on all not-yet-treated observations. The test of the parallel-trends assumption and the estimation of treatment effects are therefore two separate exercises. Ideally, I would estimate separate post-treatment coefficients for event time $k = 0$ and $k = 1$ to learn more about whether the career benefits of online social network access are markedly different for students with longer exposure during college. However, $k = 1$ is identified only for the first 36 colleges, which received access in the academic year 2004 (see Appendix Figure A1), resulting in very imprecise estimates. Therefore, I estimate the post-treatment coefficients by distinguishing college cohorts with access to Facebook in the last six months before expected graduation ($\beta^{\leq 6}$) from cohorts with at least

seven months of access ($\beta^{\geq 7}$), which splits the post-treatment coefficient into 496 and 281 treated college cohorts, respectively.

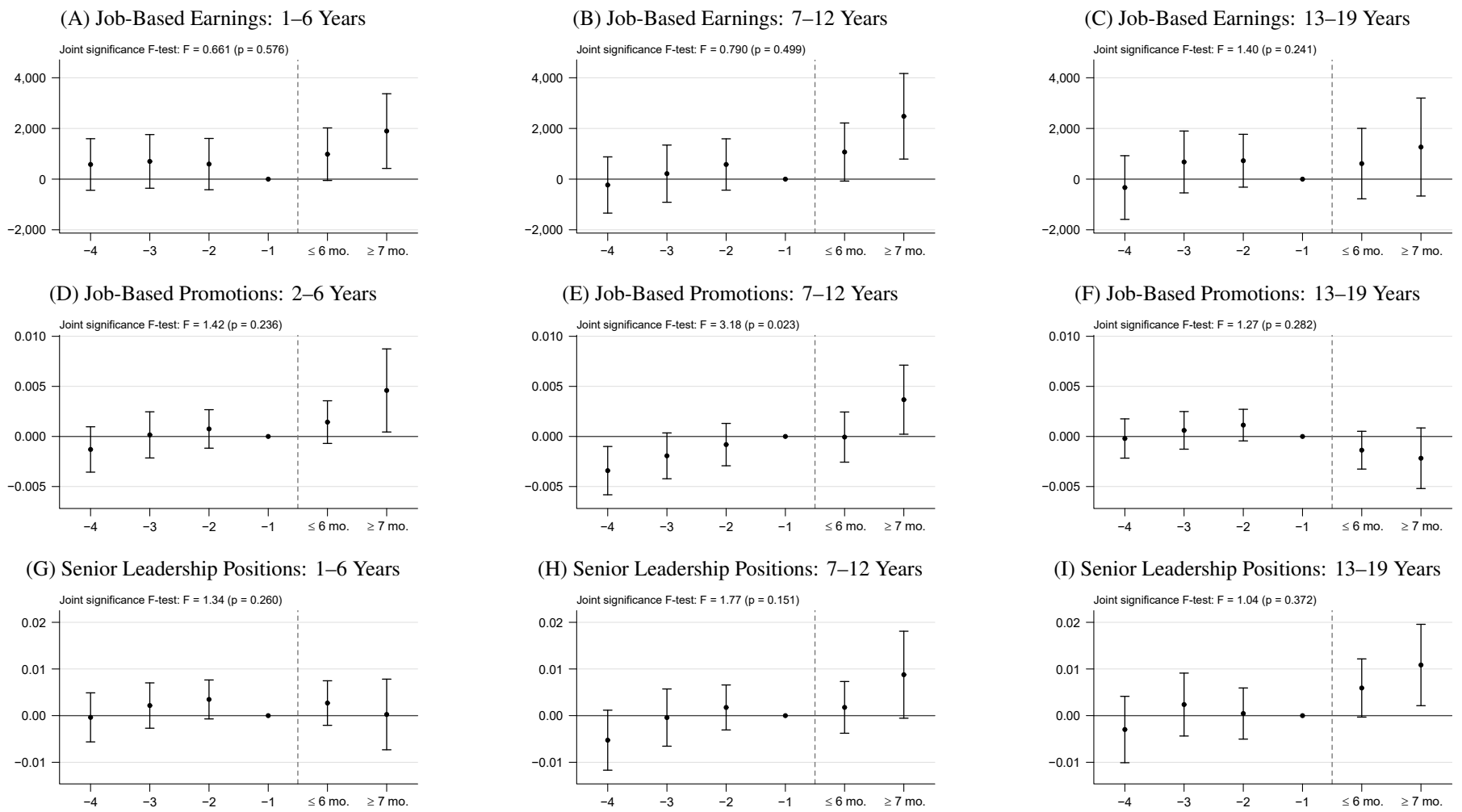
Appendix Figures B1 to B7 show the results. Across all outcome variables, almost all pre-treatment coefficients in the more than 50 event studies are individually and jointly insignificant at the 5% level, supporting the parallel-trends assumption and mitigating concerns that the documented treatment effects simply reflect pre-existing differential trends in career outcomes between treated and control cohorts.[47] The coefficients at $k = -2$ are close to zero in almost all event studies and indicate neither a dip nor a positive jump in the career outcomes of the last cohort expected to graduate before Facebook's introduction ($k = -1$), suggesting that there are no spillover (anticipation) effects in this setting.[48]

Looking at the post-treatment coefficients suggests that the career benefits are generally stronger for cohorts with longer exposure to Facebook during college. These patterns are generally consistent with Armona (2025), who provides evidence that longer exposure during college yields larger earnings effects. Nevertheless, the estimated post-treatment effects should be interpreted only as suggestive evidence, as the composition of colleges and the number of cohorts already exposed to the technology differ between the two post-treatment groups.

[47] A future version of this paper will integrate honest DiD estimates following the recommended practice by Rambachan and Roth (2023).

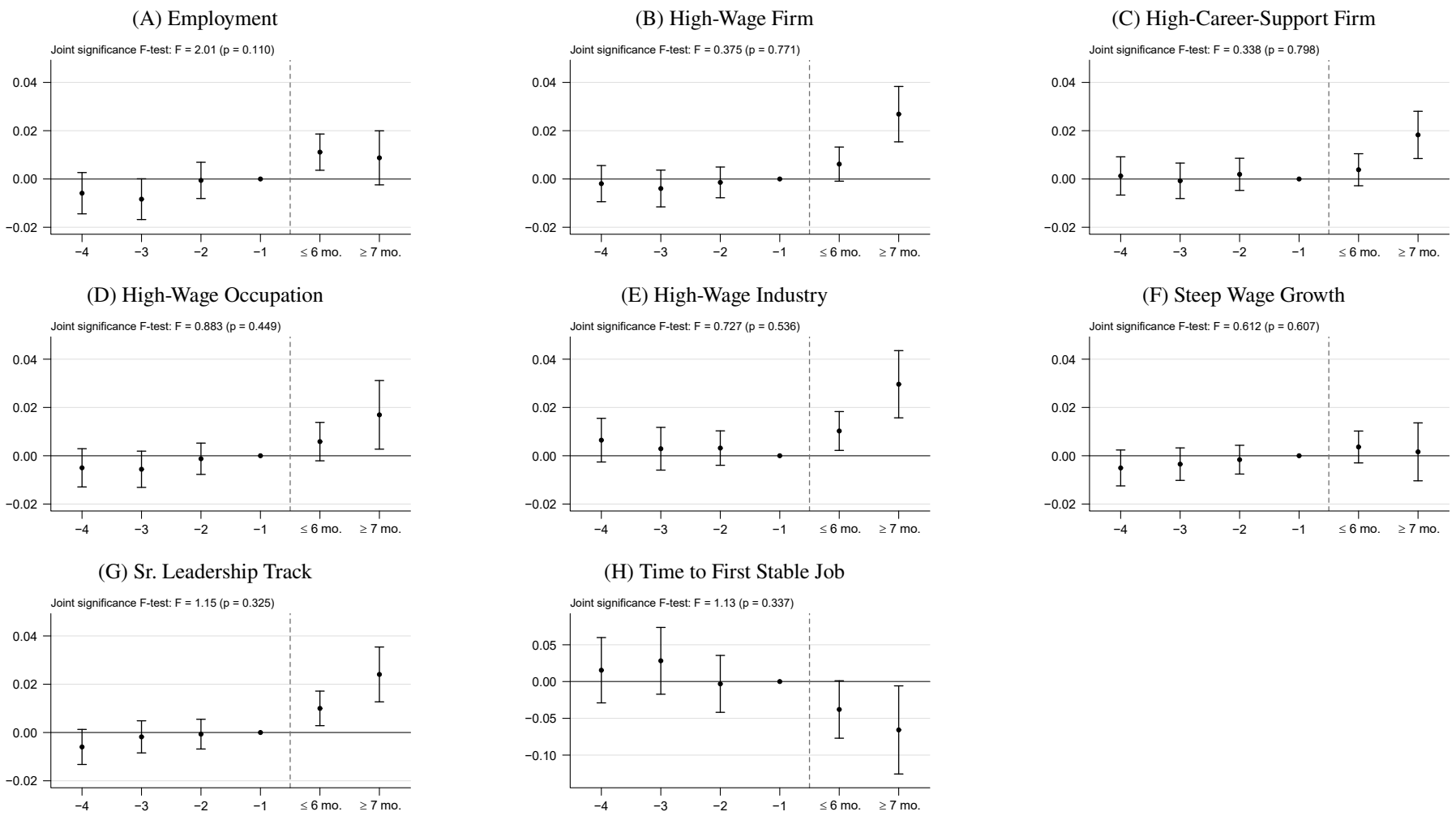
[48] Dropping the $k = -1$ cohort from the estimation produces results similar to the baseline estimates (not shown here).

Figure B1: Facebook Access and Career Development – Event Studies



Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six (≤ 6 mo.) and with at least seven (≥ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

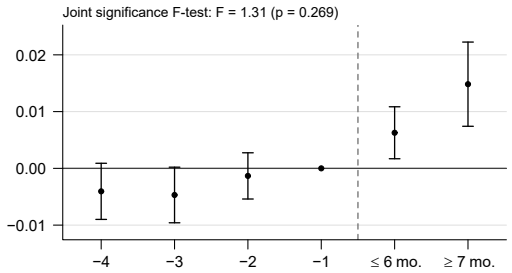
Figure B2: Facebook Access and the College-to-Work Transition – Event Studies



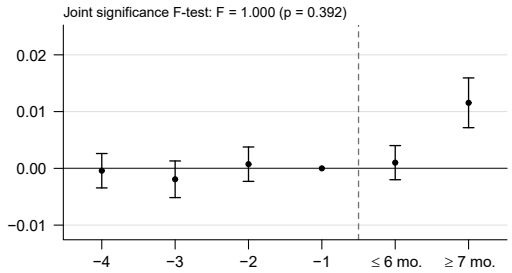
Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six ($\leq$ 6 mo.) and with at least seven ($\geq$ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure B3: Facebook Access and Early-Career Job Help in the College Network – Event Studies

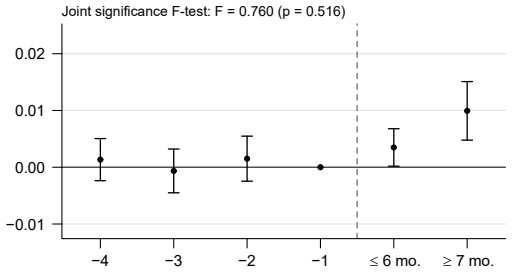
(A) Job via College Network: Any Job



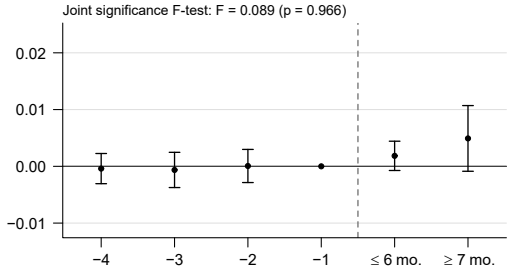
(B) Job via College Network: High-Wage Firm



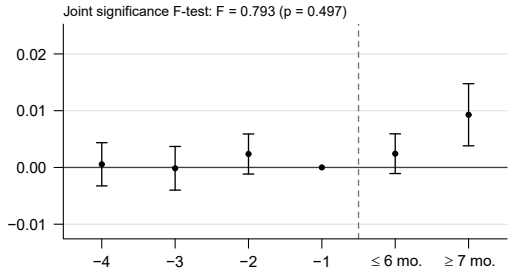
(C) Job via College Network: High-Career-Support Firm



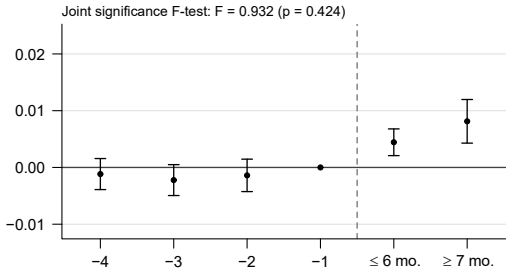
(D) Job via College Network: High-Wage Occupation



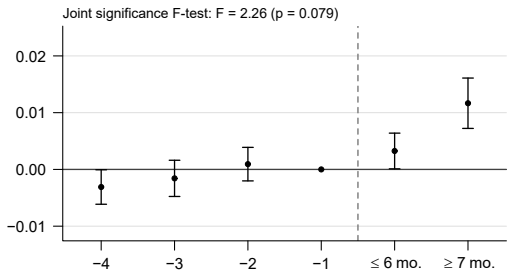
(E) Job via College Network: High-Wage Industry



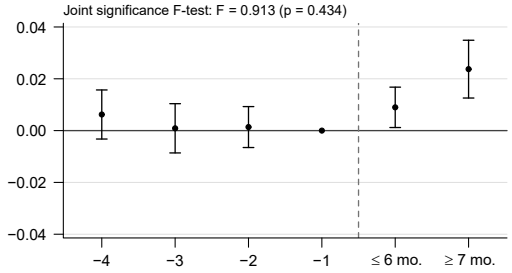
(F) Job via College Network: Steep Wage Growth



(G) Job via College Network: Sr. Leadership Track



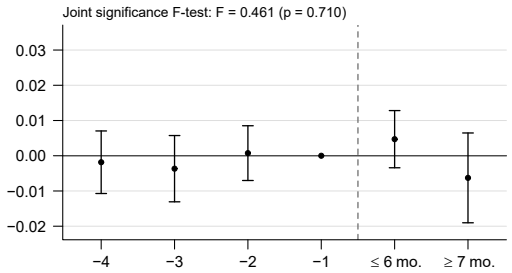
(H) Job via College Network: First Stable Job



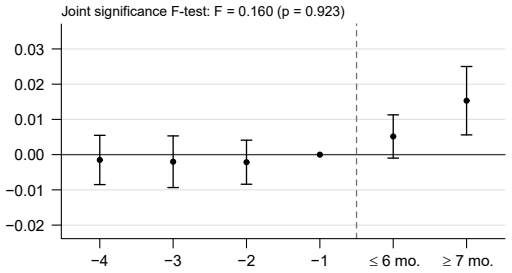
Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six (≤ 6 mo.) and with at least seven (≥ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure B4: Facebook Access and Early-Career Jobs without College Network Help – Event Studies

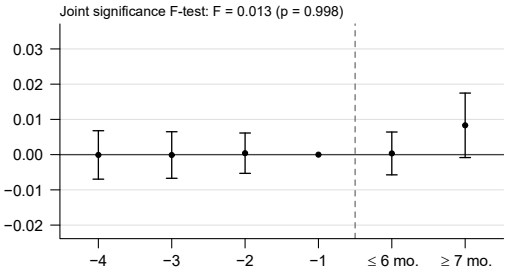
(A) Job without College Network: Any Job



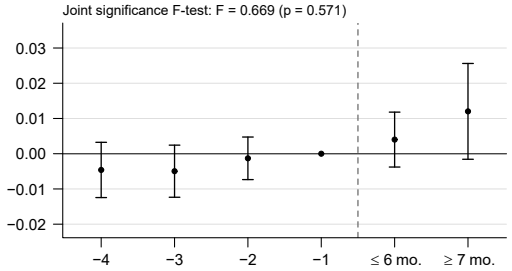
(B) Job without College Network: High-Wage Firm



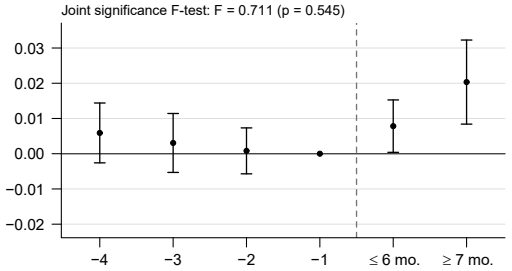
(C) Job without College Network: High-Career-Support Firm



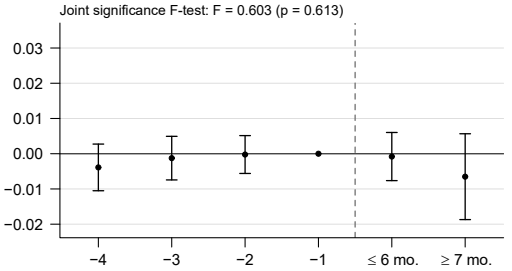
(D) Job without College Network: High-Wage Occupation



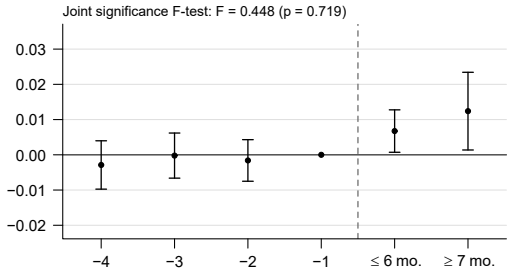
(E) Job without College Network: High-Wage Industry



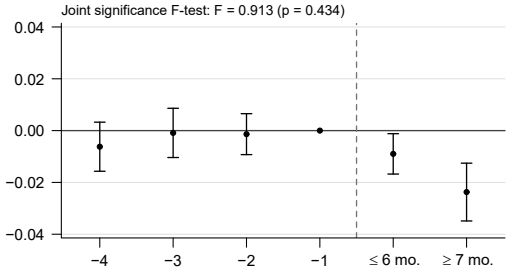
(F) Job without College Network: Steep Wage Growth



(G) Job without College Network: Sr. Leadership Track



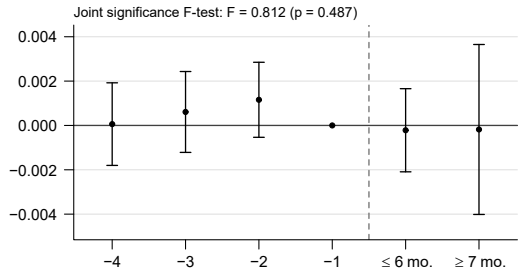
(H) Job without College Network: First Stable Job



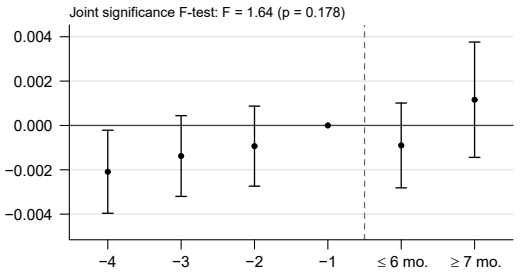
Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six (≤ 6 mo.) and with at least seven (≥ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure B5: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Event Studies

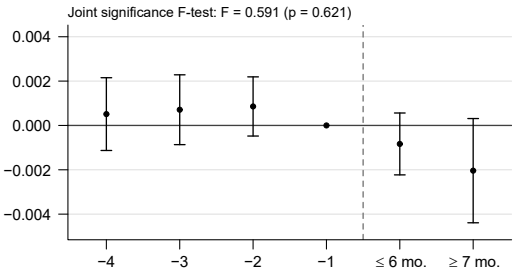
(A) Between-Firm Promotions: 2–6 Years



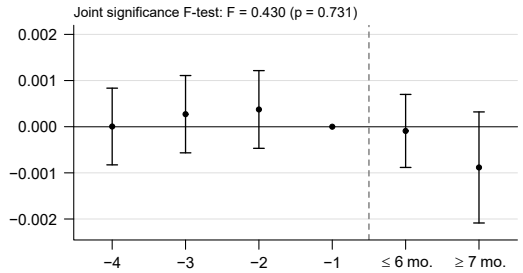
(B) Between-Firm Promotions: 7–12 Years



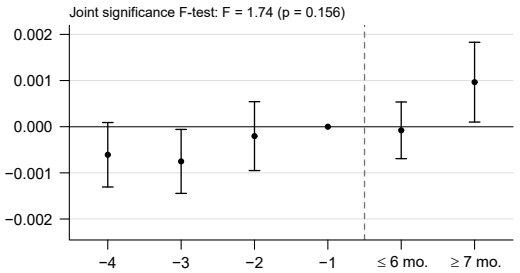
(C) Between-Firm Promotions: 13–19 Years



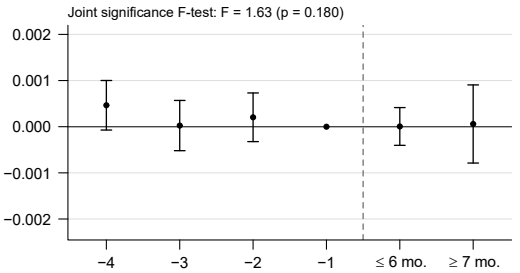
(D) Promotion via College Network: 2–6 Years



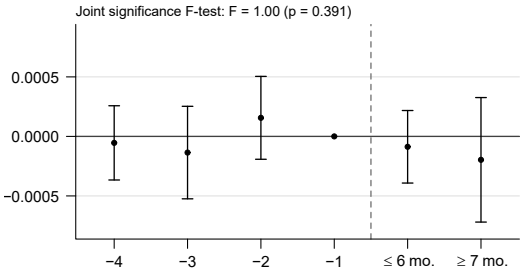
(E) Promotion via College Network: 7–12 Years



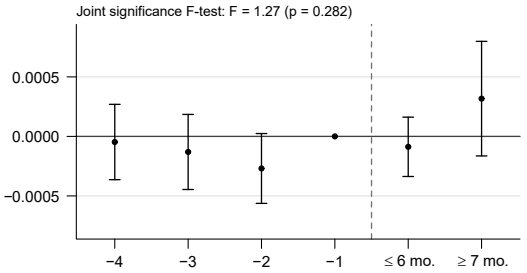
(F) Promotion via College Network: 13–19 Years



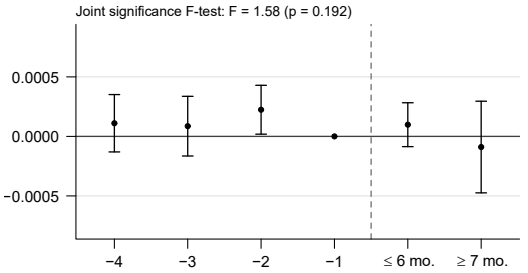
(G) Sr. Leadership Transition via College Network: 2–6 Years



(H) Sr. Leadership Transition via College Network: 7–12 Years

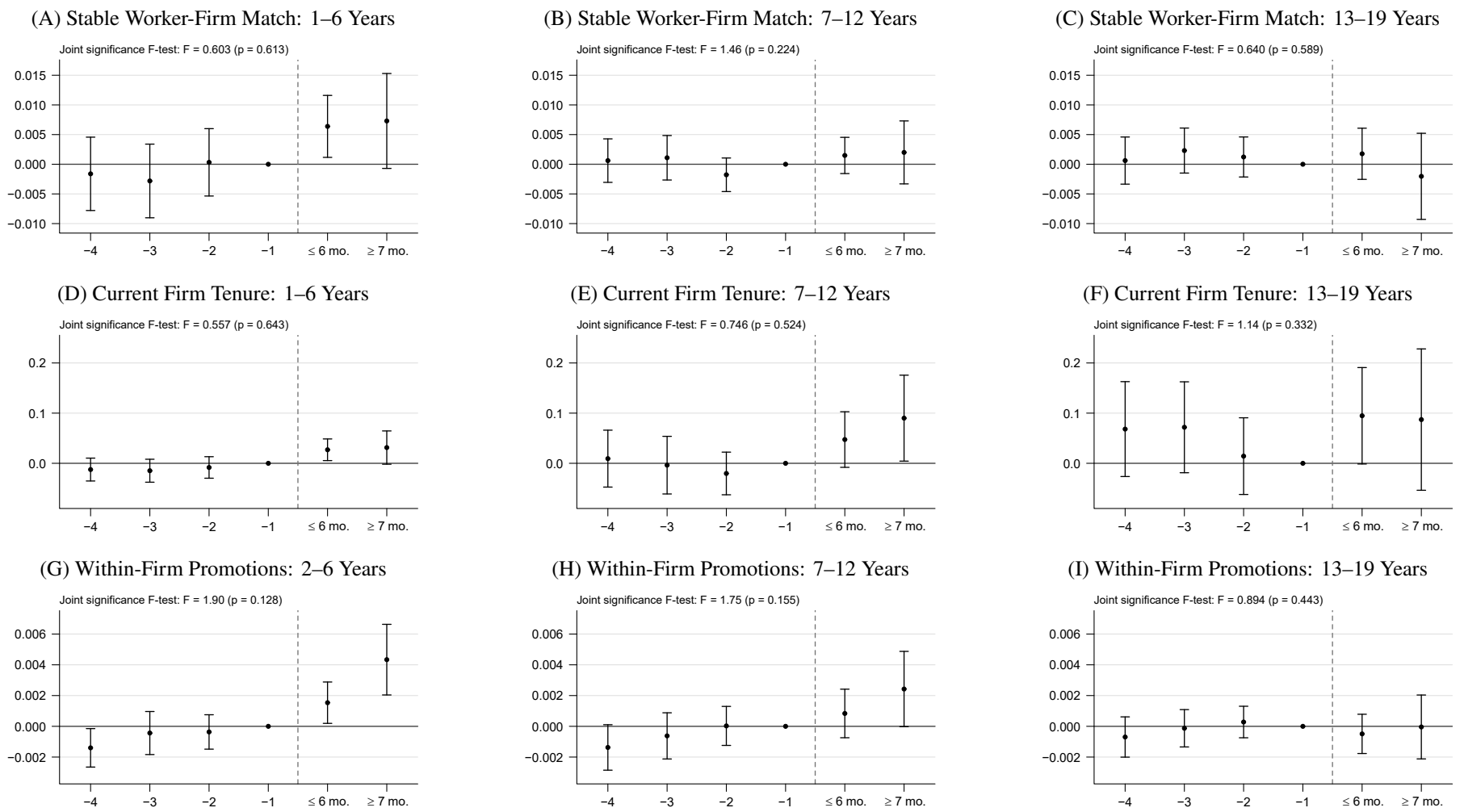


(I) Sr. Leadership Transition via College Network: 13–19 Years



Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six ($\leq$ 6 mo.) and with at least seven ($\geq$ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

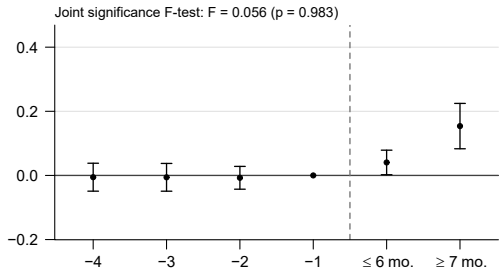
Figure B6: Facebook Access and Worker-Firm Match Stability and Productivity – Event Studies



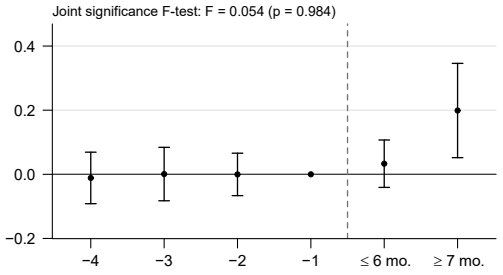
Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six (≤ 6 mo.) and with at least seven (≥ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure B7: Facebook Access, Firm Sorting, and Human Capital Accumulation – Event Studies

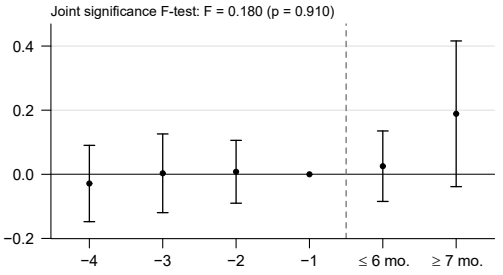
(A) Work Experience at High-Wage Firms: Year 6



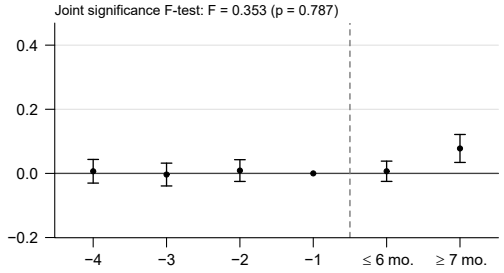
(B) Work Experience at High-Wage Firms: Year 12



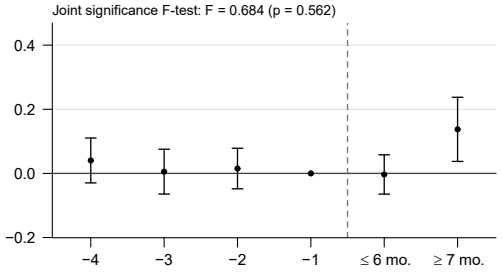
(C) Work Experience at High-Wage Firms: Year 19



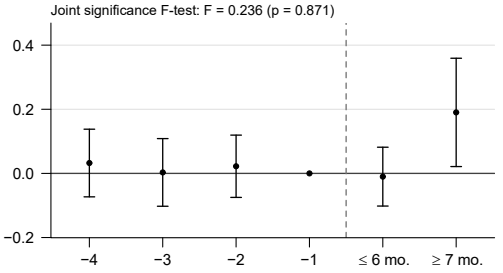
(D) Work Experience at High-Career-Support Firms: Year 6



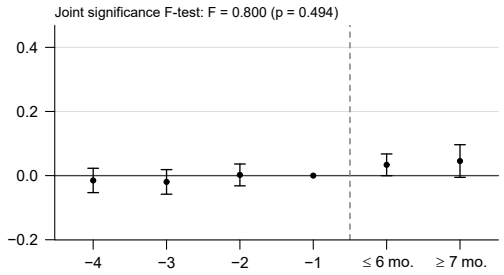
(E) Work Experience at High-Career-Support Firms: Year 12



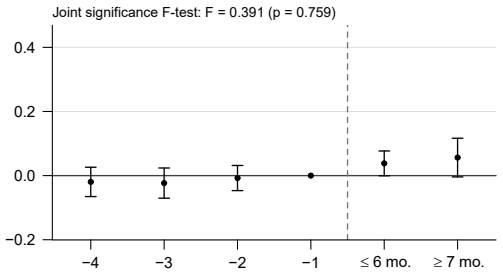
(F) Work Experience at High-Career-Support Firms: Year 19



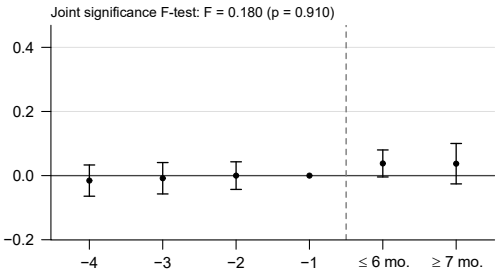
(G) Cumulative Work Experience: Year 6



(H) Cumulative Work Experience: Year 12



(I) Cumulative Work Experience: Year 19



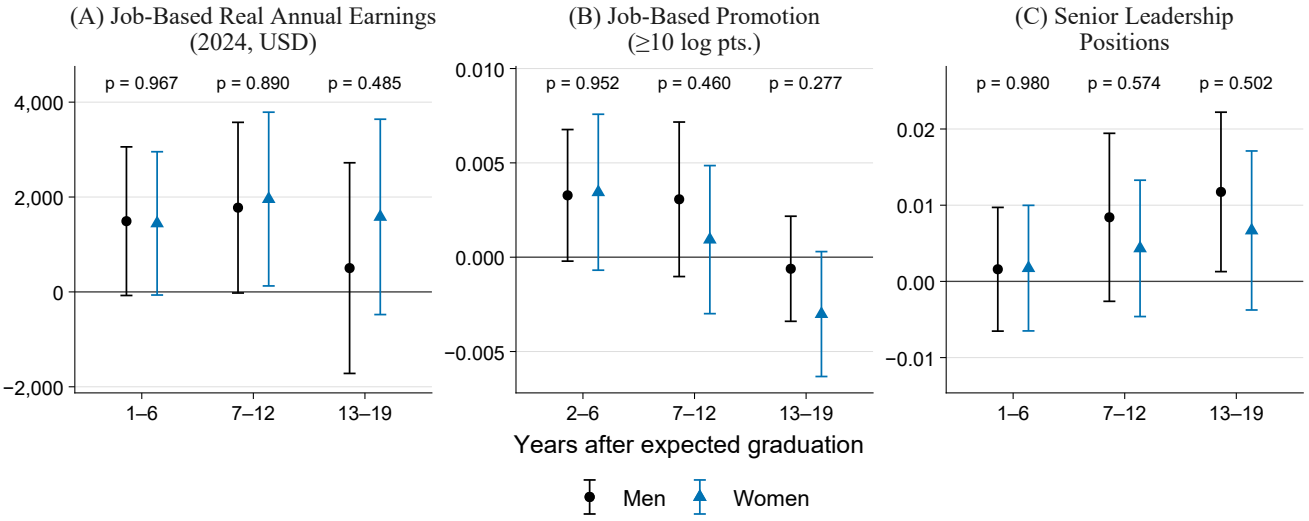
Notes: Event-study estimates with 95% confidence intervals. Pre-treatment coefficients: dynamic TWFE specification estimated on the not-yet-treated observations, relative to event time $k = -1$. The lower endpoint is binned: $k = -4$ includes $k = -5$. The joint F-test of the pre-treatment coefficients is reported above each panel. Post-treatment coefficients: imputation estimates (Borusyak *et al.* 2024) for cohorts with at most six (≤ 6 mo.) and with at least seven (≥ 7 mo.) months of Facebook access before expected graduation. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Appendix C: Gender Heterogeneity

How does having access to online social networking platforms at labor market entry affect gender gaps in career outcomes? Despite decades of progress, women still work in lower-paying jobs and are systematically underrepresented at the top of the corporate world (e.g., Goldin 2014; Hindlian *et al.* 2018; Olivetti *et al.* 2024). While several studies highlight a range of explanations, including the role of labor supply (e.g., Bertrand *et al.* 2010), family responsibilities (e.g., Kleven *et al.* 2019), or preferences for risk and competition (Niederle and Vesterlund 2007; Buser *et al.* 2014) and for job amenities (Wiswall and Zafar 2018; Mas and Pallais 2017), another part of the literature emphasizes the role of social networks, homophily, and referrals (e.g., Lalanne and Seabright 2016; Beaman *et al.* 2018; Zeltzer 2020; Mengel 2020; Cullen and Perez-Truglia 2023; von Essen and Smith 2023; Hampole *et al.* 2026). As Facebook reduces the costs of adding and maintaining social connections, this may have facilitated women's access to male-dominated networks, equalizing access to valuable labor market connections, and ultimately reducing gender gaps in career outcomes. Consistent with the first part of this hypothesis, the results by Armona and Bailey (2025) indicate that early Facebook adopters have online social networks with lower gender homophily.

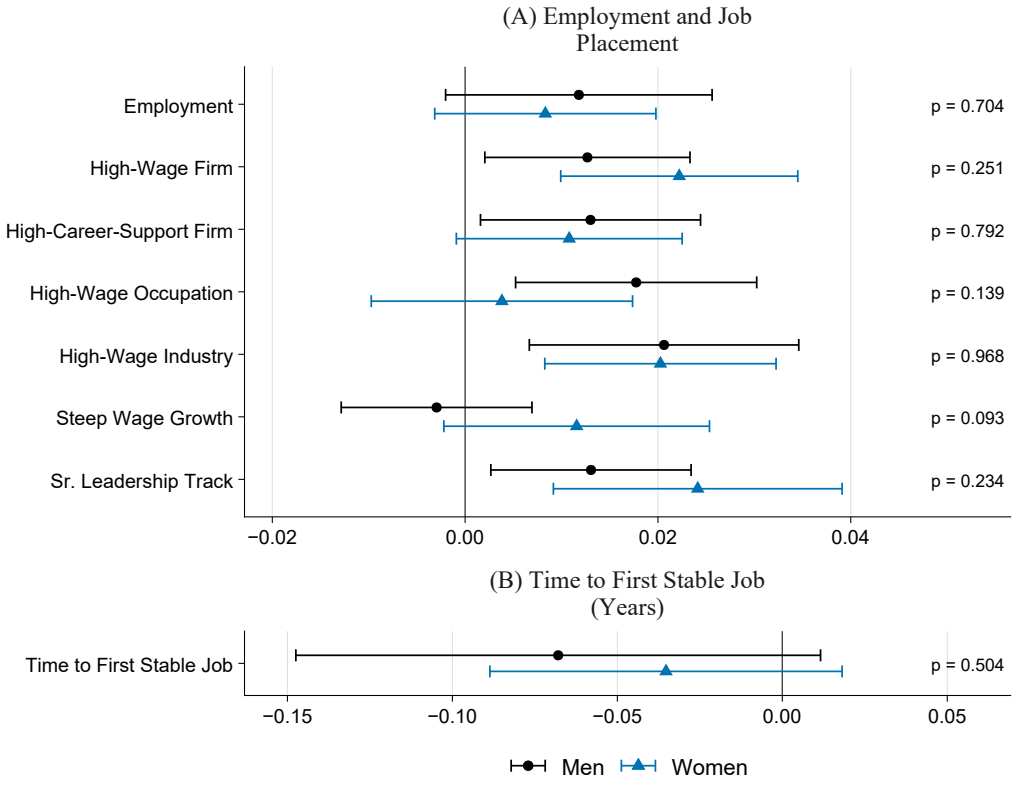
This appendix investigates to what extent this observation translates into differential career effects for men and women. Appendix Figure C1 starts by showing that the documented career benefits are homogeneously shared across gender. While the effects on job-based earnings are slightly larger for women in the medium and long run, consistent with the cross-sectional evidence in Armona (2025), the coefficients are not statistically distinguishable. Similarly, while single point estimates for the underlying mechanisms tend to differ across gender (see Appendix Figure C2 to Figure C6), the differences are, with few exceptions, not statistically significant. Nevertheless, consistent with the finding by Armona and Bailey (2025) of reduced gender homophily, Appendix Figure C3 shows that during the college-to-work transition, students with early access to Facebook, particularly women, were more likely to receive job help from opposite-sex peers. Taken together, the results indicate that early access to Facebook did not meaningfully affect the gender gap in career outcomes, but may have facilitated women's access to otherwise male-dominated labor market networks.

Figure C1: Facebook Access and Career Development – By Gender



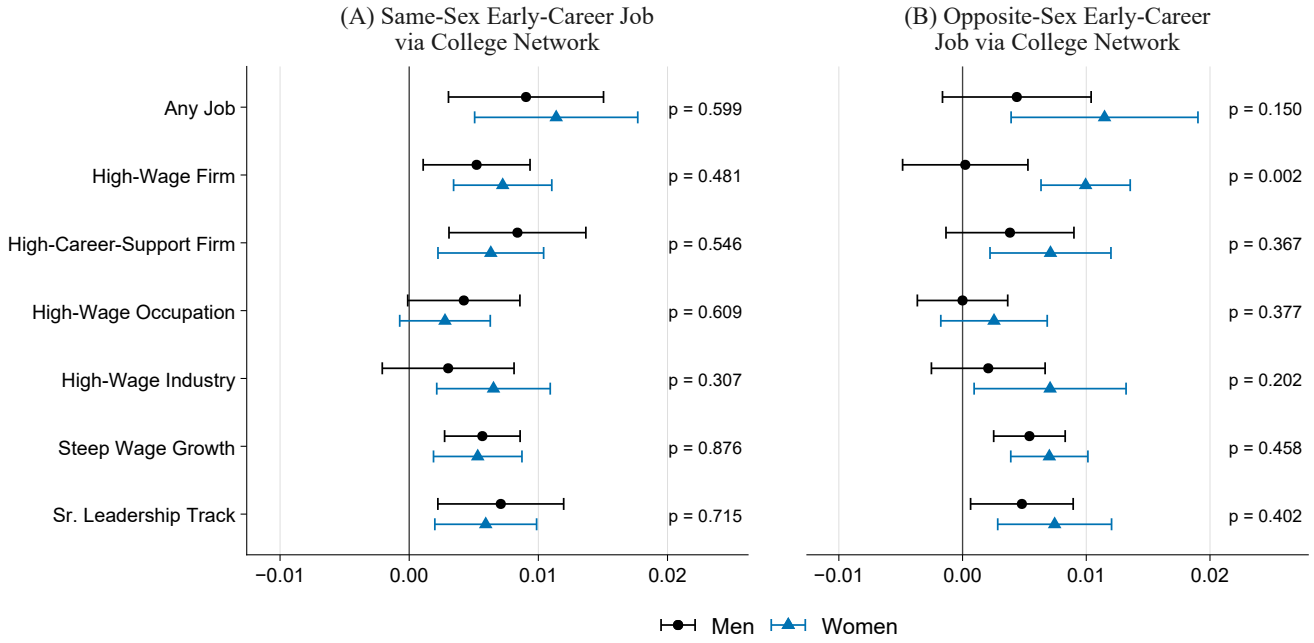
Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 1, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024), estimated separately for male and female graduates, at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. p: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. See the notes of Figure 1 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure C2: Facebook Access and the College-to-Work Transition – By Gender



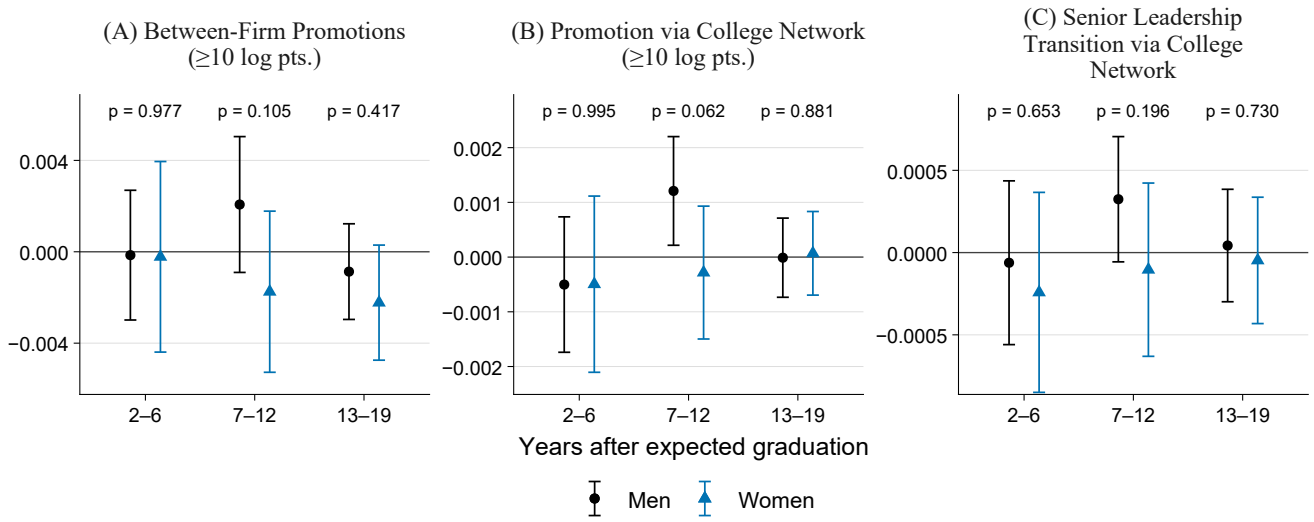
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Figure C3: Facebook Access and Early-Career Job Help in the College Network – By Gender



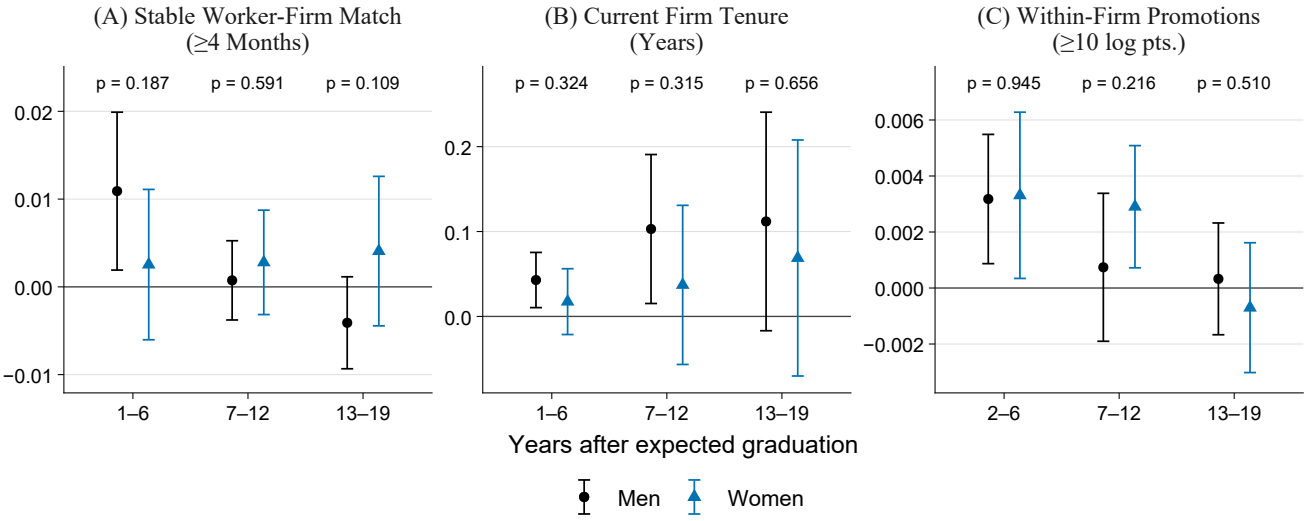
Notes: Point estimates and 95% confidence intervals, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level, estimated separately for male and female graduates. Estimates are pooled over the first two years after expected graduation. Column 8 of Table 3 (first stable job) is not split by the sex of the college peer. College peers: students up to 3 classes above and below the expected graduation cohort. Job via college network: indicator for joining a firm where a college peer had already worked for at least one year (Gee *et al.* 2017). The two panels distinguish college peers of the same sex and of the opposite sex. See the notes of Table 3 for the definitions of the job characteristics. p: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure C4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – By Gender



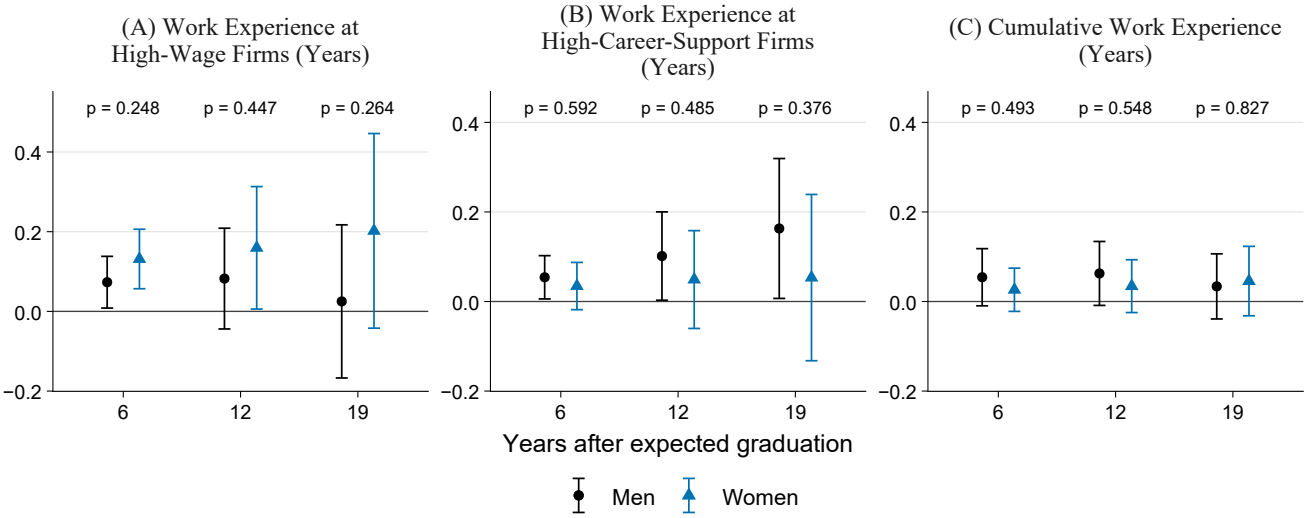
Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 2, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024), estimated separately for male and female graduates, at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. p: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. See the notes of Figure 2 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure C5: Facebook Access and Worker-Firm Match Stability and Productivity – By Gender



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 3, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024), estimated separately for male and female graduates, at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. p: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. See the notes of Figure 3 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure C6: Facebook Access, Firm Sorting, and Human Capital Accumulation – By Gender

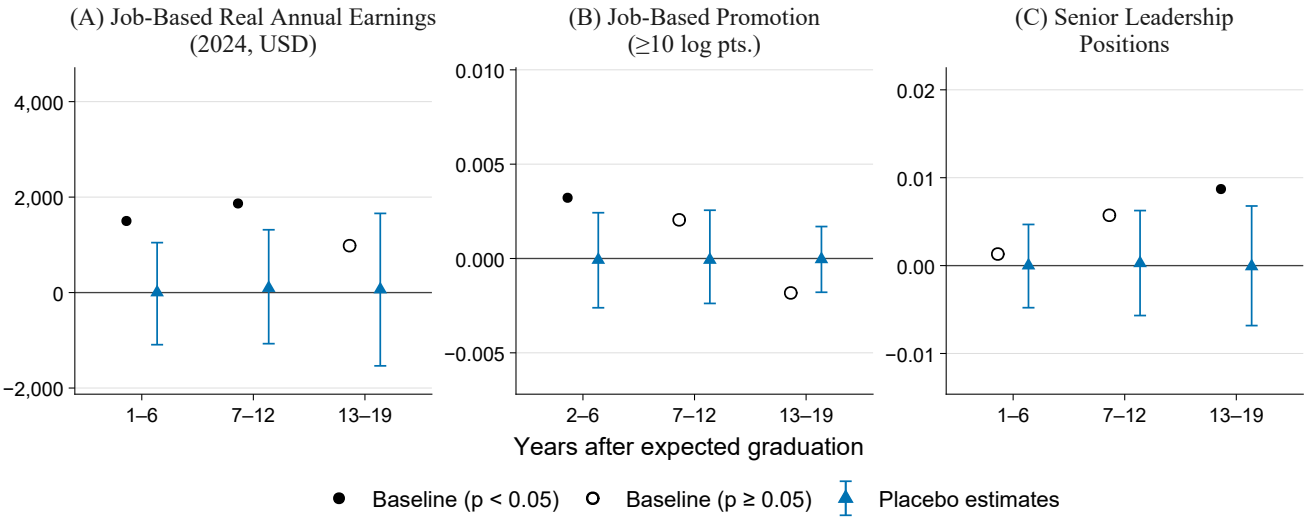


Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 4, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024), estimated separately for male and female graduates, at the college*cohort*year level. Estimates refer to 6, 12, and 19 years after expected graduation. p: p-value for the null hypothesis $H_0 : \beta_{\text{male}} = \beta_{\text{female}}$. See the notes of Figure 4 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Appendix D: Randomization Inference

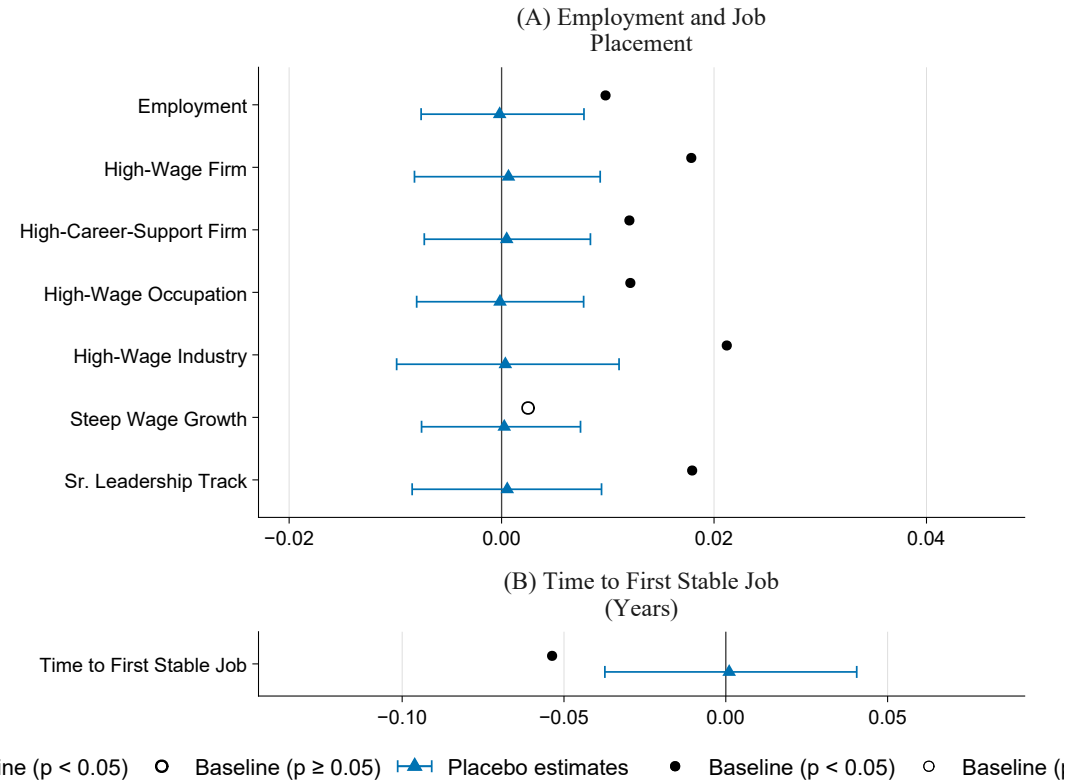
A key concern in the main empirical strategy is that even within the narrow selectivity-by-region comparisons, spurious correlations with the timing of the Facebook rollout might remain, biasing the main estimates. For example, this could be the case if Facebook (deliberately or not) expanded early to colleges *within the same selectivity and U.S. region* where the career outcomes of U.S. business and economics cohorts were on a differential trend. To formally test this, I implement a randomization inference test in which I randomly reshuffle Facebook's introduction year across colleges within each selectivity-by-region cell (N = 1,000). The test preserves the total number of treated colleges per cell and the overall distribution of treatment timing. Appendix Figures D1–D6 show the results. The placebo estimates generally have a dense distribution around zero, and virtually all of the statistically significant effects from the main specification fall above the 95th percentile of the placebo distribution. The remaining variation in the Facebook rollout is therefore unlikely to reflect spurious correlations.

Figure D1: Permutation Test: Career Development



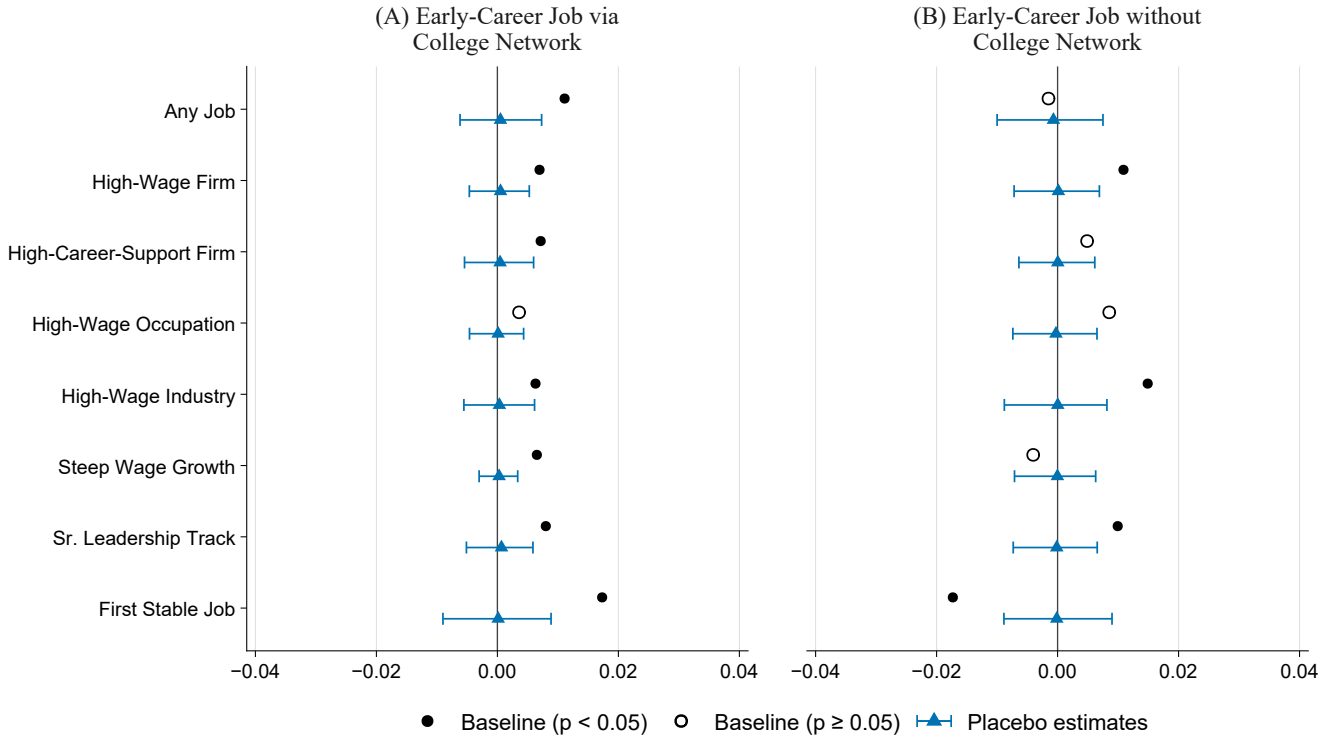
Notes: Baseline estimates of Figure 1 next to the distribution of 1,000 placebo estimates, all using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Placebo estimates: within each stratum (college selectivity*region), the vector of Facebook introduction years is randomly permuted across colleges, which preserves the number of treated colleges and the distribution of introduction timing per stratum. Triangles and bars: mean and range from the 2.5th to the 97.5th percentile of the placebo estimates. A baseline estimate outside this range is statistically different from zero at the 5% level in a two-sided randomization test. Filled circles: baseline estimates that are statistically significant at the 5% level in the baseline table. Hollow circles: baseline estimates that are not statistically significant. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. See the notes of Figure 1 for the definitions of the outcomes. *Data source:* Revelio Labs.

Figure D2: Permutation Test: College-to-Work Transition



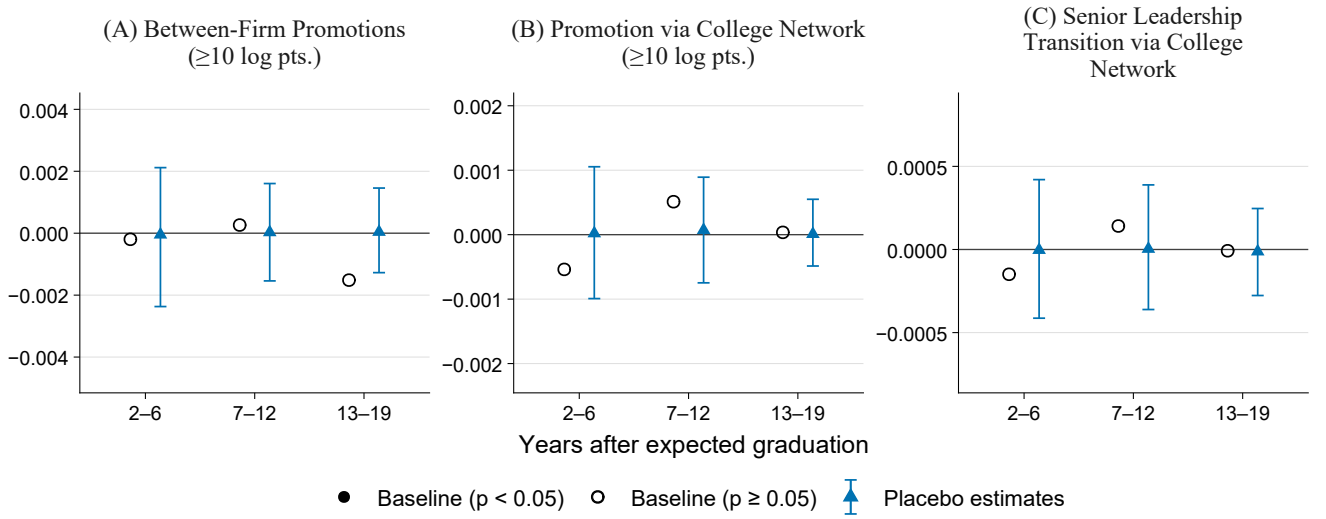
Notes: Baseline estimates of Table 2 next to the distribution of 1,000 placebo estimates, all using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Placebo estimates: within each stratum (college selectivity*region), the vector of Facebook introduction years is randomly permuted across colleges, which preserves the number of treated colleges and the distribution of introduction timing per stratum. Triangles and bars: mean and range from the 2.5th to the 97.5th percentile of the placebo estimates. A baseline estimate outside this range is statistically different from zero at the 5% level in a two-sided randomization test. Filled circles: baseline estimates that are statistically significant at the 5% level in the baseline table. Hollow circles: baseline estimates that are not statistically significant. Estimates are pooled over the first two years after expected graduation. Panel B: time to first stable job (Column 8 of Table 2). See the notes of Table 2 for the definitions of the outcomes. *Data source:* Revelio Labs.

Figure D3: Permutation Test: Early-Career Job Help in the College Network



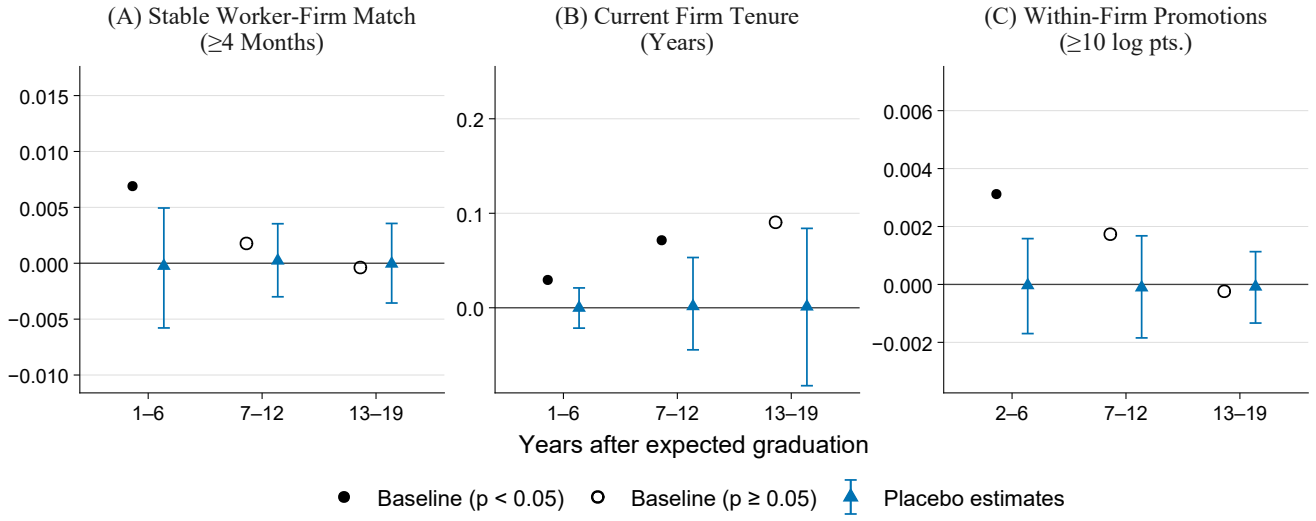
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Figure D4: Permutation Test: Promotion Ties



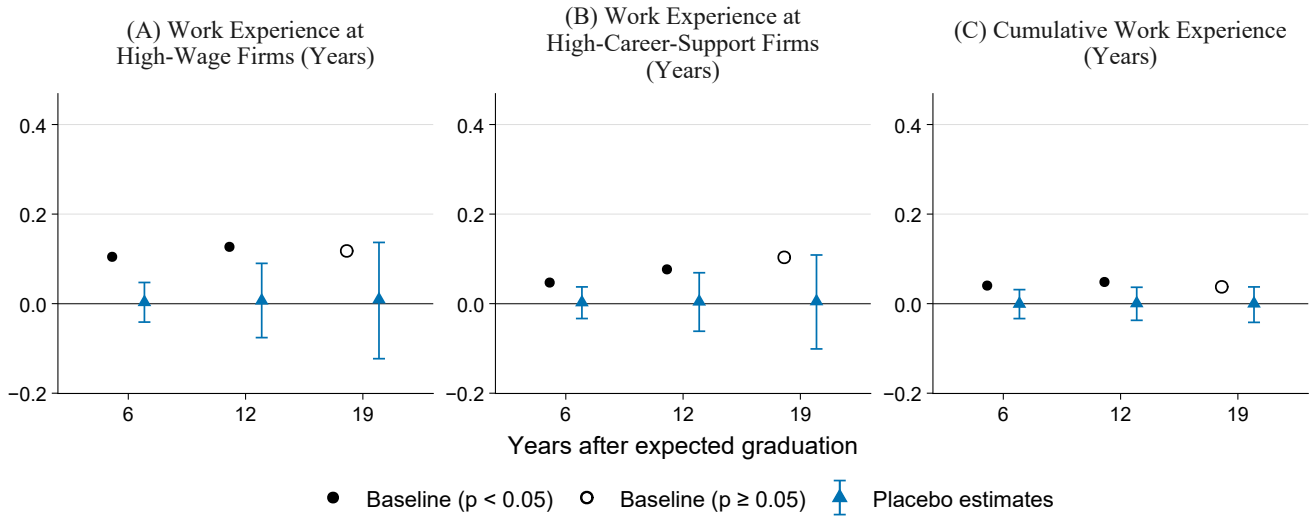
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Figure D5: Permutation Test: Worker-Firm Match Stability and Productivity



Notes: Baseline estimates of Figure 3 next to the distribution of 1,000 placebo estimates, all using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Placebo estimates: within each stratum (college selectivity*region), the vector of Facebook introduction years is randomly permuted across colleges, which preserves the number of treated colleges and the distribution of introduction timing per stratum. Triangles and bars: mean and range from the 2.5th to the 97.5th percentile of the placebo estimates. A baseline estimate outside this range is statistically different from zero at the 5% level in a two-sided randomization test. Filled circles: baseline estimates that are statistically significant at the 5% level in the baseline table. Hollow circles: baseline estimates that are not statistically significant. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. See the notes of Figure 3 for the definitions of the outcomes. *Data source:* Revelio Labs.

Figure D6: Permutation Test: Firm Sorting and Human Capital Accumulation



Notes: Baseline estimates of Figure 4 next to the distribution of 1,000 placebo estimates, all using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Placebo estimates: within each stratum (college selectivity*region), the vector of Facebook introduction years is randomly permuted across colleges, which preserves the number of treated colleges and the distribution of introduction timing per stratum. Triangles and bars: mean and range from the 2.5th to the 97.5th percentile of the placebo estimates. A baseline estimate outside this range is statistically different from zero at the 5% level in a two-sided randomization test. Filled circles: baseline estimates that are statistically significant at the 5% level in the baseline table. Hollow circles: baseline estimates that are not statistically significant. Estimates refer to 6, 12, and 19 years after expected graduation. See the notes of Figure 4 for the definitions of the outcomes. *Data source:* Revelio Labs.

Appendix E: Spillovers

One key concern with the main specification is that the results are driven by spillovers. In particular, treated students might take away jobs from students in the control group, upward biasing the treatment effect and violating the stable-unit treatment value assumption (SUTVA). To assess the seriousness of this issue, I begin by predicting each potential control cohort's exposure to spillovers using the actual career overlap of U.S. business and economics students across colleges *before* the Facebook rollout. The main motivation behind this measure is that graduate placements are highly persistent over time (e.g., Hampole *et al.* 2025) and therefore well-suited to predicting future labor market competition across college cohorts. To account for the changing set of treated colleges over time, I define the predicted spillover exposure measure for each potential control cohort as follows:

$$J_{a,T_t} = \frac{\left| C_a^{2000\text{–}2003} \cap \left(\bigcup_{b \in T_t} C_b^{2000\text{–}2003} \right) \right|}{|C_a^{2000\text{–}2003}|}$$

where J_{a,T_t} measures the share of graduates from college a working in the same job as graduates from the set of colleges T that already had access to Facebook in the academic year t. To ensure that the prediction is not contaminated by potential treatment effects, I use the cohorts from 2000–2003 *before* the Facebook rollout for computation. To closely reflect career overlap and labor market competition between college graduates, I define the same job as working in the same firm*2-digit SOC*year cell.[49]

Table E1 investigates the predicted spillover exposure measure for the potential control group cohorts in detail. In 2004, the average predicted fraction of control-group cohorts competing for the same jobs as graduates of the treated colleges was around 16%, consistent with the small number of highly selective institutions that had early access to Facebook. By 2005, this share increased to around 33% as Facebook expanded to a broader set of colleges. Table E1 regresses the exposure score on college characteristics to identify which control cohorts are most susceptible to spillovers. In 2004, institutions in the highest selectivity tiers and those in the Northeast had substantially higher predicted exposure scores, consistent with the idea that these colleges are the most likely to compete for the same jobs as the elite colleges Facebook targeted first. By 2005, as more colleges across the U.S. gained access to Facebook, differences in selectivity and region narrowed considerably relative to the average exposure, with colleges in the West and those with the lowest selectivity being slightly less exposed. Table E2 shows the twenty college cohorts

[49] Note that the ideal labor market competition measure would use information on job applications instead of realized outcomes. While this information is not available, the derived measure from LinkedIn profile data can be seen as a meaningful but lower-bound proxy for it.

with the highest predicted exposure scores. All twenty come from the 2005 graduation cohort, consistent with spillovers being more severe at the later stage of Facebook's expansion. The most-exposed college cohorts come predominantly from HBCUs (e.g., Johnson C. Smith University, Central State University, Lane College) and Catholic colleges (e.g., St. Catherine University, Mount St. Joseph University, College of Saint Benedict).

To probe the robustness of the main results to potential spillovers, I then drop every college that has at least one potential control cohort with a predicted exposure score above the population-weighted upper quartile of the 2005 exposure distribution ($Q^{\text{pop}}_{0.75}\left(J_{a,2005}\right) = 0.370$).[50] This removes 110 colleges, or 23% of the potential control student body graduating in 2005. Figures E1 to E6 present the results. Reassuringly, the results are very similar to the baseline estimates. Combined with the baseline findings, the results suggest that the treatment effects do not simply reflect a reallocation of jobs within the economy. They are rather consistent with the idea that access to Facebook reduces information frictions about job opportunities.

Table E1: Predicted Spillover Exposure Score and Observable College Characteristics

	Predicted Spillover Exposure Score (J_{a,T_t})	
	$J_{a,2004}$	$J_{a,2005}$
U.S. Region (ref.: West)		
Midwest	0.009*	0.048***
	(0.004)	(0.011)
Northeast	0.028***	0.042***
	(0.004)	(0.012)
South	0.005	0.046***
	(0.004)	(0.011)
Selectivity (ref.: Less competitive)		
Competitive	0.009*	0.001
	(0.004)	(0.007)
Very competitive	0.029***	0.014
	(0.004)	(0.011)
Highly competitive	0.055***	0.029
	(0.005)	(0.018)
Elite	0.115***	0.159*
	(0.007)	(0.066)
Dep. Var. Mean (pop.-wtd.)	0.158	0.333
Upper Quartile (pop.-wtd.)	0.177	0.370
Observations	1,129	424
R^2	0.330	0.073

Notes: OLS regressions at the college-cohort level. Sample: all potential control college-cohorts during the Facebook expansion period by academic year. Dependent variable: predicted spillover exposure score in each academic year. *p<0.05; **p<0.01; ***p<0.001. *Data source:* Revelio Labs.

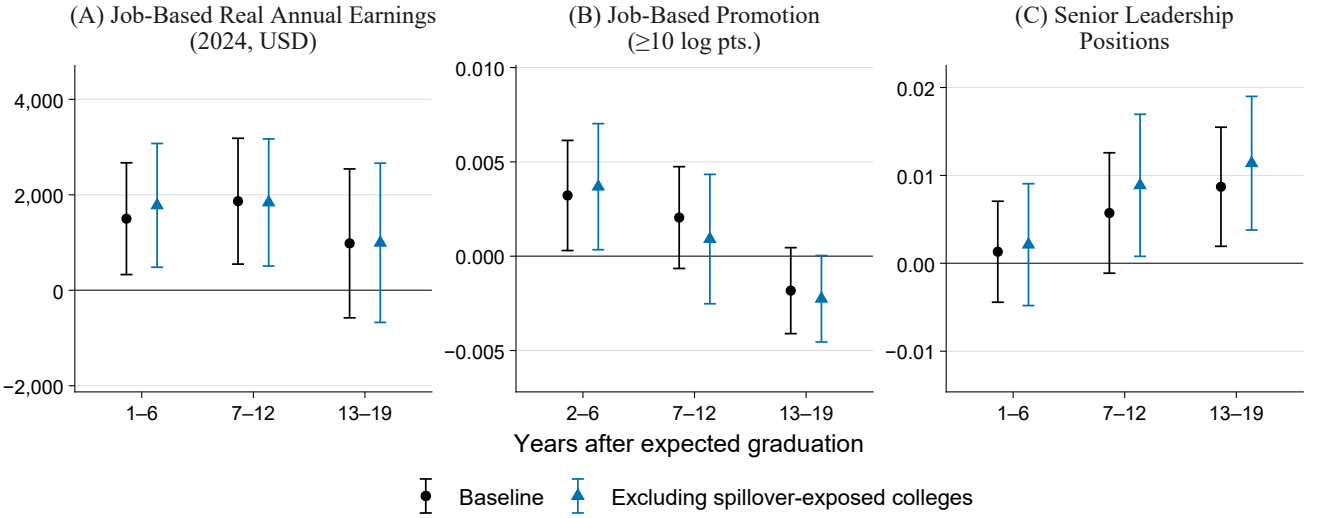
[50] The results are robust to using different thresholds.

Table E2: Top Twenty Control College-Cohorts with Highest Predicted Spillover Exposure

College	Cohort	Exposure score
johnson c smith university	2005	0.520
williams baptist college	2005	0.510
st catherine university	2005	0.492
barnard college	2005	0.485
central state university	2005	0.483
eureka college	2005	0.476
mount saint joseph university	2005	0.469
lane college	2005	0.468
missouri university of science and technology	2005	0.467
monmouth college	2005	0.460
westminster college	2005	0.458
university of arkansas at pine bluff	2005	0.456
college of saint benedict	2005	0.451
university of michigan-dearborn	2005	0.449
gwynedd mercy university	2005	0.447
university of minnesota-morris	2005	0.445
langston university	2005	0.444
saint johns university	2005	0.438
grand view university	2005	0.438
massachusetts maritime academy	2005	0.438

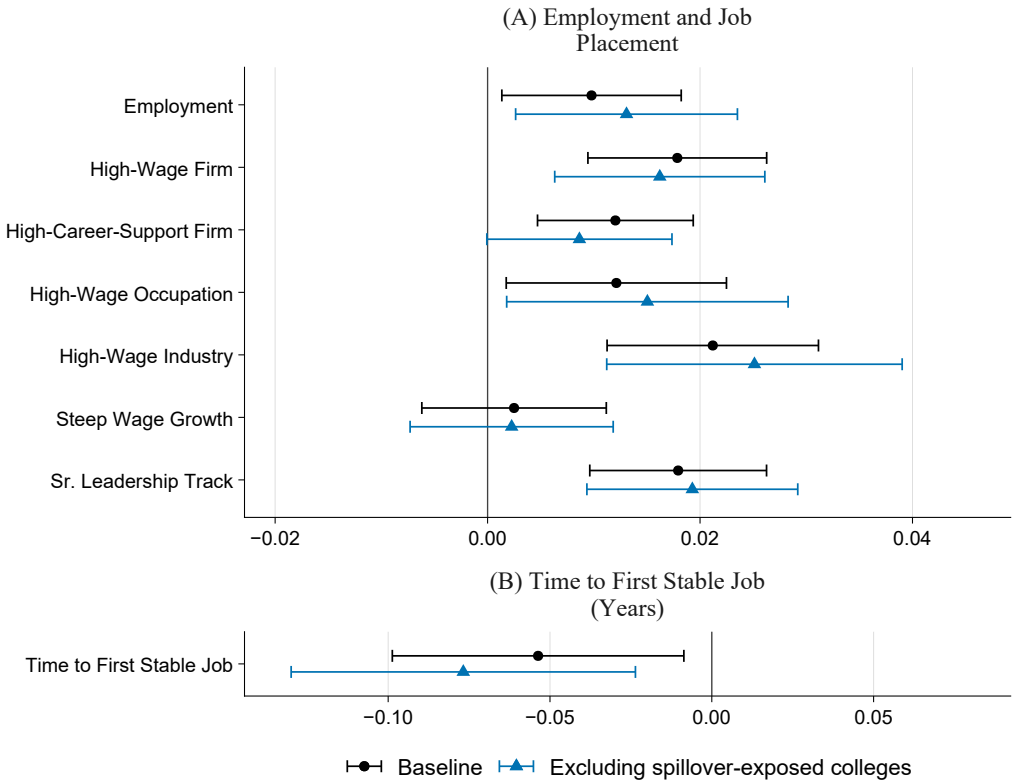
Notes: Potential control college-cohorts with the twenty highest predicted spillover exposure scores. *Data source:* Revelio Labs.

Figure E1: Facebook Access and Career Development – Excluding Spillover-Exposed Colleges



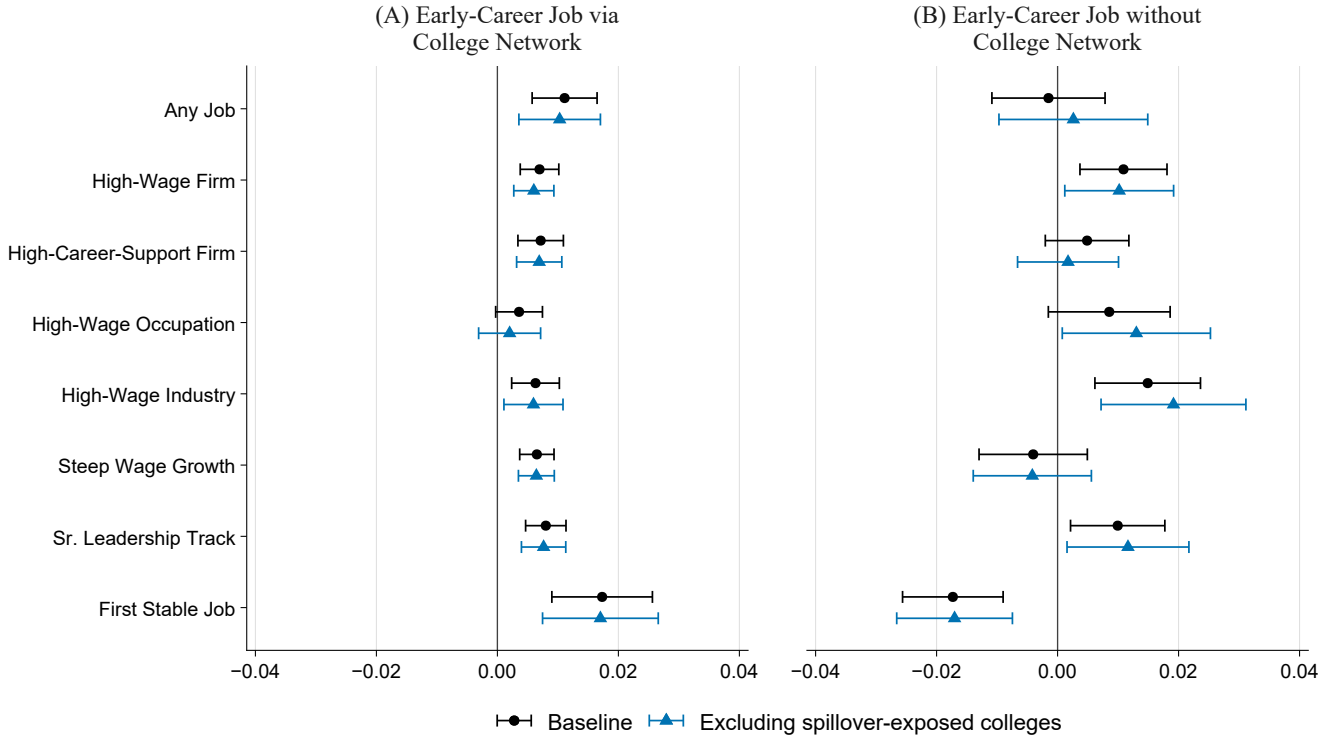
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Figure E2: Facebook Access and the College-to-Work Transition – Excluding Spillover-Exposed Colleges



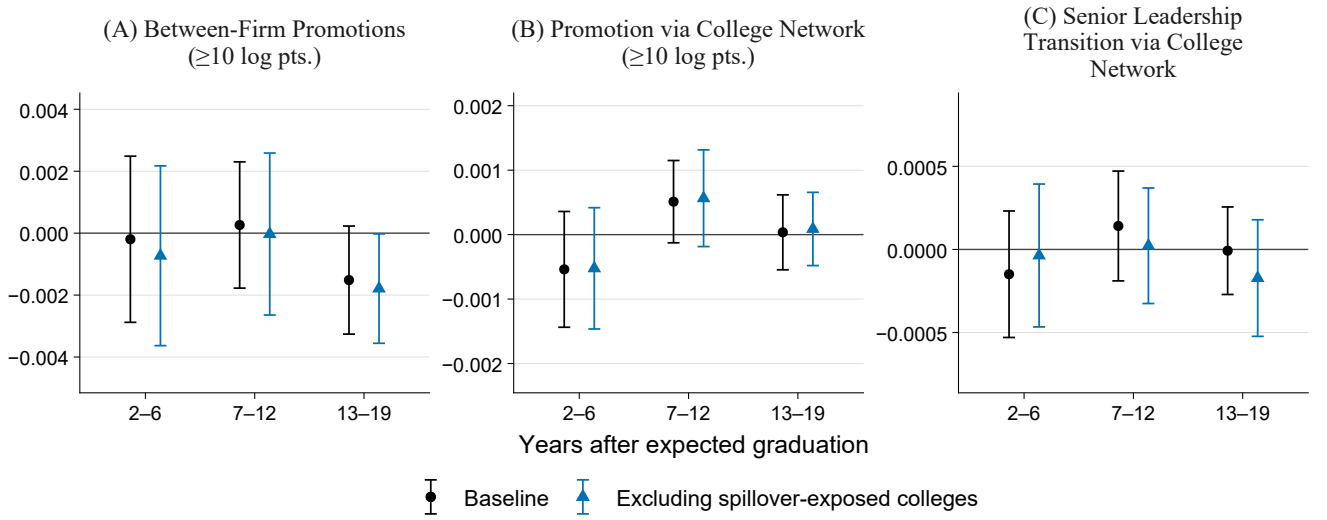
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Figure E3: Facebook Access and Early-Career Job Help in the College Network – Excluding Spillover-Exposed Colleges



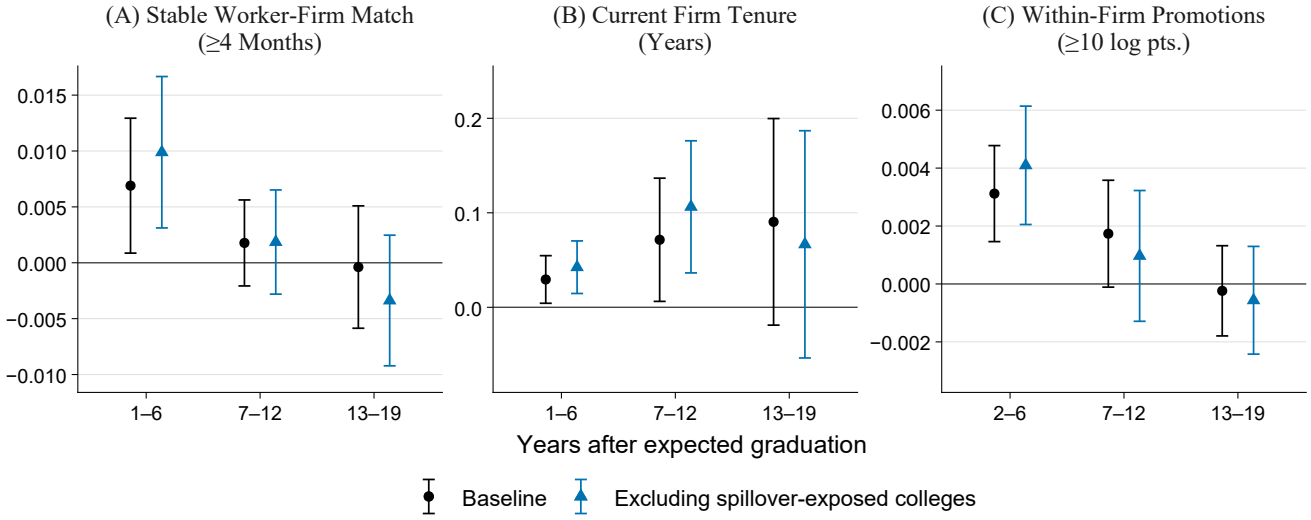
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Figure E4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Excluding Spillover-Exposed Colleges



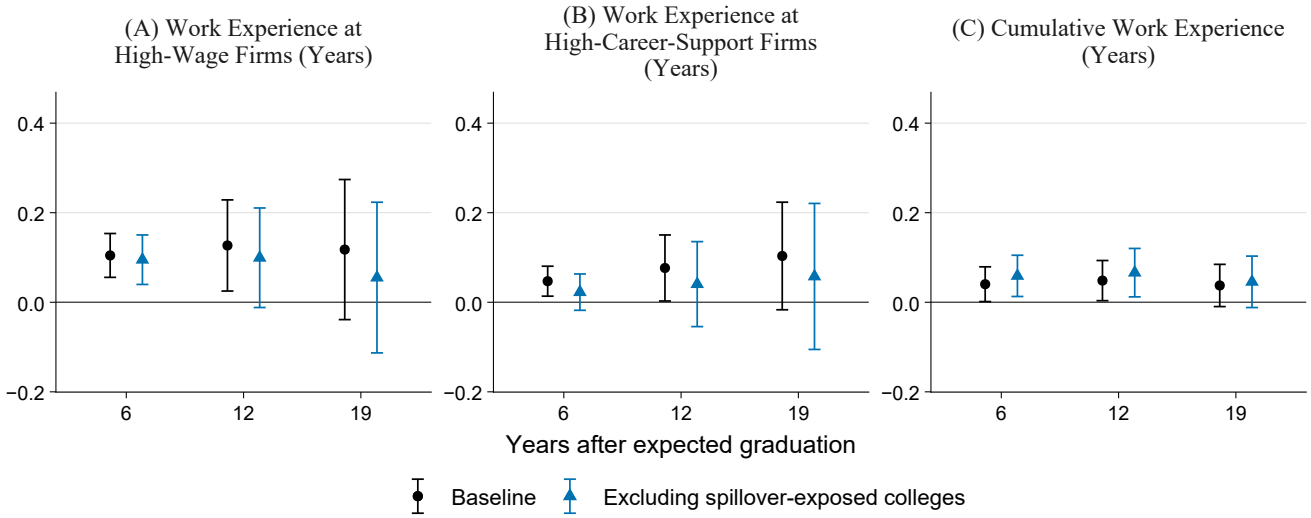
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Figure E5: Facebook Access and Worker-Firm Match Stability and Productivity – Excluding Spillover-Exposed Colleges



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 3, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. Baseline: estimates shown in Figure 3. Excluding spillover-exposed colleges: colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.370) are excluded entirely. See the notes of Figure 3 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

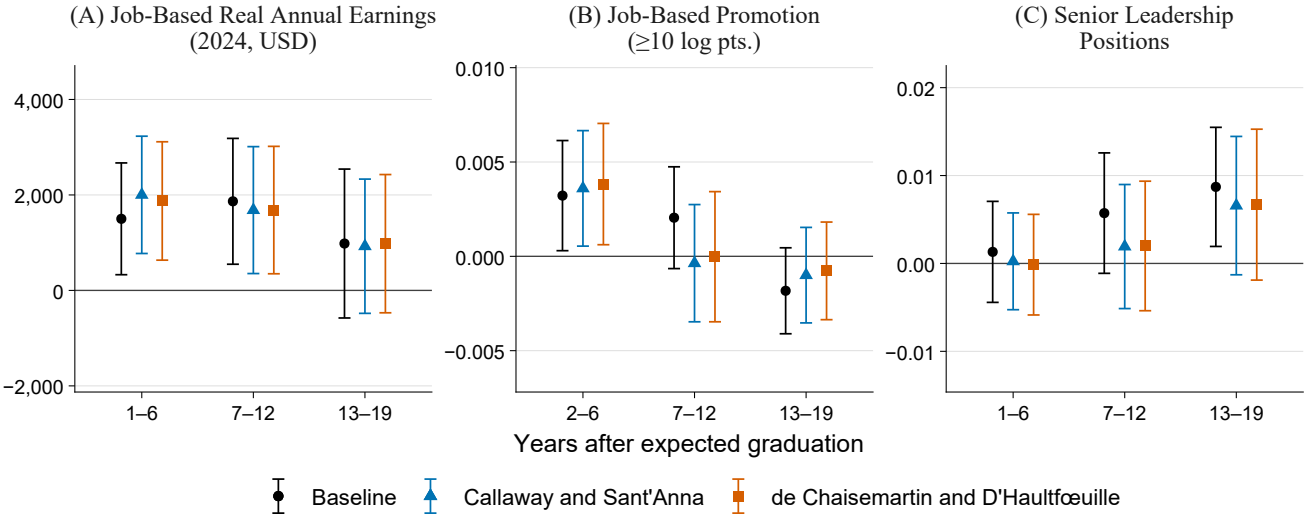
Figure E6: Facebook Access, Firm Sorting, and Human Capital Accumulation – Excluding Spillover-Exposed Colleges



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 4, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates refer to 6, 12, and 19 years after expected graduation. Baseline: estimates shown in Figure 4. Excluding spillover-exposed colleges: colleges with any control cohort whose predicted spillover exposure score exceeds the population-weighted upper quartile of the 2005 distribution (0.370) are excluded entirely. See the notes of Figure 4 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

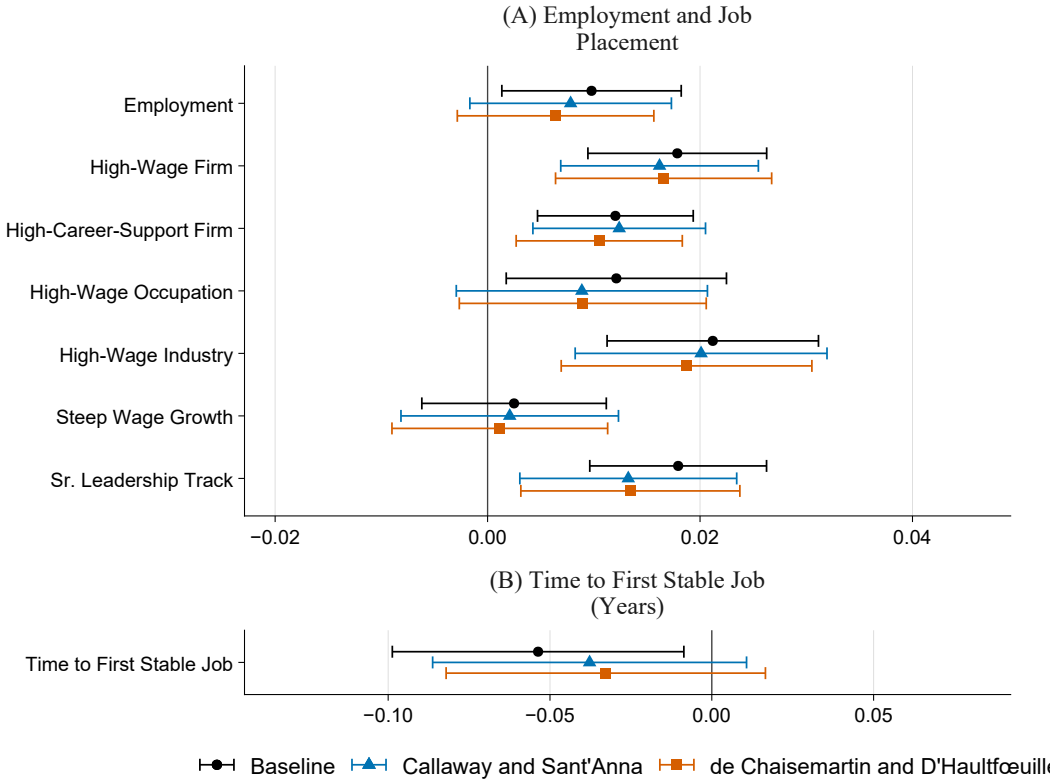
Appendix F: Alternative Heterogeneity-Robust DiD Estimators

Figure F1: Facebook Access and Career Development – Alternative Heterogeneity-Robust Estimators



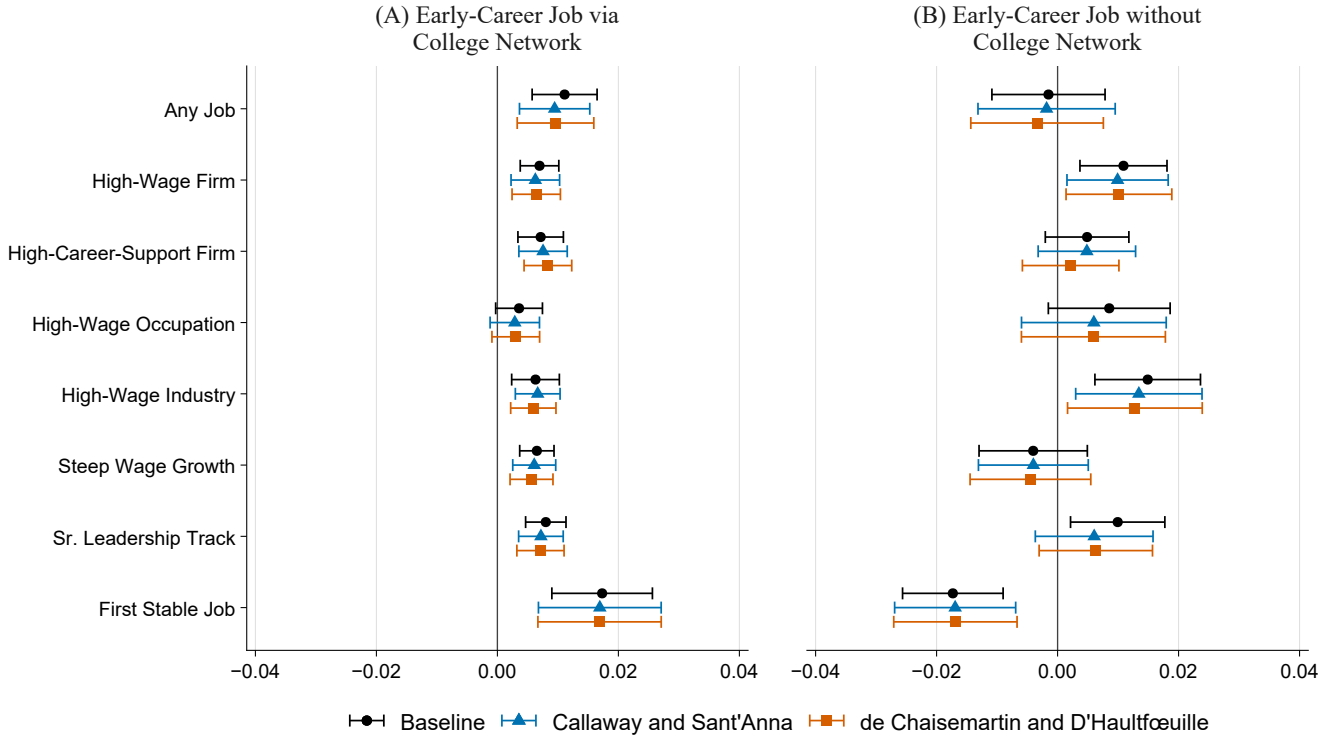
Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 1, each from a separate DiD regression estimated at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Figure 1. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Figure 1 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure F2: Facebook Access and the College-to-Work Transition – Alternative Heterogeneity-Robust Estimators



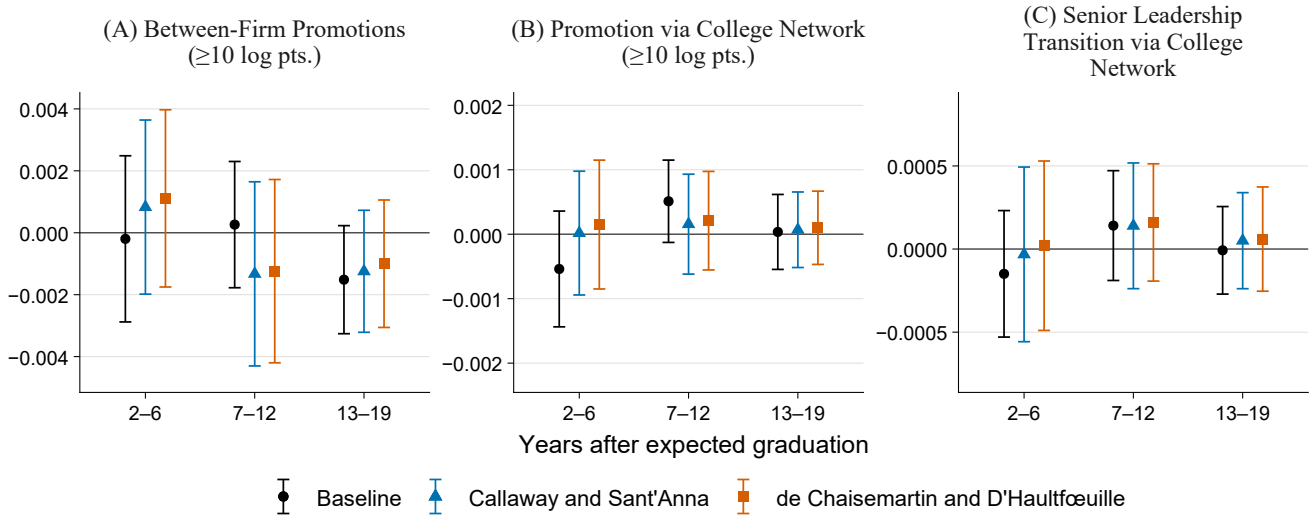
Notes: Point estimates and 95% confidence intervals for the outcomes of Table 2, each from a separate DiD regression estimated at the college*cohort*year level. Estimates are pooled over the first two years after expected graduation. Panel B: time to first stable job (Column 8 of Table 2). Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Table 2. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Table 2 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure F3: Facebook Access and Early-Career Job Help in the College Network – Alternative Heterogeneity-Robust Estimators



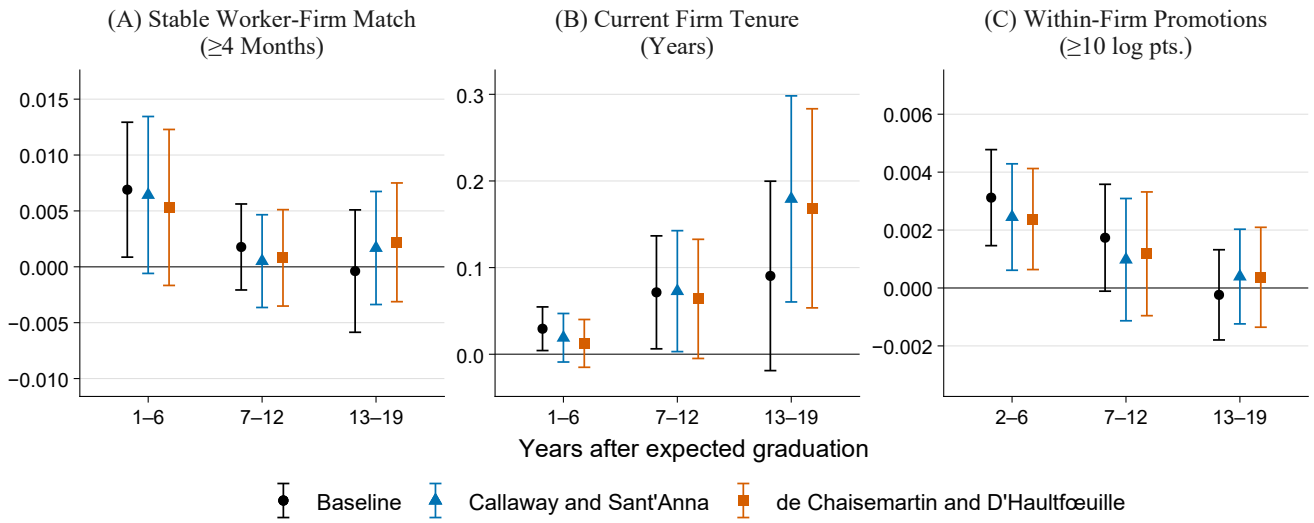
Notes: Point estimates and 95% confidence intervals for the outcomes of Table 3, each from a separate DiD regression estimated at the college*cohort*year level. Estimates are pooled over the first two years after expected graduation. Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Table 3. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Table 3 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure F4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – Alternative Heterogeneity-Robust Estimators



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 2, each from a separate DiD regression estimated at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 2–6, 7–12, and 13–19 years. Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Figure 2. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Figure 2 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure F5: Facebook Access and Worker-Firm Match Stability and Productivity – Alternative Heterogeneity-Robust Estimators



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 3, each from a separate DiD regression estimated at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Figure 3. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Figure 3 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

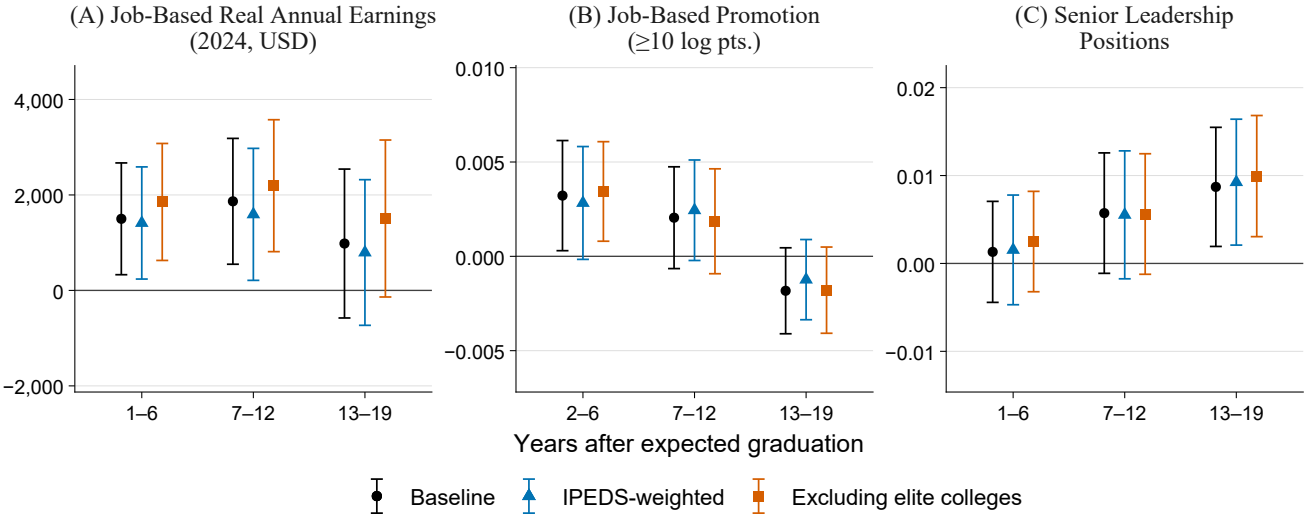
Figure F6: Facebook Access, Firm Sorting, and Human Capital Accumulation – Alternative Heterogeneity-Robust Estimators



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 4, each from a separate DiD regression estimated at the college*cohort*year level. Estimates refer to 6, 12, and 19 years after expected graduation. Baseline: imputation estimator of Borusyak *et al.* (2024), as reported in Figure 4. Callaway and Sant'Anna: estimator of Callaway and Sant'Anna (2021). de Chaisemartin and D'Haultfœuille: estimator of de Chaisemartin and D'Haultfœuille (2020). Both alternative estimators compare cohorts within the region-by-selectivity strata of the baseline specification without accounting for the differential linear cohort trends within the highly competitive and elite cluster. See the notes of Figure 4 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

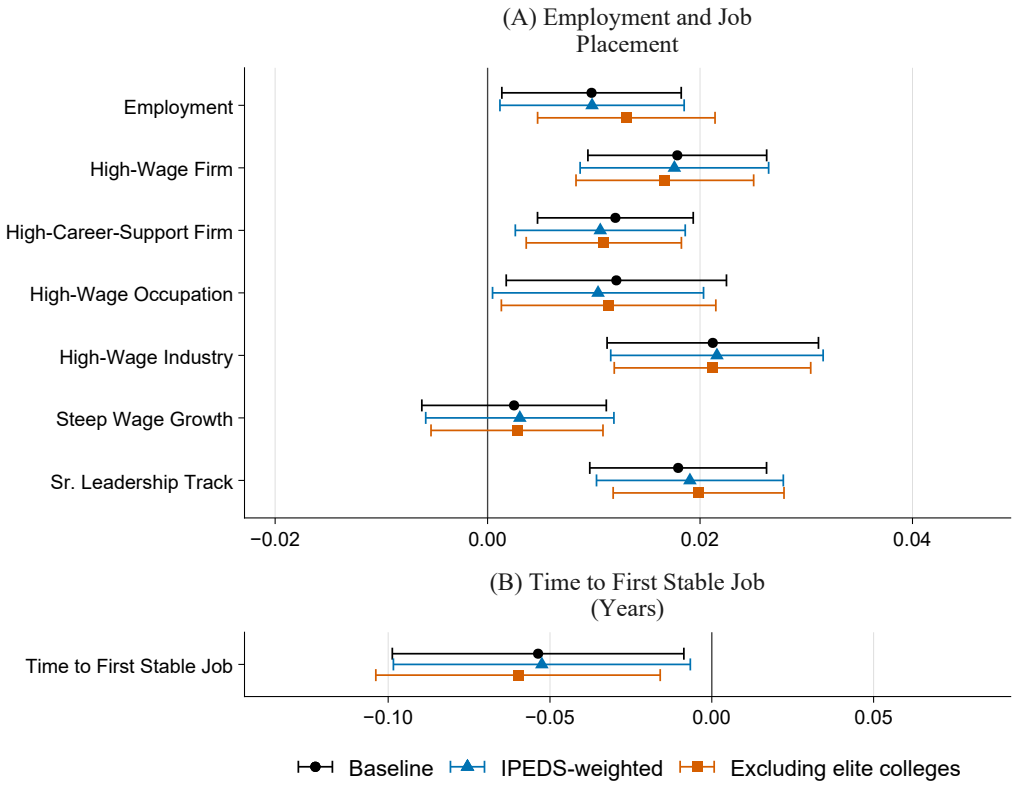
Appendix G: IPEDS Weights and Excluding Elite Colleges

Figure G1: Facebook Access and Career Development – IPEDS Weights and Excluding Elite Colleges



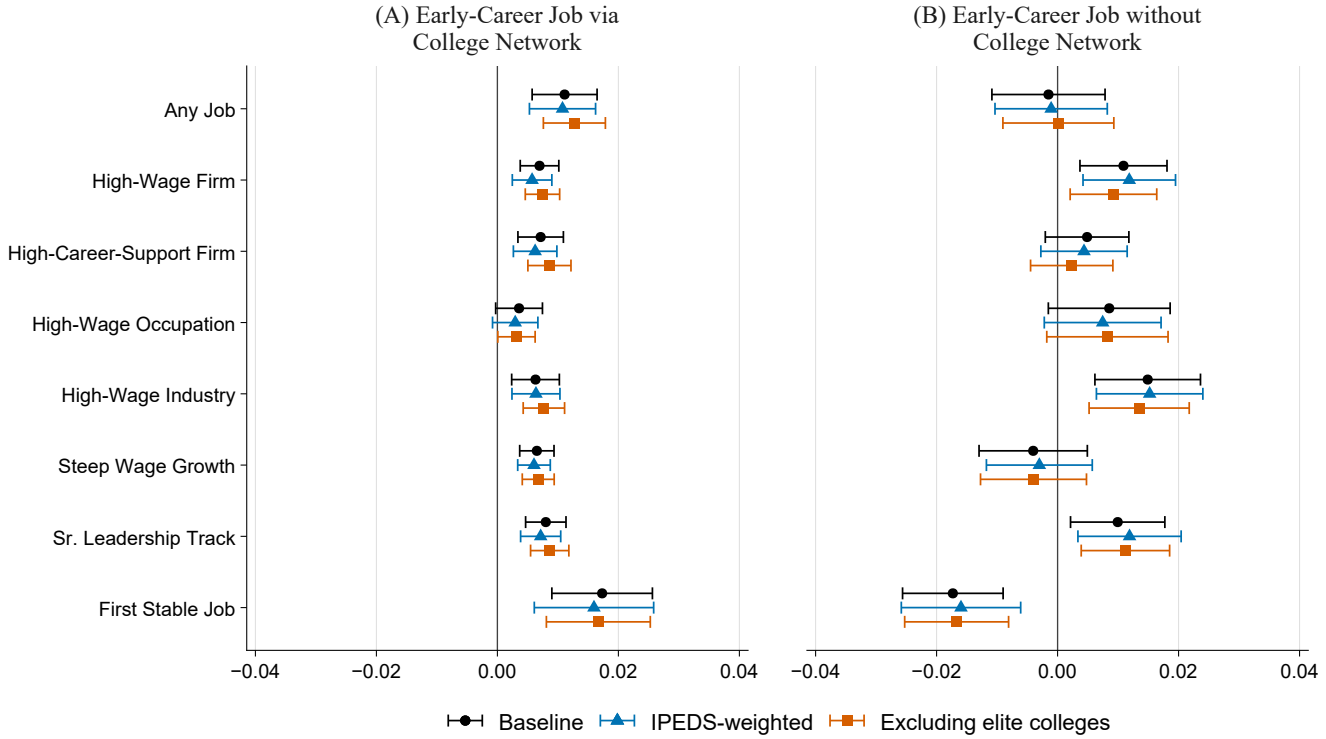
Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 1, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For the job-based promotion measure, the first time interval refers to 2–6 years after expected graduation. Baseline: estimates shown in Figure 1. IPEDS-weighted: observations are weighted by gender and cohort size using IPEDS enrollment data. Excluding elite colleges: Ivy Plus and other elite colleges are excluded from the sample. See the notes of Figure 1 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure G2: Facebook Access and the College-to-Work Transition – IPEDS Weights and Excluding Elite Colleges



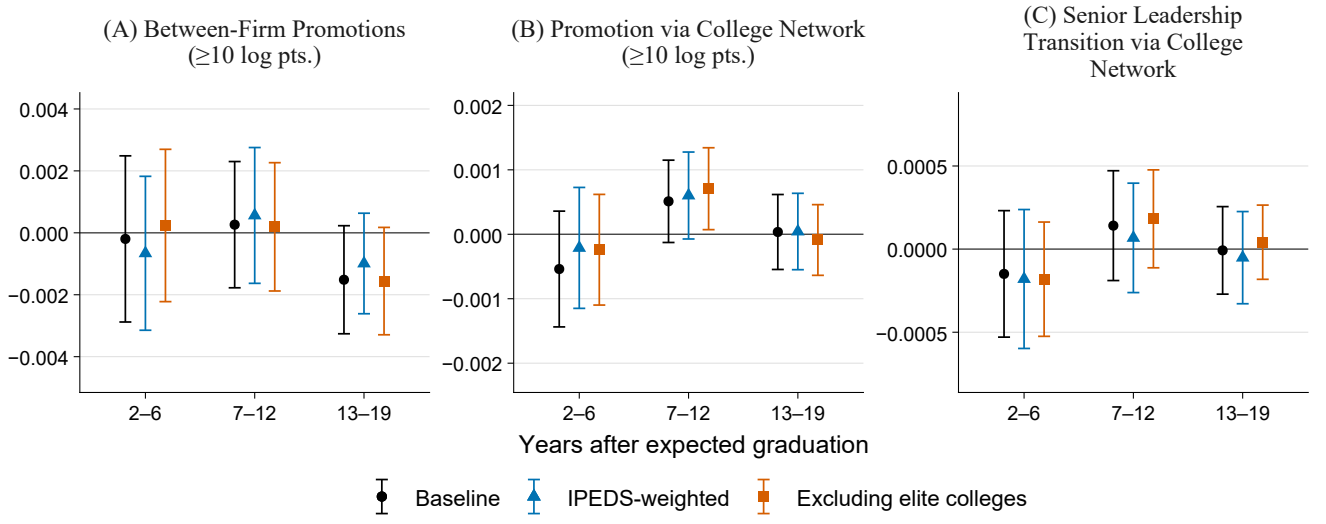
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Figure G3: Facebook Access and Early-Career Job Help in the College Network – IPEDS Weights and Excluding Elite Colleges



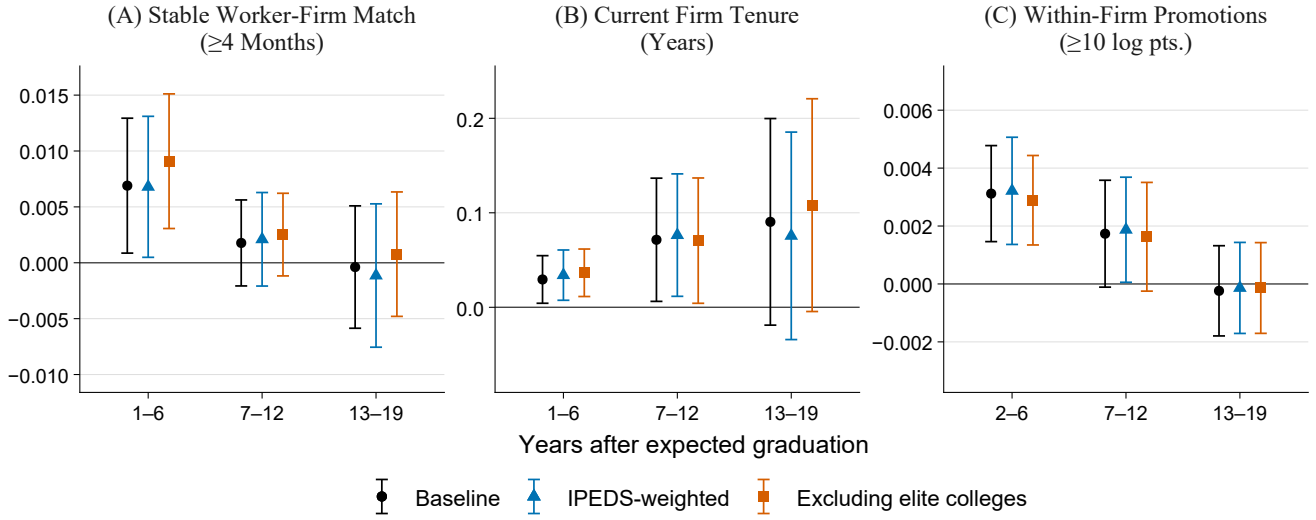
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Figure G4: Facebook Access, Firm Mobility, and Long-Run Job Help in the College Network – IPEDS Weights and Excluding Elite Colleges



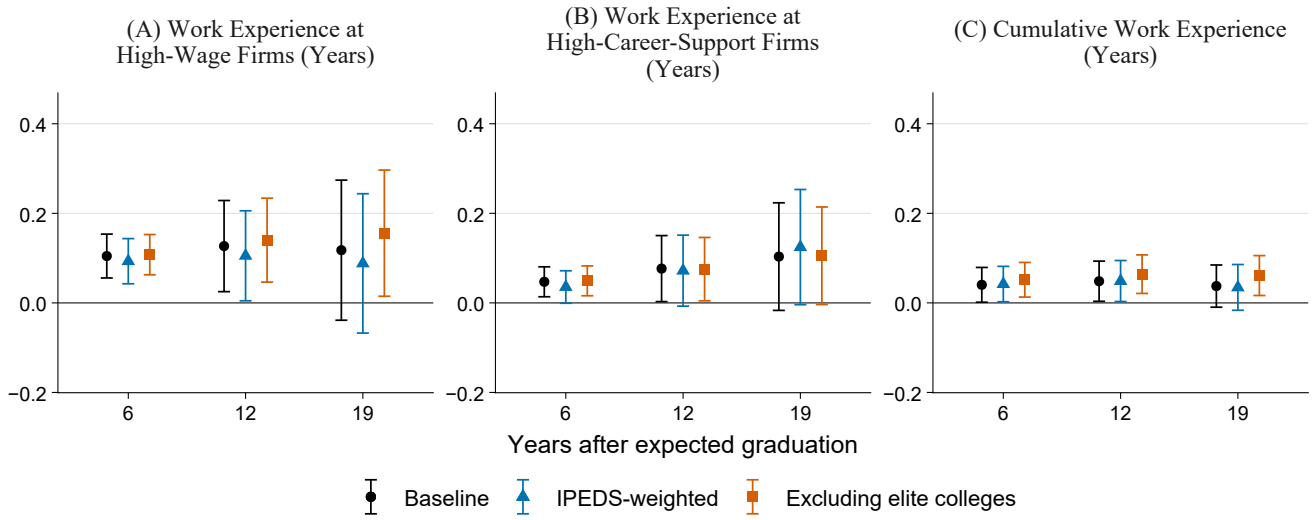
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Figure G5: Facebook Access and Worker-Firm Match Stability and Productivity – IPEDS Weights and Excluding Elite Colleges



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 3, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates are pooled over the following intervals after expected graduation: 1–6, 7–12, and 13–19 years. For within-firm promotions, the first time interval refers to 2–6 years after expected graduation. Baseline: estimates shown in Figure 3. IPEDS-weighted: observations are weighted by gender and cohort size using IPEDS enrollment data. Excluding elite colleges: Ivy Plus and other elite colleges are excluded from the sample. See the notes of Figure 3 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.

Figure G6: Facebook Access, Firm Sorting, and Human Capital Accumulation – IPEDS Weights and Excluding Elite Colleges



Notes: Point estimates and 95% confidence intervals for the outcomes of Figure 4, each from a separate DiD regression using the imputation estimator of Borusyak *et al.* (2024) at the college*cohort*year level. Estimates refer to 6, 12, and 19 years after expected graduation. Baseline: estimates shown in Figure 4. IPEDS-weighted: observations are weighted by gender and cohort size using IPEDS enrollment data. Excluding elite colleges: Ivy Plus and other elite colleges are excluded from the sample. See the notes of Figure 4 for the definitions of the outcomes. Standard errors are clustered at the college level. *Data source:* Revelio Labs.